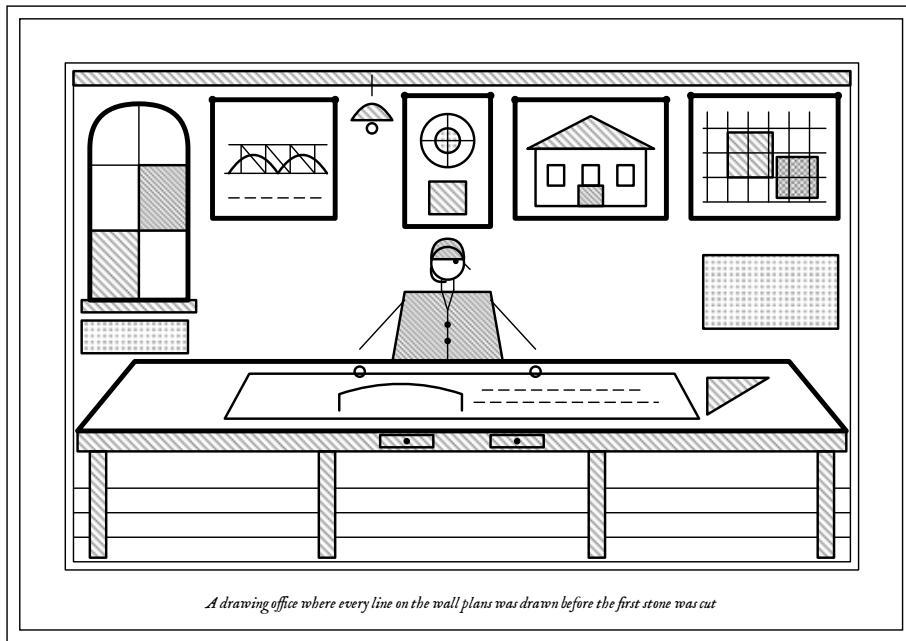




CHAPTER III

The Functions of Management and How Management Is Judged

Four kinds of work, three ways to ask: was it done well?



This chapter names the four kinds of work that make up management and shows how they repeat in a cycle. It then teaches three measures of results: productivity, effectiveness and efficiency. It ends by asking whether management is a science, an art or a profession, and by introducing the principle of management.

THE ELEMENTS OF MANAGEMENT



Being a Plain Account of Planning, Organising, Directing and Controlling

T. TAN, S. NILL

organisation. Human skill is needed at every level. Later teachers have kept this pattern in many books. It is the pattern that this chapter has used. The article appeared in volume 33, issue 1, of the journal, on pages 33 to 42. Its importance lies in its plain claim: a manager's success depends on what the manager can do, and what a manager can do can be taught.

REVIEW QUESTIONS

- 2.1 Place each job at a level: the head of a bank's whole network of branches; the manager of the loans department of one branch; the supervisor of the cashiers at the counter.
- 2.2 Explain why a top manager needs more conceptual skill than a first-line manager.
- 2.3 Name the four kinds of resource that a manager handles, and give a short example of each from a bookshop.
- 2.4 Explain the difference between a functional manager and a general manager, with one example of each.

Four questions return to the main ideas of the chapter.

WORKED CASE: SKILLS IN ONE WEEK AT MEGHNA AND SONS

This worked case applies the skills of this chapter to one documented week. It does not model a written answer. It shows one way of reasoning, step by step. *Situation.* Read the case file below first.

Case file, illustrative. Time use of Nurul, the top manager of Meghna and Sons, in a typical week of 48 hours: 20 hours planning and dealing with publishers; 16 hours on the shop floor; 12 hours on staff and money matters.

Question. Which skill does each block of Nurul's week mainly use? Does the pattern fit the usual mix for a top manager?

Reasoning. We take the three blocks one at a time.

Step 1. Planning, and dealing with publishers, 20 hours. Planning the shop's year means seeing the whole shop and how its parts connect, so this block uses mainly conceptual skill. Dealing with publishers also needs human skill, because it means working with people outside the shop.

Step 2. The shop floor, 16 hours. Selling, shelving and serving customers are the methods of specific jobs, so this block uses mainly technical skill.

Step 3. Staff and money matters, 12 hours. Working with the staff uses human skill. Keeping a record of money uses some technical skill. We call this block mainly human skill.

Common slips. Readers often give each block of the week to one skill only. Most blocks use more than one skill, so we name the main skill and note the second. Readers also forget that the pattern for a top manager is a general pattern, and not a rule that every top manager must follow.

Step 4. The usual pattern says that a top manager needs a small amount of technical skill. Nurul spends 16 of 48 hours on technical work, which is one third, because $16 \div 48 = 1 \div 3$. This is a larger share than the pattern suggests. The reason is the size of the shop. With few employees, the top level and the shop floor overlap, in the same way that Rafiq's work reaches two levels.

Step 5. The same week can be seen from other viewpoints. The employees see the 16 hours on the shop floor as help at busy times, and as a manager who knows the work. The publishers see the 20 hours as dealing with the person who decides the shop's orders. Nurul, as owner, knows that all three blocks answer to the same duty: results for the shop. Each viewpoint shows a different side of the same week.

Conclusion. Nurul's week uses all three skills. Conceptual skill takes the largest block, as the pattern predicts for a top manager. The large technical block is an exception explained by the small size of the organisation.

Check. The three blocks add to the week: $20 + 16 + 12 = 48$ hours. Each hour has been given to one skill, so nothing is missing or counted twice.

The worked case applies the three skills and the pattern by level to one week.

The case file records how Nurul used a typical week of 48 hours.

The question asks which skill each block uses and whether the pattern fits a top manager.

The reasoning takes each block in turn.

Step one: planning and publishers call mainly for conceptual skill.

Step two: the shop floor calls for technical skill.

Step three: staff and money matters call for human skill and some technical skill.

A common slip is to give every hour to one skill only.

Step four: we compare the week with the usual pattern for a top manager.

Step five: the week looks different from the owner's side and the employees' side.

The conclusion states what the week shows and where the pattern bends.

The check confirms that the three blocks add to the whole week.

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each level, the results depend on people. The table below gathers the three steps. Read each row from left to right: the level, then the size of need for each skill.

SKILL NEED BY LEVEL

LEVEL	TECHNICAL SKILL	HUMAN SKILL	CONCEPTUAL SKILL
Top manager	Small	Large	Large
Middle manager	Medium	Large	Medium
First-line manager	Large	Large	Small

The figure below shows the same pattern. Read it from top to bottom. Each row is one level, drawn wider as it goes down. The three bars on the right show how much of each skill that level needs. A longer bar means a larger need. Notice that the bar for human skill has the same length in every row.

The figure shows the same pattern as bars, and it is read from top to bottom.

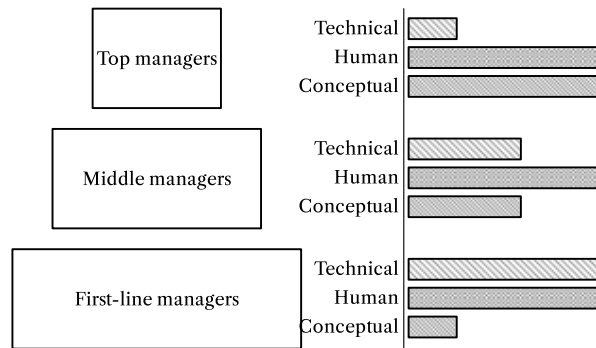


FIG. 2.1. Three levels of management and the skills each needs

IN SHORT

The need for technical skill falls as we go up the levels, the need for conceptual skill rises, and the need for human skill stays large at every level.

Fourth, the exceptions. The table shows a general pattern, and not a law. In a very small organisation, the top manager may still perform technical work every week, because there are too few people to share it. A skilled specialist who is promoted often arrives with plenty of technical skill and little human skill, and must learn the second one. Finally, skills can be learned. A person who is weak in one skill is not fixed in that weakness, and an organisation can train its managers.

Before moving on. Technical skill uses the tools and methods of a job. Human skill works with people. Conceptual skill sees the whole organisation. The mix changes with level, but human skill stays large at every level. *You should now be able to say which skill each level needs most.*

Step four: the pattern has exceptions, and a beginner should know them.

The recap repeats the three skills and the pattern by level.

At the lowest level stands the **first-line manager**. A first-line manager is a manager who directly supervises employees doing the day-to-day work. The supervisor of the cashiers at a bank counter is a first-line manager. The people under this manager are not themselves managers, so this level is the closest to the customer and to the daily work.

One can picture the same ladder in a railway company of the 1800s. The directors, meeting in a boardroom, set the direction of the whole company. A station master directed the work of one station. A foreman supervised a gang of workers on the track. Each of these three saw a different part of the business, and each answered for a different part of its results.

Case file, illustrative. Roles at Meghna and Sons in 2026. Top: Nurul sets prices, hires and approves loans. Middle: Rafiq makes the weekly stock plan and supervises 5 staff; Tariq handles accounts and online orders. First-line work: Rafiq directly supervises 3 sales assistants, 1 cashier and 1 stock clerk.

Read the case file with care, because Rafiq appears twice. His weekly stock plan is middle-level work, because it turns the owner's direction into a plan. His direct supervision of five employees is first-line work. Many students find this confusing, because they expect each person to have one level. In a small organisation the levels are thin, and one person often covers two of them. The level belongs to the work, and not only to the person.

PAUSE AND CHECK 2.P1 In the case file, at which level does Nurul work? At which two levels does Rafiq's work stand?

Levels describe height in the chain. Managers also differ in width. A *functional manager* is in charge of one kind of work, such as accounts, sales or stock. A *general manager* is in charge of several kinds of work together, and answers for a whole unit. A department head in a large company is a functional manager, and the head of a whole branch is a general manager. At Meghna and Sons, Tariq's work lies in two specialised areas, accounts and online orders, while Nurul answers for the whole shop. Height and width are separate ideas, so a middle manager can be either kind.

The levels also look different from different viewpoints. An employee takes daily directions from a first-line manager and may rarely meet the top manager. A customer meets the first-line level most often, at the counter or on the shop floor. The owner expects the top manager to answer for the whole organisation, as Chapter 1 explained when it described responsibility.

Before moving on. A top manager sets the direction of the whole organisation. A middle manager turns the direction into plans for departments. A first-line manager directly supervises the employees who do the daily work. A functional manager looks after one kind of work, and a general manager looks after several. *You should now be able to say what each of the three levels does.*

* * *

First-line manager. A manager who directly supervises employees doing the day-to-day work.

The first-line manager directly supervises the employees who do the daily work.

A railway company of the 1800s gives a historical picture of the three levels.

The case file shows all three levels in one small shop.

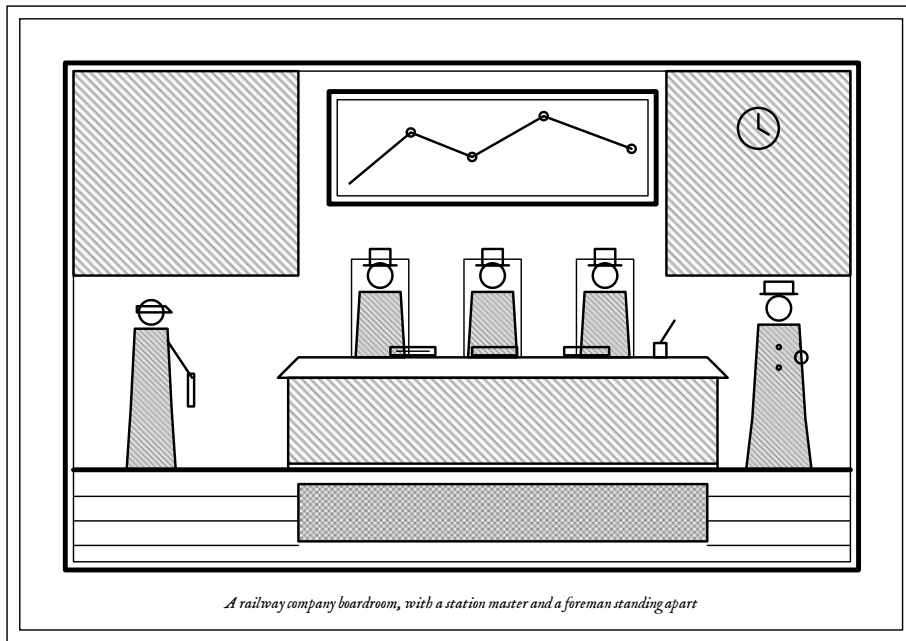
One person can stand at two levels, as Rafiq does in the case file.

Managers also differ in the width of their area, apart from their level.

Each level looks different from the employee's side, the customer's side and the owner's side.

The recap repeats the three levels and the two kinds of width.

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CHAPTER 11

Managers and Their Skills

Does one kind of ability fit every kind of manager?



This chapter shows that managers work at different levels, and that each level calls for a different mix of abilities. It names three levels of manager, three kinds of skill, and the resources and roles that managers handle. It also asks what makes a manager effective.

1.T3 Why does a one-person stall still need management, although it has no employees?

1.T4 A village cricket club has members, a small fund, a ground and a wish to win its league. Name the organisation, two resources, and the person who would be the manager.

WHERE PEOPLE GO WRONG

Wrong: Management means giving orders. *Correct:* Giving orders is one small part of leading. Management also includes deciding what to do, arranging the work and checking the results.

Wrong: Only large businesses need management. *Correct:* Management is universal. A family shop, a stall and a village club all have resources and goals, so they all need it.

Wrong: Anyone whose job title includes the word manager is a manager. *Correct:* The test is the definition. A manager is responsible for the work of other people and for the results of their work. A title is only a clue.

Wrong: Responsibility and authority are the same thing. *Correct:* Responsibility is a duty to perform a task and to answer for it. Authority is the right to give orders and make decisions.

Wrong: A manager who works hard has done the job. *Correct:* A manager answers for results. Hard work that does not move the organisation toward its goals does not meet the manager's responsibility.

IN HISTORY

Robert Owen (1771–1858) was a Welsh-born manufacturer. On 1 January 1800 he became manager and part-owner of the New Lanark cotton mills in Scotland, beside the River Clyde near the town of Lanark. David Dale had founded the mills in the 1780s.

Owen did not manage only machines and output. He managed the people who worked there. He limited working hours and raised the standards of housing and cleanliness in the village. He opened a village store and a fund for the sick. He also established schools, including an infant school in 1816. When the mills stopped for four months, he continued to pay wages. His mills became famous, and visitors from many countries came to see how they were run.

Owen's reforms cost money, and some of his partners disagreed with the spending. Yet the mills continued to succeed as a business. His story shows a manager who answered for output and for people at once, and who also had to answer to the owners. It also shows that responsibility has more than one direction. A manager's duty reaches upward to owners, sideways to the work, and downward to the employees.

REVIEW QUESTIONS

1.1 Name four resources that a hospital uses, and say which kind of

Five common misunderstandings are stated and corrected.

A second misunderstanding limits management to large businesses.

A third misunderstanding trusts the job title.

A fourth misunderstanding confuses a duty with a right.

A fifth misunderstanding confuses effort with results.

Robert Owen managed a mill and its people together, and the case still teaches.

Owen managed people as well as output.

His partners disliked the cost, yet the business continued to succeed.

Three questions return to the main ideas of the chapter.

This chapter needs no earlier knowledge.

A family wedding shows that willing helpers are not enough.

A bookshop meets the same puzzle every day, and the answer has a name.

The book has a fixed way of teaching, and you can learn it now.

GO SLOWLY HERE.

A household already contains the idea we need.

Management. The work of planning, organising, leading and controlling people and resources to reach an organisation's goals.

Management is the name for this careful deciding, arranging, guiding and checking.

Management. Word in use. In daily life, to manage can

You will need: nothing. This is the first chapter, and it assumes no earlier knowledge of the subject.

Think of a family wedding. Someone must choose the date and the guest list. Someone must hire the hall, buy the food and collect the money. Many relatives are willing to help, and each one works hard. Yet on the morning of the wedding, nobody can say who decides what happens next, or who answers if the food is not enough. Willing people alone do not make the day succeed. Something more is needed.

Now picture a small bookshop with seven employees. Every day the same questions return. Which books should we order? Who serves the customers, and who counts the cash? What happens when a school asks for a large order by Friday? If nobody has the task of answering such questions, the employees work hard but not together. The shop loses time, money and customers. If someone has that task, the same seven people achieve far more. The work that answers these questions has a name: management. This chapter builds the idea slowly, one step at a time.

You will find the same signs in every chapter. A new word appears in bold, and its meaning follows at once. A short note in the side column repeats that meaning. At the end of each section, a paragraph beginning *Before moving on.* repeats the main points. Short questions marked *Pause* let you test yourself before you continue. Cases about a bookshop called Meghna and Sons are labelled *Case file, illustrative.*, because the shop is an illustration and not a real organisation. Where you see the note *Go slowly here*, read each sentence twice.

* * *

1.1 THE IDEA OF MANAGEMENT

We begin with something every reader knows: a household. A family has a limited income, a few members who can work, a home and some wishes. They want food on the table, school fees paid and the rent settled. Someone must decide how to spend the money. Someone must decide who does which task. Someone must also check, at the end of the month, whether the plan worked. When nobody does these things, bills are missed and tasks are repeated. When someone does them with care, the same income lasts longer.

The same work takes place in a business, a school or a hospital, and there it has a special name. **Management** is the work of planning, organising, leading and controlling people and resources to reach an organisation's goals. In a bookshop, management includes choosing which books to order and checking that the day's takings are correct.¹

You may know the verb *to manage* from daily speech, as in "she manages on a small income". That meaning is about coping. The ¹ People also use the word management for the managers as a group. This book uses *management* only for the work, and *manager* for the person.

is responsible for the till must count the cash correctly and explain any difference.

A manager answers for results. It is not enough to say that the employees worked hard. If the sales target is missed, the manager must explain why, and say what will change. The manager's responsibility covers the work of other people, as the definition of a manager says.

A manager answers to owners, employees and customers. The table below shows what each group expects. Read each row as: the group, then what the manager answers for.

WHAT A MANAGER ANSWERS FOR

TO WHOM	WHAT THE MANAGER ANSWERS FOR
Owners	Results, and the care of the owners' money and goods
Employees	Clear tasks, fair treatment and pay on time
Customers	Goods and service as promised

Responsibility looks different from the manager's own chair. The manager cannot say that a mistake belongs to the assistant who made it, because the manager answers for the assistant's work. Many people find this unfair at first. It becomes reasonable when you remember the definition: a manager is a person who gets results through other people, and the results belong to the person who directs the work.

Responsibility is not the same as authority. Authority means the right to give orders and make decisions. Chapter 14, 'Authority, Delegation and Coordination', teaches it. A manager may be given a duty without receiving the right to act. That gap causes trouble, and the later chapter explains how to avoid it.

Before moving on. Responsibility is a duty to perform a task and to answer for it. A manager answers for results to owners, employees and customers.

* * *

WORKED CASE: MEGHNA AND SONS

This worked case applies the ideas of the chapter to one documented organisation. It does not model a written answer. It shows one way of reasoning, step by step. *Situation.* Read the case file below first.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna, the owner-manager. Position on 1 January 2026: seven staff are Rafiq Meghna, store manager; Tariq Meghna, accounts and online orders; three sales assistants; one cashier; and one stock clerk. Resources: stock Tk 3,500,000; fittings Tk 600,000; seven staff. The sales target for 2026 is Tk 15,840,000, which is 10% above the 2025 sales of Tk 14,400,000.

Question. Which parts of the case file are the organisation, the resources, the manager and the goal? How was the target reached?

A manager answers for results, and not only for effort.

A manager answers to three groups, and each expects something different.

From the manager's own side, responsibility can feel heavy, and it needs support.

Responsibility is different from authority, which later chapters explain.

The recap repeats responsibility and its three audiences.

The worked case applies the definitions to a documented shop.

The case file records the shop's people, resources and sales target.

The question asks us to identify the organisation, the resources, the manager and

results through other people. *You should now be able to say that management is the work of directing people and resources toward an organisation's goals.*

* * *

1.2 NATURE OF MANAGEMENT

Is the management of a bookshop a different kind of work from the management of a hospital, a bank or a school? The question is practical. If the work were different each time, a book like this one would have little to teach. The answer is that the *nature* of management, meaning its main features, is shared. Four features appear in every organisation. We take them one at a time.

The first feature is that management is universal and continuing. *Universal* means found everywhere. A family shop, a government office, a factory in Chattogram and a hospital in Edinburgh all need someone to decide, arrange, guide and check. The details differ, and we return to that point below, but the work is of the same kind. *Continuing* means never finished. A manager does not plan once and stop. Customers change, prices change and employees change, so the manager returns again and again to the same four kinds of work, for as long as the organisation lives. Think of a household cook: the meal is eaten, and tomorrow the cook plans again. A bookshop is the same. The books ordered on Monday are sold by Friday, and on Saturday the order must be placed again.

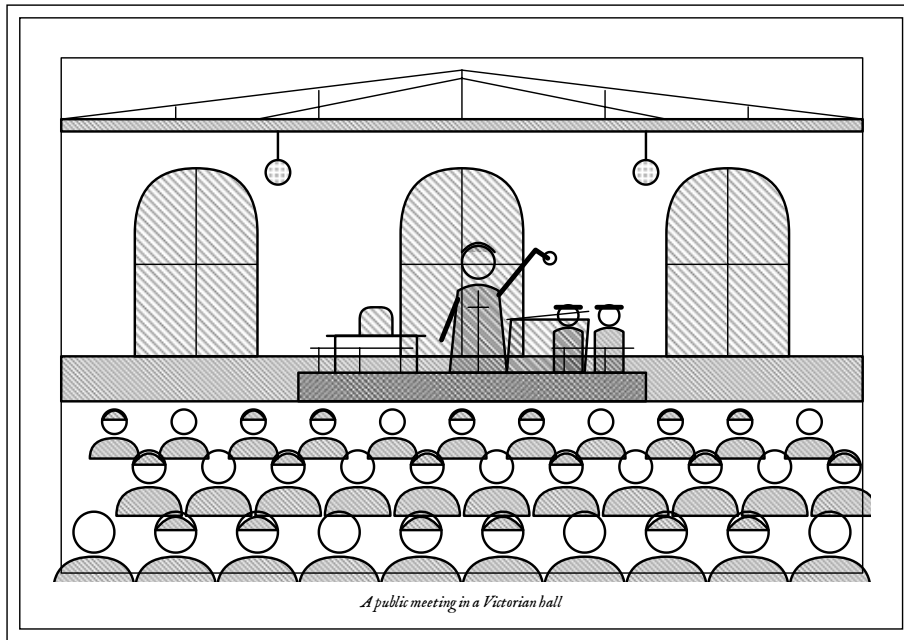
The second feature is that management is a group activity. A person who works entirely alone has nobody to direct. A group is different. Tasks must be shared, effort must be combined, and disagreements must be settled. Management is the work that does all three. In the bookshop, the sales assistants, the cashier and the stock clerk each perform a part. Management is what joins the parts into one shop. Without it, there is a collection of individuals and not an organisation.

The third feature is that management is goal-directed. It does not exist for its own sake. A manager always works toward the goals of the organisation, such as a sales target for the year. A manager who is busy but moves toward no goal is not performing management.

The table below shows the four features together. Read each row from left to right: the feature, its meaning, and an example from a bookshop.

FOUR FEATURES OF MANAGEMENT

FEATURE	MEANING	BOOKSHOP EXAMPLE
Universal	Found in every organisation	The owner decides which books to order
Continuing	Never finished	Each new day brings new orders to place
Group activity	Needs people working together	Assistants, cashier and clerk each perform a part
Goal-directed	Aims at goals	The shop aims to sell more



CHAPTER VI

Social Responsibility, Ethics and Sustainability

What does an organisation owe to people it never invited to its table?



This chapter asks what an organisation owes to society beyond its profit. It explains the duty of organisations to society, the moral rules that guide business choices, and the goal of leaving the world fit for later generations. A bookshop in Dhaka and a disaster at Savar serve as cases.

YOU WILL NEED

Chapter 1 "What Management Is", section 1.2 "Nature of Management"; Chapter 2 "Managers and Their Skills", section 2.1 "Levels and Kinds of Manager".

Two cooks, two different failures: which one managed better?

Each cook failed in a different way, and each way needs its own name.

Behind the feast stand four kinds of work, and a question: do they happen once, or repeat?

The chapter answers three questions: what managers do, how results are judged, and what kind of knowledge management is.

GO SLOWLY HERE.

A family journey contains all four kinds of work, though nobody in it has studied management.

Recall: nature of management, Chapter 1, section 1.2 "Nature of Management".

Chapter 1 said what management is; this chapter says what managers do.

Two cooks are asked to prepare a wedding feast for 200 guests. The first cook finishes on time, and all 200 guests are served. Afterwards the kitchen shows a large loss: rice thrown away, trays burnt, and many hours of paid overtime. The second cook wastes nothing. The costs are low and the pots are clean. But the food runs out after 150 guests, and 50 guests go home hungry. Which cook managed better?

Most readers answer that neither did well. The first cook reached the goal at a very high cost. The second cook used resources with great care but did not reach the goal. These are two different failures, and a manager who cannot tell them apart will repair the wrong one. To judge management, we need exact words for the different ways in which it can succeed or fail.

A second puzzle stands behind the first. Before the feast, somebody decided the menu and the day. Somebody divided the tasks among the helpers. Somebody kept the helpers working willingly through a hot afternoon, and somebody checked the pots while the cooking was in progress. These are four different kinds of work. Do they happen once, in order, like the steps of a recipe? Or do they come round again and again?

There is also a third question. Is cooking a feast a science, with fixed rules that give the same result each time? Is it an art, learnt by practice and feel? Or is it a profession, like medicine, with a licence and a code of conduct? This chapter answers all three questions. It names the four kinds of work, gives three measures for judging results, and asks what kind of knowledge management really is.

* * *

3.1 THE MANAGEMENT PROCESS

Before we use the technical words, consider a family travelling to a village for a wedding. Someone chooses the day, the route and the bus. Someone assigns the tasks: who buys the tickets, who packs the sweets, who looks after the grandparents. Someone keeps everyone cheerful when the bus is late. And someone counts the family at every stop, to be sure that nobody is left behind. No member of this family has studied management. Yet the family has used all four kinds of work that this section names.

In Chapter 1 you learned that management means reaching an organisation's goals through people and resources. That statement is correct, but it is broad. It does not tell us what a manager does between nine in the morning and five in the afternoon. This section divides the broad statement into four kinds of work and shows how they join together.

environment at once. The groups are tools for noticing, not boxes with rigid walls.

Some students believe that scanning predicts the future. It does not. It collects information about changes that have begun, so that managers meet them early. The value of scanning is a smaller surprise, not certainty.

IN HISTORY

Nokia is a company from Finland. For many years it was the largest maker of mobile phones in the world by number sold. In 2007 the American company Apple released the iPhone, a phone that worked as a small computer with a touch screen. Phones that used Google's Android software soon followed. Nokia chose Microsoft's Windows Phone software for its phones, and its first Windows Phone devices reached customers in late 2011.

On 3 September 2013, Nokia announced that it would sell its devices and services business to Microsoft, the American software company. The total price was €5.44 billion. Of that sum, €3.79 billion paid for the business and €1.65 billion paid for a licence to use Nokia's patents. About 32,000 Nokia employees were to move to Microsoft. On 19 November 2013, Nokia's shareholders approved the sale by a very large majority of the votes cast.

Writers still debate why the change happened. Some stress the company's software and the speed of its decisions. Others stress the force of the new competitors. We do not need to settle that debate here. The case shows three ideas from this chapter. A technological change altered what customers wanted. Rivals in the task environment moved quickly. A strategic response that comes late may leave few choices. Nokia remained in business after the sale, with its network equipment business and its patents.

REVIEW QUESTIONS

- 5.1 Sort these forces as internal or external: a rival's price cut; a new store manager; a change in school book rules; the shop's own opening hours.
- 5.2 In your own words, state the difference between the internal environment and the external environment. Explain how the boundary decides which is which.
- 5.3 Name the five groups of wider forces. Give one example of each for a bookshop.
- 5.4 Explain the difference between scanning, adapting and influencing. Say which one the price-match list of 3 March 2026 is.

Mistake four: believing that scanning predicts the future.

Nokia, a leader in mobile phones, sold its phone business in 2013 after a shift in its market.

The sale was announced on 3 September 2013 and approved by shareholders on 19 November 2013.

Writers still debate the causes, but the case shows the environment at work.

person. At Meghna and Sons, the sales target of Tk 15,840,000 gives a measure. When 2026 ends, actual sales can be compared with it, and any gap shows what needs correction.⁴

A CYCLE, NOT A LINE

The four functions do not run once, in a straight line, and then stop. Controlling produces information, and that information changes the next plan. In daily work the functions also overlap: a manager who holds a short meeting may be organising, leading and checking in the same ten minutes. Their weight differs by level. A top manager usually spends more time on planning, and a first-line manager more time on leading and checking daily work.

Other books use different lists, for example adding a fifth function called staffing. The four used here cover the same ground, because the work of finding and keeping employees falls within organising and leading. The list is a tool for thinking, not a law of nature.

Each person sees the cycle from a different position. The owner asks whether the cycle ends in the goals. The manager lives inside the cycle. The employee feels mostly its leading and controlling. The customer sees none of it directly and sees only the results: books on the shelf, a fair price and a correct bill.

Before moving on. The management process is the repeating cycle of four functions: planning, organising, leading and controlling. Each function is a main kind of work. They overlap and repeat, and their weight changes with the level of the manager. *You should now be able to say* what the four functions are, and why they form a cycle and not a line.

* * *

3.2 PRODUCTIVITY

Imagine two stalls in a market. The first stall has 2 sellers and sells 40 melons in a morning. The second stall has 3 sellers and sells 54 melons. The second stall sold more, so is it better run? Many students answer yes at once and then feel unsure, because the sellers are not equal in number. The unease is correct. We must compare the results with what was used to obtain them.

We need two plain words. *Output* is what an organisation produces or sells, such as melons, loaves or books. *Input* is what it uses to obtain that output, such as people's time, money, materials and machines. Now the puzzle resolves. The first stall sold 40 melons using 2 sellers: $40 \div 2 = 20$ melons for each seller. The second sold 54 melons using 3 sellers: $54 \div 3 = 18$ melons for each seller. The smaller stall obtained more output for each unit of input.

This ratio has a name. **Productivity** is the amount of output an organisation obtains for each unit of input it uses. At the stalls, productivity is 20 melons per seller for the first and 18 for the second. In words: productivity equals output divided by input. In symbols, $P = O \div I$, where P is productivity, O is output and I is input.

⁴ Controlling is taught in Chapter 18, "Controlling", section 18.1 "What Controlling Is".

Recall: levels of management, Chapter 2, section 2.1 "Levels and Kinds of Manager".

The four functions repeat, overlap and carry different weight at different levels.

Other books use other lists of functions; the four used here cover the same ground.

Owner, manager, employee and customer each see the cycle from a different position.

The management process is a repeating cycle of four functions.

GO SLOWLY HERE.

Two market stalls show why a bigger total is not always a better result.

Output is what an organisation produces; input is what it uses to produce it.

Productivity is output divided by input.

Productivity. The amount of output an organisation obtains for each unit of input it uses.

management taught earlier. A response that wins customers but loses money has not solved the problem.

Before moving on. Scanning means watching the outside world on a regular basis. Adapting means changing the organisation to fit an outside force. Influencing means acting on the outside force itself. A strategic response is the chosen combination of these, matched to the strength of the force and the power of the organisation.

* * *

WORKED CASE: SORTING THE OUTSIDE CHANGES OF 2025 AND 2026

This section applies the ideas of the chapter to the records of Meghna and Sons. It does not give a model answer. It shows one way of reasoning, step by step, so that you can follow each decision and then repeat it on a new situation.

Situation. The shop's records for 2025 and 2026 list four changes. First, online price cuts of about 15% on textbooks since March 2025. Second, textbook sales of Tk 4,300,000 in 2025 against Tk 5,100,000 in 2024. Third, a curriculum change from January 2026. Fourth, the departure of 2 of the 7 employees in 2025. A fifth record is a response: the price-match list for 200 titles, issued on 3 March 2026.

Question. For each change, decide whether it is internal or external. For each external change, name the group or groups to which it belongs. Then decide whether the response of 3 March 2026 is adapting or influencing.

Reasoning. Begin with the boundary test from section 5.1: can the managers change this condition directly? The departure of 2 employees fails the test as a cause, but passes it as a result, because the owner can hire and train. It is internal. The price cuts do not pass the test. The owner cannot command a rival's price list, so they are external. The curriculum change is external for the same reason. The fall in textbook sales is a result recorded in the shop's own books, but its cause lies partly outside the door, so we treat it as the effect of external forces.

Now we sort each external change into groups, using the five groups and the task environment. The table below shows the result. Read each row from left to right: the change, the inside or outside decision, and the groups that fit.

SORTING THE RECORDED CHANGES

CHANGE	INSIDE OR OUTSIDE	GROUPS THAT FIT
Online price cuts of about 15%	Outside	Economic, technological, task environment
Curriculum change from January 2026	Outside	Political and legal; task environment (schools)
Fall in textbook sales	Result in the shop's books	Effect of the forces above

The recap fixes the three ways of meeting the environment.

The worked case applies the chapter's ideas to the records of Meghna and Sons.

The situation lists every change in the record.

The question asks for three sortings of each change.

Reasoning begins with the boundary test: can the manager change it directly?

The table records the sorting of each change.

FOUR SITUATIONS COMPARED

SITUATION	GOAL REACHED?	INPUT USED
Effective and efficient	Yes	Little
Effective, not efficient	Yes	Much, with waste
Efficient, not effective	No	Little
Neither	No	Much, with waste

Two short questions separate the words: was the goal reached, and what did it cost?

A short test separates the words. For effectiveness, ask: was the goal reached? For efficiency, ask: how much input did reaching it cost? A manager can be effective without being efficient, as the first cook was. A manager can also be efficient without being effective, as an organisation is when it does the wrong job very cheaply. Good management aims at both.

IN SHORT

Effectiveness asks whether the goal was reached. Efficiency asks how much input was used to reach it. Productivity measures output for each unit of input.

PAUSE AND CHECK 3.P2 A charity delivers all 500 food parcels it planned to deliver, but spends twice its budget. Is it effective, efficient or both?

Before moving on. Effectiveness is how far goals are reached, whatever the cost. Efficiency is how little input a result needs. The two can differ, and a manager needs to judge each.

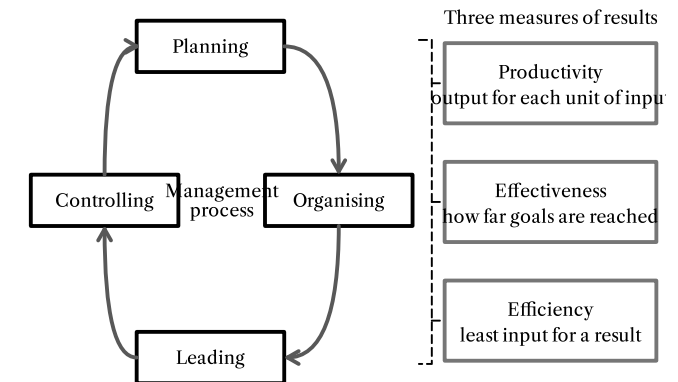


FIG. 3.1. Functions as a cycle, with the three measures beside it

To read the figure, begin with the left side. The four boxes are the functions, and each arrow leads to the next, so that the last arrow returns to the first. The right side lists the three measures with which the results of that cycle are judged.

* * *

The figure shows the cycle on the left and the three measures on the right.

The table below places the five groups side by side. Each row names a group, says in plain words what it covers, and gives an example for a bookshop. Read across a row to see one group, and down a column to compare the groups.

THE FIVE GROUPS OF WIDER FORCES

GROUP	WHAT IT COVERS	EXAMPLE FOR A BOOKSHOP
Economic	Money, prices, incomes, trade	A fall in family spending on books
Social and cultural	Tastes, beliefs, habits, education	A shift from printed books to screens
Political and legal	Rules and actions of public bodies	A new tax or a change in school books
Technological	Methods and machines	Online selling and phone payment
Global	Forces from beyond the country	The price of foreign money

One caution applies to the table. The groups overlap. A rival's online price cut is an economic force, because it changes prices. It is also a technological force, because online selling made it possible. It also belongs to the task environment, because the rival competes for the same buyers. A manager does not need to decide which single box is correct. The sorting is useful when it makes the manager notice every way in which the event reaches the organisation.

Case file, illustrative. Outside changes recorded at Meghna and Sons: online price cuts of about 15% on textbooks since March 2025; textbook sales of Tk 4,300,000 in 2025 against Tk 5,100,000 in 2024; and a curriculum change from January 2026.

Each recorded change reaches the shop through a different route. The online price cuts come from rivals and from technology. The fall in textbook sales shows how those forces reached the shop's own results. The curriculum change is a decision of public authority, and it falls within the political and legal group. In the Worked Case at the end of the chapter we shall sort these changes step by step.

Before moving on. The task environment is the set of outside groups that deal directly with the organisation. The wider forces fall into five groups: economic, social and cultural, political and legal, technological, and global. One event may belong to several groups at once.

* * *

5.3 MANAGING THE ENVIRONMENT

A manager who knows that a rival has cut prices has learned something useful. The next question is what to do about it. Managers have three broad ways of meeting the external environment: they can watch it, they can fit the organisation to it, and they can try to change it. We take them one at a time, and then we join them into a single plan of response.

The table sorts the five groups by what each covers.

The groups overlap, so one event can touch several of them.

The case file records the outside changes at Meghna and Sons.

Each outside change reaches the shop through a different group.

The recap fixes the five groups and the task environment.

Knowing the outside world is useful only if managers act on what they learn.

Management is partly a science, largely an art, and only partly a profession.

Management matters because it directs effort, saves resources and prepares an organisation for change.

A principle of management is a broad guide drawn from experience, to be adapted and not obeyed like a law.

Principle of management.
A broad guide to action drawn from experience, to be adapted and not applied like a law.

Management is important, and its principles are guides to adapt.

The first example is small and clean, so that only the method is new.

Before moving on. Management uses tested knowledge as a science does, and personal judgement as an art does. It is only partly a profession, because it usually needs no licence and no governing association.

* * *

3.5 IMPORTANCE AND PRINCIPLES IN BRIEF

Management is important for three plain reasons. It directs the effort of many people toward common goals. It helps an organisation use limited resources with less waste. And it prepares an organisation to meet change, such as new laws, rivals and technologies. Without it, the same people and materials produce less, and the cost falls on owners, employees and customers.

Over many years, managers have collected broad guides drawn from experience. Each is a **principle of management**. A principle of management is a broad guide to action drawn from experience, to be adapted and not applied like a law. Water boils at 100 °C at sea level every time. A principle such as "divide work among specialists" may suit a factory and fit badly in a shop with seven people.⁵

Before moving on. Management directs effort, saves resources and prepares for change. A principle of management guides action and must be adapted to the situation.

* * *

WORKED EXAMPLE AND WORKED CASE

WORKED EXAMPLE: PRODUCTIVITY

Many students find rounding confusing here, because a result such as 5,857.14 cannot be a number of copies in daily life. We round at the end and never in the middle. The table below sets out the working in six parts. Read it from the top: the first column names the part, and the second column gives its content.

WORKED EXAMPLE 1: COPIES PER EMPLOYEE

PART	WORKING
Given	2025: 41,000 copies sold by 7 employees. 2026 forecast: 45,100 copies by 7 employees.
Find	Copies per employee in each year, and the change.
Formula	In words: productivity = output ÷ input. In symbols: $P = O \div I$, where P is productivity, O is output in copies and I is input in employees.
Steps	2025: $P = 41,000 \div 7 = 5,857.14$, so 5,857 copies per employee. 2026: $P = 45,100 \div 7 = 6,442.86$, so 6,443 copies per employee. Change: $6,443 - 5,857 = 586$ copies per employee. Percentage change: $586 \div 5,857 = 0.1000$, which is 10.0%.

⁵ A well-known list of such principles is taught in Chapter 4, "The Evolution of Management Thought", section 4.2 "Administrative Principles".

The departure of 2 employees belongs to the internal environment. The owner and the store manager can respond by hiring, by training, or by dividing the work again. Notice the word *directly*. Managers did not cause the departures, but they can act on the result with their own authority. That power to act directly is what marks a condition as internal.

The forces outside the door form the **external environment**. The external environment means all outside forces, such as customers, rivals, laws and technology, that affect an organisation. A manager cannot give these forces an order. A manager can, however, study them and prepare. A customer who changes her tastes, a rival who lowers a price and a government that issues a new rule all belong here. Each of them reaches the organisation from beyond the door.

IN SHORT

Internal environment: a manager can change it directly. External environment: a manager can study it, prepare for it, and sometimes influence it, but cannot command it.

PAUSE AND CHECK 5.P1 A new bookshop opens two streets away from Meghna and Sons. Is this change internal or external? Give one reason.

The line between the two environments is called the boundary. It is not a wall, because goods, money, information and people cross it every day. It also follows control, not distance. A supplier in the next street is outside the boundary, because the owner cannot direct that supplier. A branch in another town is inside the boundary if the same organisation owns it and directs it.

Before moving on. The internal environment is what managers can change directly. The external environment is what they can only study, prepare for or influence. The boundary between them follows control, not distance.

* * *

5.2 COMPONENTS OF THE EXTERNAL ENVIRONMENT

Suppose a manager writes down every outside force that touches a bookshop in one year. The list would include tastes, prices, laws, machines, rivals, weather and world events. A list that long is hard to use. Managers therefore sort outside forces into groups, and watch each group in turn. Writers divide the groups in different ways. This book uses five, and the groups are a convenience for thinking, not a law of nature.

Before we meet the five groups, one more distinction helps. Some outside forces are people and bodies that deal directly with the organisation. They form the **task environment**. The task environment means the outside groups that deal directly with an organisation, such as customers and suppliers. For a bookshop, the

An employee leaving is an internal change, because managers can act on it directly.

External environment. All outside forces, such as customers, rivals, laws and technology, that affect an organisation.

The external environment is every outside force that affects the organisation.

The line between inside and outside is a boundary, and it follows control, not distance.

The recap fixes the two environments and the boundary between them.

The outside forces are too many to study as one heap, so managers sort them.

Task environment. The outside groups that deal directly with an organisation, such as customers and suppliers.

The task environment holds

Rung 4 joins the cycle and the three measures in one chain of events.

The chapter in brief.

Mistake: the functions happen once, in a fixed order.

Mistake: more output always means higher productivity.

Mistake: effectiveness and efficiency are two names for one idea.

Mistake: management is a profession like medicine.

Mistake: a principle of management is a law.

The Toyota Production System

Rung 4, a case that combines earlier ideas. Nurul sets the target (planning). Shelves are relabelled (organising and leading). Search times are compared month by month (controlling). The target tests effectiveness, the relabelling raises efficiency, and the monthly comparison closes the cycle, so that the next plan can use the result.

* * *

SUMMARY

Management work falls into four functions: planning, organising, leading and controlling. They form a repeating cycle called the management process. Results are judged by three measures: productivity (output for each unit of input), effectiveness (how far goals are reached) and efficiency (how little input a result needs). Management uses a science and an art, but it is only partly a profession. A principle of management is a guide to adapt, not a law.

THINK IT THROUGH

3.T1 A delivery firm promises to deliver 300 parcels in a day. It delivers all 300, but 4 vans each drive twice the distance needed, because the routes were not planned. Explain why the firm was effective but not efficient, and name the function whose improvement would help most.

3.T2 Meghna and Sons is forecast to reach 6,443 copies per employee in 2026. Suppose the shop hires 1 more employee and sells 45,100 copies. Find the new productivity and say whether it is higher or lower than in 2025.

WHERE PEOPLE GO WRONG

Mistake: "The four functions are steps in a recipe, done once in order." Correction: they repeat and overlap, and controlling gives the information for the next round of planning.

Mistake: "More output always means higher productivity." Correction: productivity is output for each unit of input. If input grows faster than output, productivity falls.

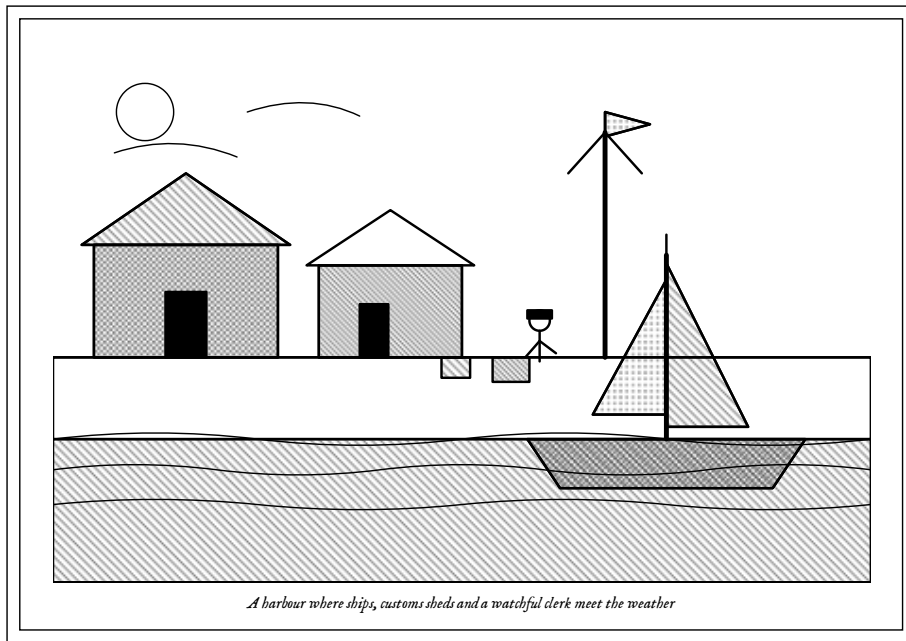
Mistake: "Effective and efficient mean the same." Correction: effectiveness asks whether the goal was reached; efficiency asks how little input was used.

Mistake: "Management is a profession, like medicine." Correction: in most countries a manager needs no licence and answers to no single governing association.

Mistake: "A principle must be followed in every case." Correction: a principle is drawn from experience and must be adapted to the situation.

IN HISTORY

Toyohi Ohno (1912 to 1990) joined the Toyota company and managed



CHAPTER V

The Environment of Organisations

Can any organisation live apart from the world around it?



Every organisation stands inside a larger world of customers, rivals, laws and machines. This chapter divides that world into its parts, shows how managers study it, and explains the ways in which they respond to it.

Conclusion. The shop saves 50 minutes a day, which is 5/6 of an hour. The change belongs to scientific management, supported by a rule list that applies Fayol's discipline and order.

Check. Before the change, 10 cartons took $10 \times 14 = 140$ minutes. After the change, they took $10 \times 9 = 90$ minutes. The difference is 140 minutes minus 90 minutes, which is 50 minutes. The answer agrees with the first calculation.

* * *

SUMMARY

Management thought grew as factories grew. Taylor's scientific management measured work to find one best method, using time and motion study. Fayol looked at the whole organisation and gave fourteen principles in three groups. Weber described bureaucracy, and Mayo's followers formed the human relations school. Later thinkers added the management science, systems and contingency approaches. No approach replaced the others; managers choose by situation.

THINK IT THROUGH

These four questions start with the shop's case and move to new settings. Each builds on the one before.

4.T1 In the shop's method change, which of Taylor's four principles does the fixed shelf-position list best show, and why?

4.T2 Suppose the unloading time after the change had been 11 minutes and not 9 minutes. How many minutes a day would the shop save?

4.T3 A busy kitchen must prepare many orders in a rush, the cooks differ in skill, and orders change every minute. Which school of thought suits it best, and what would you borrow from another school?

4.T4 A shop times its employees for every task and also wants them to feel part of one team. Where might scientific management and the human relations school pull against each other, and how could a manager reduce the conflict?

WHERE PEOPLE GO WRONG

Some students think Taylor wanted only faster work at any cost. This is half true. He also proposed careful selection, training and higher pay for employees who reached the standard. The fair criticism is different: his method treated employees as parts of a machine and ignored their feelings.

Some students think bureaucracy always means delay and waste. In daily speech it often does. In Weber's sense it is a design with written rules, clear levels and appointment by skill, meant to give fairness and steady work. Delay appears when such a design is applied without judgement.

The conclusion gives the saving and the school.

The check repeats the arithmetic from the other side.

The chapter in a few lines.

Four questions that change the situation Step by Step.

Taylor is wrongly remembered as caring only for speed.

Bureaucracy in Weber's sense is not the same as slow paperwork.

YOU WILL NEED

This chapter uses two earlier ideas. The first is whether management can claim to be a science, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.4 'Science, Art or Profession'. The second is the list of activities that every manager performs, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

Two shops with the same books and staff can earn very different results.

For most of history nobody studied management; people learned by watching.

Thinkers who share one main idea are placed together in one school.

School. Word in use. In daily life, a place where children learn. In management, a group of thinkers who share one main idea about how to manage.

Picture two bookshops in the same street. They stock the same titles, charge the same prices and employ the same number of people. Yet at the end of the year one shop has sold far more books than the other. The books have not changed, so something else has. What is it? Many students first answer "luck" or "a better owner". Both answers may hold some truth, but neither can be taught, and nobody can learn luck.

For thousands of years, people managed armies, farms, trading houses and building sites without a book of management to read. A young person learned by watching an older one, and good methods stayed secret inside families. This worked while work was done in small groups. It failed when factories gathered hundreds of people under one roof, because no one could learn fast enough by watching. Owners needed ideas that could be written down, tested and passed on. This chapter tells how such ideas were found, and in what order.

Historians place thinkers who share one main idea in the same school of thought. We shall meet several schools, one after another. The timeline below is a map of the road. Read it from top to bottom: each row is one school, each bar shows the years when its ideas were most active, and the scale along the bottom gives the dates. Do not worry about the names yet; each is explained in its own place.

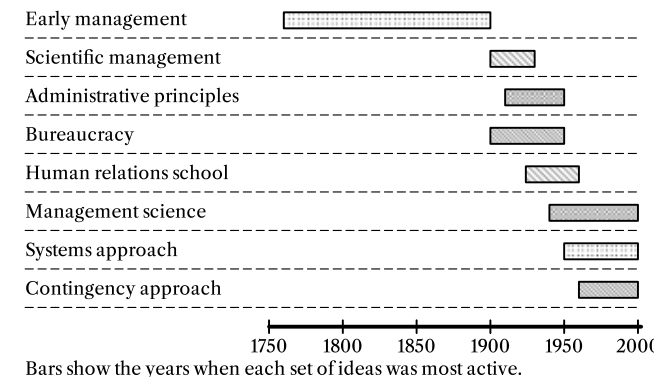


FIG. 4.1. *Timeline of the schools of management thought*

4.3 BUREAUCRATIC AND BEHAVIOURAL THOUGHT

Max Weber (1864 to 1920), a German scholar of society, studied how large organisations could treat people fairly. He described **bureaucracy**. Bureaucracy means a system of management built on written rules, clear levels of authority, and appointment by skill and not by family favour. In daily speech the word often means slow paperwork, but Weber meant a design for fairness and steady work. A shop with a written rule list and clear roles takes a small step in this direction.

Rules and measurement were not the whole story. Studies at the Hawthorne Works of the Western Electric Company, near Chicago, led thinkers such as Elton Mayo to the **human relations school**. The human relations school is a school of thought holding that employees' feelings and social bonds strongly affect output. For example, an employee who feels respected by the group may work harder than one who feels ignored, even when pay is the same. The studies are told in the section 'In History' at the end of this chapter.

Behavioural science grew from this school. Behavioural scientists are psychologists, sociologists and others who study how people behave, and they test their ideas by careful research and not by opinion. Their findings help managers understand why employees act as they do. Taylor trusted method and pay; Mayo and the behavioural scientists looked at people.

Before moving on. Bureaucracy rests on written rules, clear levels of authority and appointment by skill. The human relations school holds that feelings and group bonds affect output. Behavioural science studies human behaviour by research.

* * *

4.4 MODERN APPROACHES

During the Second World War (1939 to 1945), teams of scientists in Britain and the United States used mathematics to plan the use of ships, aircraft and supplies. After the war, businesses adopted the same methods, and the **management science approach** was born. The management science approach means using mathematical models and data to solve management problems. For example, a shop can use past sales data to decide how many copies of a title to order. It works well when a problem can be measured, but it cannot measure feelings or judgement.

The systems approach, popular from the 1950s, views an organisation as a set of connected parts that take in resources, change them, and deliver results. In a bookshop, buying, shelving, selling and accounts are parts of one chain: a delay in one part appears later in another. The approach reminds managers to look at the whole and at the links between the parts.

The earlier thinkers each looked for one best way. From the 1960s, researchers found that no single method suited every situ-

Weber described a way of organising by written rules, levels and skill.

Bureaucracy. A system of management built on written rules, clear levels of authority and appointment by skill.

Studies of employees' feelings and group bonds led to a new school.

Human relations school. A school of thought holding that workers' feelings and social bonds strongly affect output.

Behavioural science tests ideas about human behaviour by careful research.

Section 4.3 in three lines.

Management science uses mathematics and data to solve problems.

Management science approach. The use of mathematical models and data to solve management problems.

The systems approach sees an organisation as connected parts.

The contingency approach says the best way depends on

a task is repeated many times, as in unloading cartons, and works poorly when every task is different. The owner sees higher productivity; the employee may feel constantly watched.

Before moving on. Scientific management studies work by measurement to find one best method. A time and motion study times each movement. Taylor's four principles replace habit with science, train the employee, share the work with the employee, and divide responsibility. Its limit is that it treats employees as parts of a machine.

* * *

4.2 ADMINISTRATIVE PRINCIPLES

Taylor looked at one employee at one bench. But who looks at the whole organisation: the owner's decisions, the way departments connect, the lines of authority? A French mining engineer asked exactly this question, and his answer has lasted more than a hundred years.

WHO WAS FAYOL?

Henri Fayol (1841 to 1925) trained as a mining engineer in France. In 1888 he became managing director of a French mining and iron company that was in difficulty. In 1916 he published *Administration industrielle et générale* in French; the English translation, *General and Industrial Management*, appeared in 1949. Fayol wrote from his own experience as a senior manager, so his view starts at the top, where Taylor's view started at the bench.

Fayol saw six kinds of activity in any industrial organisation: technical (making things), commercial (buying and selling), financial (finding and using money), security (protecting people and property), accounting (recording and counting), and managerial. He said that managing is a separate activity that needs its own study. He described it through five activities: planning, organising, commanding, coordinating and controlling.² These five are an early form of the management process.

Fayol held one belief that mattered greatly: management can be taught. If it can be taught, it needs principles, which are general guides that managers can learn and apply. He listed fourteen and said that the list was not complete. He called them principles and not laws, because a law in science holds every time, while a principle must bend with the situation.

THE FOURTEEN PRINCIPLES

The table lists Fayol's fourteen principles with a plain meaning for each. Read across a row: the name is on the left, and what it asks of a manager is on the right.

² The management process is set out in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

Section 4.1 in four lines.

Taylor studied the single task; Fayol studied the whole organisation.

Fayol was a mining engineer who ran a large company and wrote down what he learned.

Fayol sorted a company's activities into six groups, and managing is one of them.

Fayol believed management can be taught, so he wrote principles, not laws.

Law. Word in use. In daily life, a rule made by the state and enforced by courts. In science, a statement that always holds. Fayol's principles are guides that bend with the situation.

Fayol's fourteen principles are short rules about work, authority, fairness and order.

An **objective** means a specific result that an organisation or a person intends to reach within a stated time. For example, "sell 500 copies of one school textbook before 30 June" is an objective, because it names a result, a number and a date. In everyday talk people say goal or aim. In this book, objective is the formal word, and goal appears only in plain talk. An organisation usually holds several objectives at once, and together they decide which plan it chooses from the alternatives open to it.

Why does an organisation need objectives? There are four reasons. Objectives give direction, because every person knows the destination. They join effort, because the cashier and the stock clerk steer towards the same place. They motivate, because people work harder towards a result they can see and reach. They supply a measure: when the year ends, the result can be set beside the objective. Chapter 18, 'Controlling', teaches how that comparison is used.

Objectives share several features, and each feature helps you test whether a statement deserves the name. First, an objective names a result, not an activity. "Hold a sale" is an activity. "Sell 2,000 copies in the sale" is a result. Second, an objective states a time. A result with no date can always be postponed. Third, an objective carries a measure where one is possible; section 9.2 returns to this. Fourth, objectives form a network. The organisation's objectives lead to objectives for each department, and these lead to objectives for individual employees. Fifth, objectives can conflict. A shop that cuts its stock to save money may find that it cannot also promise every customer that every title is on the shelf. The manager balances the objectives and does not pretend that the conflict is absent.

Each person sees an objective from a different position. The owner sees a result that must be paid for and protected. The manager sees a task to be shared among people. The employee sees a measure of what is expected, and so of what counts as a good year. The customer rarely sees the objective itself, yet feels its effect, for example in better service at the counter.

An organisation does many things, and it cannot give an objective to every action. Where should it place its effort? The answer is the **key result area**, which means a part of the organisation's work where results are vital to success. For a bookshop, sales are one key result area, because a shop that does not sell cannot survive. Stock is another, because too little stock loses sales and too much stock ties up money. Customer service, returns and profit are further

¹ Tk is the symbol for the taka, the currency of Bangladesh. Large amounts are written with commas in groups of three digits.

Recall: planning, Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'.

An objective is a specific, dated result that an organisation intends to reach.

Objective. Word in use. In daily life, "objective" can describe a fair opinion that does not depend on feelings, and it is also the lens of a microscope. In management, it is a noun: a result to be reached.

Objective. A specific result an organisation or person intends to reach within a stated time.

Objectives give direction, join effort, motivate and supply a measure for later checks.

Objectives have five features: a result, a time, a measure, a network and a possible conflict.

Objective. Word in use. See the note above: in management, a result to be reached.

The same objective looks different to the owner, the manager, the employee and the customer.

Key result areas are the few parts of the work where results decide success.

Key result area. A part of the organisation's work where results are vital to success.

You will need: Chapter 5, 'The Environment of Organisations', section 5.1 'Inside and Outside the Organisation'; Chapter 1, 'What Management Is', section 1.4 'A Manager's Responsibility'.

Imagine a cheap offer that no customer could detect, and ask whether you would accept it.

The easy answer becomes hard once we ask who else is affected.

Every organisation lives among people whom it affects without asking them.

The chapter asks three questions and ends at a building in Savar.

A family acts freely inside its flat but shares the stairs with its neighbours.

Recall: Chapter 5, 'The Environment of Organisations', section 5.1 'Inside and Outside the Organisation' — external forces from society.

An organisation lives in a larger block of flats, which is society.

The duty to act for society's good as well as for profit has a name.

Corporate social responsibility. An organisation's duty to act for the good of society as well as

Imagine that you own a small shop. On Monday a supplier offers you goods at a price far below the usual one. The goods look the same as those you buy now. Your customers would not notice any difference, no inspector will visit, and your profit would rise. Do you accept the offer?

Many people answer at once, but the question is harder than it first appears. Who made these goods, and were those people paid fairly? Who would lose if you bought them? Is the low price lawful? If nobody ever discovers your choice, does it still matter? Each of these questions reaches people who never signed a contract with you.

Every organisation meets this kind of question. It buys from some people, sells to others and employs others. It also uses air, water, roads and land that belong to everyone. Its choices travel far beyond its own door. A manager who reads only the price list sees a small part of the result.

This chapter asks three questions. First, what does an organisation owe to society? Second, how do we decide what is right and wrong when a choice is hard? Third, how can an organisation continue without using up what later generations will need? We shall meet a small bookshop in Dhaka that faces such a choice on 18 June 2026. We shall also visit Savar, near Dhaka, where in 2013 a disaster showed the cost of failing to ask.

* * *

6.1 SOCIAL RESPONSIBILITY

Begin with an everyday picture. A family lives in a block of flats. Inside its own door, the family decides what to cook, when to sleep and how loud to play music. The stairs, the roof and the water tank, however, are shared. A family that leaves rubbish on the stairs harms neighbours who never agreed to it. A good neighbour keeps the stairs clean without being forced to.

An organisation lives in a much larger block of flats, which we call society. Chapter 5 showed that society is one of the outside forces that press on every organisation. This chapter turns the picture round. It asks what the organisation owes to society in return.

The picture has a name in management. **Corporate social responsibility** means an organisation's duty to act for the good of society as well as for profit. A bookshop that refuses to sell stolen goods, pays its employees what it promised and sends its waste paper to recycling is acting on this duty. The word *corporate* once pointed to large companies. The duty, however, applies to every

Steps 3 to 5: open dealing, costs counted and priced, and help beyond the organisation's own business.

Step 3: the organisation deals openly with society. It listens to people outside and tells them what it does. Step 4: it counts the costs and benefits to society before it acts, and it adds the social costs to its price, so that those who gain from a product also pay for its effects. Step 5: it helps to solve social problems beyond its own activity, where it has the ability to do so.

From power (left) to responsibility and action (right)

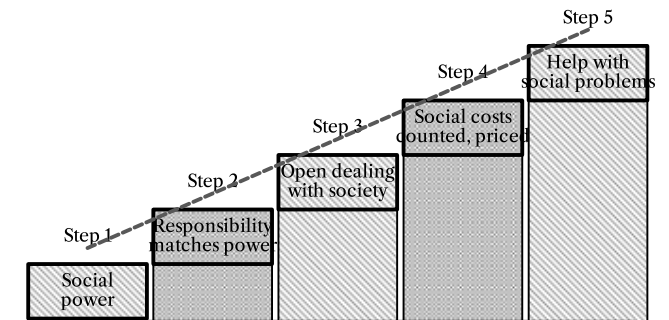


FIG. 6.1. Davis model as five steps from power to responsibility

The figure is read from the lowest step on the left to the highest on the right.

The model guides thought but does not measure results.

The recap of section 6.1.

Read the figure from the lowest step on the left to the highest on the right. Each step is one part of the model, and its height shows that it rests on the steps below. The dashed line joins the steps in order. The numbers and words carry the meaning, so the figure reads the same in grey.

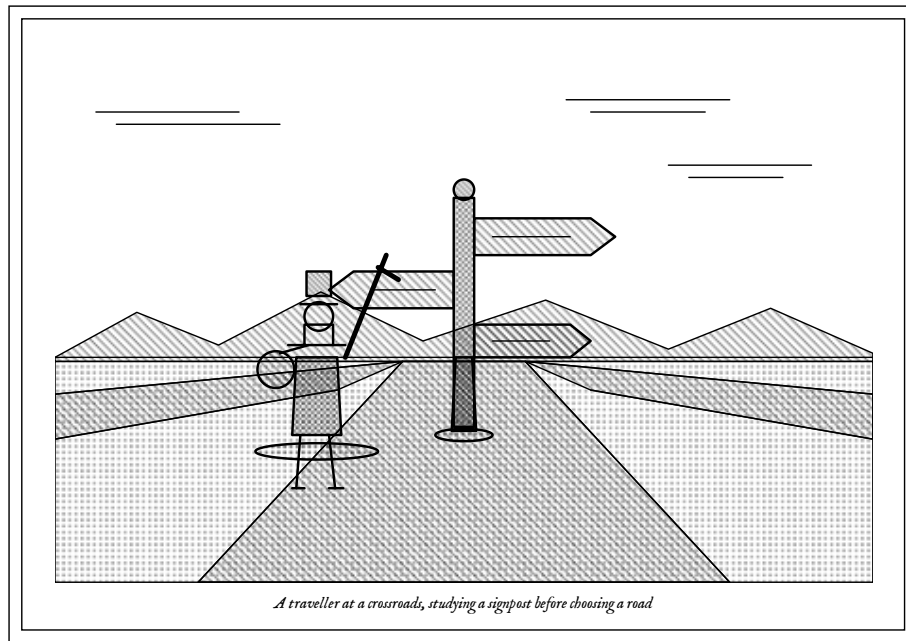
The Davis model has limits. It does not say how much cost is too much, or which social problem an organisation should help first. It assumes that society can see what an organisation does, and in many places society cannot. It gives a smaller organisation a lighter duty, which is fair, but not a duty of zero. A bookshop with seven employees has little power, yet it still owes something to the people it employs and to the neighbours on its street.

Before moving on. Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. Thinkers disagree about it. One side stresses the owners' money, and the other stresses society's permission and the long-term good name of the organisation. The Davis model climbs five steps, from power to responsibility and on to help beyond the organisation's own business.

* * *

6.2 BUSINESS ETHICS

Ask two questions about any choice. Is it lawful? Is it right? Most people see that the answers can differ. Telling a friend a hurtful lie breaks no law, yet it may be wrong. In the other direction, an old law may require something that many people now think unfair.



CHAPTER IX

Objectives and Management by Objectives

Where are we going, and how will we know when we have arrived?



This chapter explains what an objective is, how to write one that can be measured, and how managers and employees can agree objectives together. It follows one bookshop through the objectives set for 2026 and ends with the system that Peter Drucker described in 1954.

A first question: can an act be lawful and still be wrong?

IN HISTORY

On 6 June 1944, during the Second World War, Allied forces landed on the coast of Normandy in northern France. The operation was called Operation Overlord, and it began with the largest landing from the sea in history. The landing was not repeated. It was a single-use plan on a very large scale.

Planning began in 1943, when a joint British and American staff under Lieutenant-General Frederick Morgan was told to prepare a plan for an invasion in 1944. In late 1943, General Dwight D. Eisenhower was named Supreme Commander. The first target date was in the spring of 1944, and it was later moved to June, and the final day was set by the commanders after studying conditions.

The operation also needed a second plan, a campaign of deception called Operation Bodyguard, meant to hide where and when the landing would be. Both plans were prepared together. The history shows what this chapter has taught: a very large single-use plan needs exact objectives, support from the top, communication among many groups, and the ability to change.

REVIEW QUESTIONS

- 8.1 Sort these plans into standing and single-use: a returns policy, a book fair stall, a yearly budget, the opening of a new branch.
- 8.2 Define policy, procedure and budget, and give one shop example of each.
- 8.3 What is the difference between a long-range plan and a short-range plan, and between a strategic plan and an operational plan?
- 8.4 Explain why support from the top matters to a plan.
- 8.5 State two ways in which a manager can reduce the limits of planning.

The Allied landings in Normandy were planned as a single operation on a very large scale.

Planning began more than a year before the landing.

The plan shows how a single-use plan joins many standing and single-use parts.

Five questions that review the chapter.

Good reasoning about a dilemma follows steps that any manager can use.

Organisations can promote ethical conduct in several practical ways.

Reporting channels, rewards and checks complete the set of ways.

Ethics seen by owner, manager, employee and customer.

The recap of section 6.2.

A farmer who cuts every tree gains one season and loses all later harvests.

Sustainability is the wiser farmer's rule applied to an organisation.

Sustainability. Running an organisation so that it meets today's needs without harming later generations' needs.

A manager can reason through a dilemma in steps. First, state the facts and the choices. Second, list every person affected and the cost each choice puts on that person. Third, ask which moral rules apply, such as honesty or fairness. Fourth, ask whether the law allows each choice. Fifth, choose, and be ready to explain the choice to the people who bear its cost. The steps do not remove the difficulty, but they stop a manager from choosing by habit.

Managers cannot order people to be good, but they can build conditions in which good conduct is likely. Several ways are in common use. The first is a written code of ethics, which states the rules the organisation expects its people to follow. The second is the example of leaders, because employees watch what managers do far more closely than what managers say. The third is training, in which employees practise real dilemmas before they meet them.

Three more ways support the first three. A reporting channel lets an employee raise a concern about wrong conduct without fear of punishment. The organisation should reward ethical conduct and discipline wrong conduct in the same way for every person, whatever their rank. Regular checks of accounts and records then show whether the rules are followed in practice. Each way is weak alone. A code that leaders ignore teaches employees that the code is for show.

Each person sees ethics from a different place. The owner wants a name that customers trust. The manager wants rules that are clear enough to apply on a busy day. An employee wants to know that a refusal to do wrong will not cost a job. A customer wants to know that the price, the product and the promise are honest.

Before moving on. Business ethics is the set of moral rules that decide what is right and wrong in business conduct. Law and ethics overlap but are not the same, and a lawful act can still be wrong. An ethical dilemma is a choice in which every option has a cost to someone, and it differs from a temptation. Organisations promote ethics through codes, example, training, reporting channels, fair rewards and checks.

* * *

6.3 SUSTAINABILITY

Picture a farmer with an orchard of mango trees. In one hungry year the farmer cuts every tree and sells the wood. The money feeds the family this year, but there will be no fruit next year or in any year after it. A wiser farmer picks the fruit, plants new seedlings and leaves the orchard as healthy as it was found.

The wiser farmer's rule has a name. **Sustainability** means running an organisation so that it meets today's needs without harming later generations' ability to meet theirs. The United Nations asked a group called the World Commission on Environment and Development to study the question. Its report *Our Common Future*, published in 1987, made this meaning widely known.³ A bookshop

³ The commission is often called the Brundtland Commission, after its chair, Gro Harlem Brundtland of Norway.

OVERCOMING THE LIMITS OF PLANNING

The fourth habit is to accept that planning has limits and to reduce them. The future cannot be known, so every plan rests on assumptions that may prove wrong. Planning takes time and money, and a plan that is too detailed may cost more than it saves. Plans can also lead managers to follow them after they stop fitting the facts.

Three actions help. First, review plans on a fixed date, for example at the start of each season, and ask whether the planning premises still hold. Second, plan in proportion: give more detail where an error costs much, such as a large order, and less detail where the cost is small. Third, keep a margin, which means a small reserve of money or time to meet surprises. A shop that plans for a large order with no reserve is one price change away from trouble.

A short test helps a manager to decide whether a plan is ready. Ask four questions. Do the senior managers support it? Is the objective exact, with a date? Do all the affected employees know it? Is there a fixed day to review it and change it? If any answer is no, the plan has a weak point, and it is cheaper to repair the weak point before the plan starts than after it fails.

PAUSE AND CHECK 8.P2 The shop's owner makes a new plan for the book fair but tells nobody except the store manager. Which of the four habits is weak, and what should the owner do?

Before moving on. Planning works when the top supports it, objectives are exact and shared, employees know the plan, and the plan can change. The limits of planning are reduced by regular review, proportion and a margin.

* * *

WORKED CASE: SORTING THE SHOP'S PLANS FOR 2026

This worked case follows the standard order for a descriptive chapter: Situation, Question, Reasoning, Conclusion and Check. It applies the ideas of the chapter to the shop's own record, and it is a study of the idea and not a model answer.

Situation. The shop has three plans for 2026. The first is a returns policy: exchange within 7 days with a receipt. The second is a 28-day book fair stall in February 2026, with a budget of Tk 240,000 and expected sales of Tk 900,000. The third is a yearly budget with sales of Tk 15,840,000.

Question. Which plans are standing and which are single-use, and what do the figures show about the book fair?

Reasoning. First, sort the plans by repeat use. The returns policy applies every day and has no end date, so it is a standing plan. The book fair stall has a start and an end and will not be repeated, so it is a single-use plan. The yearly budget is renewed each year in form, so it behaves as a standing plan, although its figures change. Second, compare the figures of the fair. The expected sales exceed the planned spending by Tk 900,000 – Tk 240,000 = Tk 660,000.

Planning has limits, and a manager can reduce them but not remove them.

Three actions reduce the limits: review often, plan in proportion and keep a margin.

A manager should test a plan against the four habits before using it.

Section 8.3 in four lines.

This worked case sorts the shop's plans and tests their figures.

The situation lists the three plans in the 2026 case file.

The question asks for the kind of each plan and what the figures show.

The reasoning sorts the plans first, then compares the figures.

Tests 3 and 4: the model and the triple bottom line weigh the gain against the risk.

The conclusion and the check agree that the shop should decline.

Rung 2 reshapes the case: the books are genuine old stock.

Rung 3 moves the case to a garment buyer in another setting.

Rung 4 combines the ideas into a choice with costs to several people.

A short summary of the chapter's five defined terms and their links.

sale, because they receive nothing from it. The customers may also receive a poorer product than they believe they are buying. The shop would bear the cost of a damaged name.

Reasoning, tests 3 and 4: the Davis model and the triple bottom line. Davis step 3 asks whether the shop could tell its customers openly how it obtained the books. Step 4 asks that the social costs be counted, and here they fall on the authors and the publisher. For profit, the saving of Tk 40,000 compares with the 2025 profit of Tk 750,000 as $40,000 \div 750,000 = 0.053$, or about 5.3 per cent. Against this, schools bought Tk 3,600,000 of the 2025 sales of Tk 14,400,000, which is 25 per cent of sales. A school that learned of unauthorised copies might stop buying.

Conclusion. The shop should decline. The offer probably fails the law, fails the moral rules, hides a cost from the authors and the publisher, and risks a much larger share of sales than it gains. *Check.* All four tests point the same way, and the case file records that the shop declined. The shop's own record therefore agrees with the reasoning.

CHANGING THE SITUATION

The same ideas help in new situations. In the second rung, suppose the supplier offers genuine copies of an earlier edition at Tk 180 because the edition is being cleared. The law test is passed. The moral rule of honesty now asks whether customers know that the edition is old.

In the third rung, a clothing buyer in another country is offered garments at a low price from a factory that no one has inspected. The Davis model says that the buyer's power creates a duty, and step 4 says that the safety cost must be counted and not passed to the factory's employees.

In the fourth rung, a small organisation learns that its supplier keeps prices low by dismissing employees without notice. Dropping the supplier may raise its own costs. The choice is a dilemma, so we list those affected, apply the Davis model and judge the result by the triple bottom line.

* * *

SUMMARY

Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. The Davis model links that duty to power in five steps. Business ethics gives the moral rules for judging conduct, and an ethical dilemma is a choice in which every option has a cost to someone. Sustainability asks an organisation to meet today's needs without harming later generations. The triple bottom line judges an organisation by profit, people and the natural environment.

THINK IT THROUGH

6.T1 A small printing shop is offered a rush order. It can finish on

8.2 TIME SPAN AND LEVEL

A second way to sort plans is by time. How far into the future does the plan reach? A long-range plan looks several years ahead, often three to five years, or more. A short-range plan looks ahead for a year or less. The shop's plan to open a second branch in five years is long-range. Its plan for next week's staff timetable is short-range. The two are linked, because the short-range plans are the steps that lead toward the long-range plan.

A third way to sort plans is by level. A strategic plan sets the main direction of the whole organisation, such as which customers to serve and which new business to enter. An operational plan states how one part of the organisation will carry out its daily work, such as how many people stand at the counter. Strategic plans are usually made by the owner or top managers and look far ahead. Operational plans are made by lower managers and employees and look a short way ahead. The planning steps of Chapter 7 apply at every level.

The figure arranges the plans of this chapter in a grid. Read it as follows: the two columns show whether a plan is used again or used once, and the two rows show whether it looks far ahead or only a short way. Find a plan by choosing its column and its row. The placing is a guide and not a law, because a policy may last for a few months and a project may last for years.

	Used again	Used once
Long-range	Policy Yearly budget	Programme Opening of a branch
Short-range	Procedure Rule	Project Book fair stall

Dotted and hatched squares only separate the four cells; the words in each cell carry the meaning.

FIG. 8.1. Plans arranged by time span and by repeat use

Why does the shop need both? A long-range plan alone is too distant to guide the week, and a short-range plan alone has no direction. Suppose the owner wants to open a second branch in five years. That aim is long-range. To reach it, the shop needs a yearly budget, a plan for saving money, and a plan for training a manager for the new branch. These are short-range plans, each a step on the road. A manager checks that every short-range plan points the same way as the long-range one.

The same fitting is needed between levels. The owner's strategic plan says which customers the shop wishes to serve, for example

Plans can be sorted by how far ahead they look.

Plans can also be sorted by level: Strategic plans set direction, operational plans do the daily work.

The grid places the main plans by repeat use and by time span.

Short-range plans are the steps that lead toward long-range plans.

Strategic and operational plans fit together like a frame and its pictures.

6.2 Give one argument for and one argument against the idea that organisations should carry a duty to society. State which owner, manager or employee would most likely make each argument.

6.3 Name the five steps of the Davis model in order. For step 2, say what the iron law of responsibility warns.

6.4 Explain the difference between an ethical dilemma and a temptation, with one example of each.

6.5 Sort each act as legal, ethical or both, and give a reason: (a) paying employees on the day promised; (b) selling a stolen book; (c) giving employees paid leave for a funeral when no law requires it.

6.6 A rival bookshop cuts the safety of its storeroom to lower its price. Using ideas from this chapter, explain why this is a problem for society and for the rival, and say what a manager at your own shop should do.

SINGLE-USE PLANS

The second group is the **single-use plan**. A single-use plan is a plan made for one occasion or project and not used again. A book fair that happens once, a move to a new building and the opening of a branch are all examples. Once the occasion is over, the plan has no further use, although the manager may learn from it when making the next one.

The two groups answer different needs. Think of the owner's viewpoint. The owner wants the repeated questions answered cheaply and the rare event answered carefully. The employee's viewpoint is different: the employee wants to know quickly what to do about a returned book, and has little need to know about a book fair that will not return. A well-run shop therefore keeps a small collection of standing plans for daily work, and writes a new single-use plan for each large event.

PAUSE AND CHECK 8.P1 A shop plans a one-day sale to clear old stock. Is its plan a standing plan or a single-use plan? What would change your answer?

POLICY, PROCEDURE AND RULE

Standing plans come in three common forms. They differ in one respect: how much freedom each leaves to the person who follows it. A policy leaves the most freedom, a procedure leaves less, and a rule leaves none. The table below sets them side by side. Read it row by row: the left column names the form, the middle column says what it does, and the right column gives a shop example.

THREE FORMS OF STANDING PLAN

FORM	WHAT IT DOES	SHOP EXAMPLE
Policy	Gives a general guide and sets limits for choices	Returns are accepted only within a set time
Procedure	Lists the steps in order for a repeated task	Check the receipt, check the book, record the exchange
Rule	Requires one action in one situation, with no choice	No exchange without a receipt

A **policy** is a general guide for decisions, setting limits within which choices are made. For example, a shop that decides "we exchange books within a fixed number of days" has set a policy. Within that limit the employee can decide: a customer who returns a damaged book on the last day may be helped, while one who returns an old book may be refused. A policy gives direction and leaves judgement.

A **procedure** is a fixed series of steps to be followed for a repeated task. Where a policy says what is allowed, a procedure says how to carry it out. For example, the exchange procedure may say: ask for the receipt, check the date, check that the book is undamaged, give the new book, and write the exchange in the record. The steps are always the same, and the order always matters.

A rule is the narrowest standing plan. It says that in one situation

A single-use plan is made for one occasion and then set aside.

Single-use plan. A plan made for one occasion or project and not used again.

Single-use plans fit rare events, while standing plans fit repeated ones.

Three kinds of standing plan differ in how much freedom they leave.

A policy is a general guide that sets limits within which choices are made.

Policy. A general guide for decisions, setting limits within which choices are made.

A procedure is a fixed series of steps for a repeated task.

Procedure. A fixed series of steps to be followed for a repeated task.

A rule requires one action and

YOU WILL NEED

This chapter builds on two earlier ideas. The first is the list of activities that every manager performs, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'. The second is how output is compared with the resources used, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.2 'Productivity'.

A family must reach a wedding in another city before the ceremony starts.

GO SLOWLY HERE.

Wanting to arrive is not the same as knowing how to arrive.

Plan. Word in use. In daily life, an intention, as in "I plan to rest." In management, a statement of what will be done, by whom and by when.

A shop faces the same puzzle on a larger scale, with money and deadlines.

Answering that question is the work of planning, which this chapter teaches.

A plan states what will be done, by whom and by when.

Plan. A written or spoken statement of what will be

Imagine that a family has been invited to a wedding in a city 300 kilometres away. The ceremony starts at noon on Saturday. Nobody has decided how to travel, what to carry, or when to leave. On Friday evening, one person says, "We will manage somehow." Is that sentence enough? Think about the questions that remain. Which road? Which vehicle? Who pays for fuel? What happens if the road is closed by rain? Each question has an answer, but the family has not found the answers yet.

The family wants one thing, which is to arrive on time. Wanting is the easy part. The difficult part is deciding, before the journey begins, what will be done and in what order. A family that sits down on Wednesday, chooses the road, counts the hours, and agrees who will drive has made a plan. A family that waits until Saturday morning has only a wish.

Now move from a family to a shop. A bookshop must order thousands of textbooks weeks before the school season opens. If the owner waits until the shelves are empty, the books will arrive too late, and customers will buy elsewhere. If the owner orders too many, money is locked in books that do not sell. The puzzle is the same as the family's, but the cost of a wrong answer is greater. This chapter answers a simple question that has no simple answer: how does a manager decide, in advance, what to do?

We shall build the answer in layers. First we define planning and see what it does. Then we learn five steps that take a manager from a wish to a decision. Last we study why plans fail, because a plan that fails teaches as much as one that works. Many students find this chapter slow, because the ideas feel obvious after they are explained. They are not obvious before. Take your time.

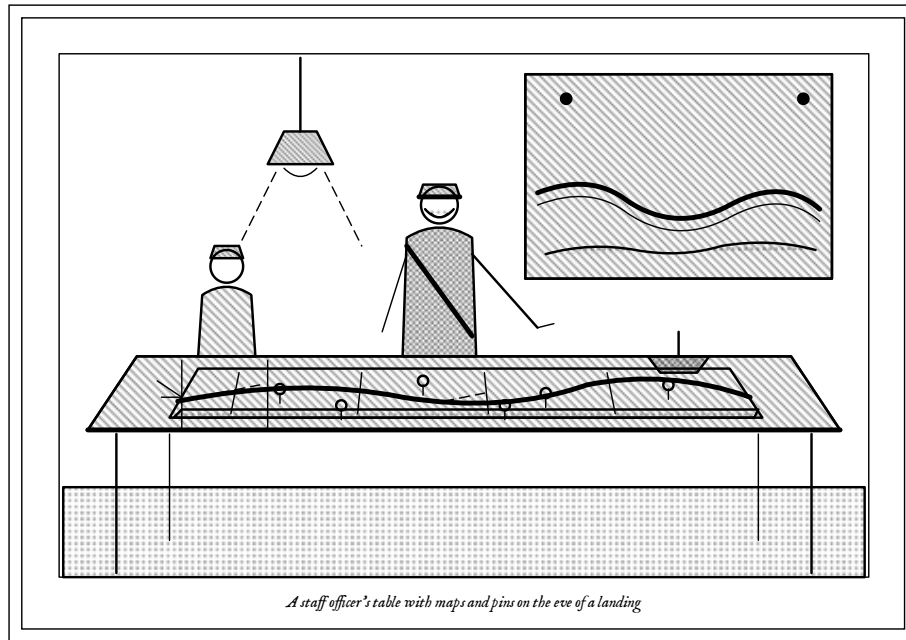
* * *

7.1 WHAT PLANNING IS

PLANNING DEFINED

Begin with the plan itself, because planning is the act of making one. A **plan** is a written or spoken statement of what will be done, by whom and by when, to reach an objective. Here an objective

¹ A full account of objectives is given in Chapter 9 'Objectives and Management by Objectives', section 9.1 'Objectives'. Until then, read objective as the result that you have chosen to reach.



CHAPTER VIII

Kinds of Plans and Making Planning Work

Must a manager think everything out again each morning?



This chapter sorts plans into kinds: those used again and again, and those used once. It shows how plans differ by length of time and by level, and it explains what makes planning work in practice. The examples come from a bookshop's returns, a book fair and a yearly budget.

You should now be able to name the five features.

Planning serves several purposes, from reducing guesswork to making control possible.

GO SLOWLY HERE.

Planning improves productivity and effectiveness by directing effort to the aim.

Planning has limits, and the last section of this chapter returns to them.

You should now be able to list the purposes of planning.

Section 7.1 in three lines.

Planning follows five steps in order, and each step needs the one before it.

in size, but each one answers the same three questions of action, person and time.

Recap. Planning looks forward, serves a purpose, chooses, happens at every level and never ends. You should now be able to say: "Planning is a reasoned decision about the future, taken in advance."

THE PURPOSES OF PLANNING

Why plan at all? The family could drive and see what happens. Four purposes answer the question. First, planning reduces uncertainty, because it makes the manager look at the future before it arrives. Second, planning gives direction: every employee knows the aim and the part each person plays. Third, planning saves effort, because it removes wasted work and repeated decisions. Fourth, planning makes control possible, because a manager can check results only against something that was planned.

These purposes connect to two ideas from Chapter 3. Planning raises productivity, which compares output with the resources used.³ It also raises effectiveness, which asks whether the right aim was reached. A shop that orders the correct number of books wastes less money, so it is more productive. It also meets its customers' needs, so it is more effective.

Planning has its limits too, and an honest account must say so. It costs time and money. It can make an organisation rigid if the plan is followed after conditions have changed. These limits do not remove the value of planning. They warn the manager to plan with care, and section 7.3 examines them.

Recap. Planning reduces uncertainty, gives direction, saves effort and makes control possible. You should now be able to say: "Without a plan there is nothing to check results against."

PAUSE AND CHECK 7.P1 Think of a task you did recently without a plan. What would the plan have contained: the action, the person and the time?

Before moving on. A plan states what will be done, by whom and by when. Planning is the advance decision that produces it, and it is forward-looking, purposeful, choosing, shared and continuous. Its purposes are less uncertainty, direction, saved effort and the possibility of control.

* * *

7.2 STEPS IN PLANNING

Is there a method for planning, or does each manager invent one? Managers have found that good plans usually pass through five steps in the same order. The ladder below shows them. Read it from the bottom rung to the top: each rung is one step, and a manager climbs one rung before reaching the next. Missing a rung is like missing a stair in the dark; the next step becomes unsafe.

³ Measures of performance are explained in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.2 'Productivity'.

THINK IT THROUGH

The questions begin with the shop and move to new settings.

7.T1 The shop's season starts on 2 January 2026 and the supplier lead time is 21 days. By what date must the order be placed, and why?

7.T2 Suppose the shop built its plan on the premise that prices stay the same, but publishers raise prices by 5%. Show how much more the order costs, and say what the manager should have done.

7.T3 A family restaurant plans to open a second branch. Name the five steps of planning and give one question the owner should ask at each step.

7.T4 A school plans a sports day outdoors in the rainy season, and the plan assumes dry weather. List two alternatives the school could weigh and say how each handles the premise.

WHERE PEOPLE GO WRONG

Some students think that a plan promises what will happen. It does not. A plan is a decision about what to do, based on premises. When the premises change, the plan must change.

Some students think that only owners plan. In fact an employee who decides the order of tasks for the morning is planning too. The size of the plan changes with the level; the three questions of action, person and time stay the same.

Some students jump from the first idea to action. The five steps exist because each one removes a kind of error. Without objectives the aim is unclear; without premises the future is ignored; without alternatives the choice is the first idea and not the best one.

Some students believe that writing the plan completes the work. A plan that is not carried out, and not reviewed, is only a document. Planning ends when results have been compared with the plan.

IN HISTORY

On 25 May 1961, President John F. Kennedy told the United States Congress that the nation should commit itself to landing a person on the Moon and returning that person safely to Earth before the decade ended. At that time the country had sent a person into space for only a few minutes. The goal was bold, and it was stated with a date.

A goal of this size cannot be reached by wishing. NASA, the space agency of the United States, had to study possible methods, estimate costs and time, build rockets and train astronauts. It listed and weighed different ways of reaching the Moon and chose the method of lunar-orbit rendezvous, in which a small landing craft leaves a larger spacecraft that stays in orbit. The work followed the same pattern that this chapter has described: an objective, premises, alternatives, a choice and then action.

Four questions that change the situation step by step.

A plan is not a guarantee about the future.

Planning is not only for senior managers.

Skipping steps produces weak plans.

A written plan is not useful if nobody follows it.

A president set a goal and a national team built the plan to reach it.

Planning turned the goal into many steps carried out by many people.

A planning premise is an assumption about future conditions on which a plan is built.

Premise. Word in use. In daily life, a building and its land, as in "the shop premises". In planning, an assumption stated at the start of an argument or a plan.

Planning premise. An assumption about future conditions on which a plan is built.

Premises are a choice of which assumptions matter, so a manager picks the few that count most.

A forecast is a reasoned estimate of what will happen, used as a base for a plan.

Forecast. A reasoned estimate of what will happen in the future, used as a base for a plan.

The shop's case file records its main premise and its supplier lead time.

The lead time is the gap between ordering and receiving, and it fixes the deadline.

You should now be able to say what a planning premise is and why it is written down.

The fourth step is to list the possible ways to reach the objective and compare them.

GO SLOWLY HERE.

premise. A family might assume that the road will be open and that fuel prices will stay the same. These assumptions are the ground on which the plan is built.

A **planning premise** is an assumption about future conditions on which a plan is built. Suppose a family assumes that it will rain on Saturday. That assumption changes the plan: umbrellas are packed and extra time is allowed. If the assumption is wrong and the day is dry, little harm is done. If the assumption is the opposite, and the family expects sun in a monsoon, the plan fails on the road.

The manager cannot list every possible condition, so the manager picks the few assumptions that would change the plan most. Of all the things that might change, a shop owner asks which ones would make the order too large or too small. Prices, customer numbers and delivery time usually matter most. Choose the few important assumptions and write them down, so that everyone who uses the plan shares them.

One of the best sources of premises is a **forecast**. A forecast is a reasoned estimate of what will happen in the future, used as a base for a plan. For example, a shop may estimate next season's sales from the sales of the last three seasons. The word "reasoned" matters: a forecast rests on evidence such as past records, and it is more than a hope. Even so, a forecast can be wrong, and a careful manager reviews the forecast when facts change.

Case file, illustrative. Premise for the textbook season: publishers raise prices by 5%. Supplier lead time: 21 days.

Read the record carefully. The lead time of 21 days is the gap between placing an order and receiving the books. Count 21 days from the order deadline of 12 December: December has 31 days, so $12 + 21 = 33$, which is 2 January. The books ordered on the last day arrive on the first day of the season. This shows why a deadline must be set backward from the date when the result is needed.

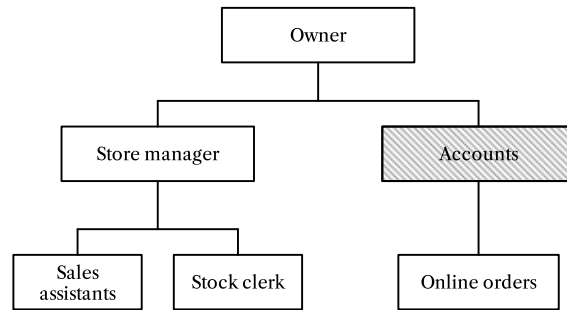
Recap. A planning premise is an assumption about future conditions on which a plan is built, and a forecast is a good source for one. You should now be able to say: "A premise is what we assume about the future, written down so that everyone shares it."

PAUSE AND CHECK 7.P2 The shop's plan assumes that publishers raise prices by 5%. Say what would happen to the order value if publishers raised prices by 10% and the shop did not change its plan.

STEP FOUR: LISTING AND WEIGHING ALTERNATIVES

With the objective and premises written down, the manager asks how the objective can be reached. There is almost never one way. A family can drive, take a bus or hire a car. We call each possible course of action an alternative. Listing alternatives forces the manager to think beyond the first idea, which is often not the best.

When the four steps are complete, the manager can draw the result. An **organization chart** is a diagram showing the jobs in an organisation and who reports to whom. Each job is shown in a box, and lines join the boxes to show the reporting lines. The figure below shows a small example. To read it, begin at the top box, which holds the highest authority. Follow a line downward from a box to meet the jobs that report to it. A line always joins two boxes and means *reports to*, read from the lower box to the upper box.



Each line means reports to, read upward

FIG. 12.1. An organization chart with lines of authority

An organization chart has two limits. It shows only the official lines and nothing of the friendships and habits that also guide behaviour; we shall meet those in the next section. It can also become out of date, because a chart drawn in January may not describe the shop after two people leave in March. A chart is a record of a decision, and the manager must update the record when the decision changes.

Students often make two slips in this process. The first is to skip step one and begin with the people, which makes the list of tasks fit the people instead of the goals. The second is to draw the chart first and decide the duties later, which produces boxes with no real work in them. Keeping the order of the steps prevents both slips. **PAUSE AND CHECK 12.P1** A shop has three sales assistants and one cashier. Name the step in which the manager decides that all four report to the store manager.

Before moving on. The four steps are to identify the tasks, group the tasks, assign the duties and set the relationships. Assigning a duty means giving the person the means and the authority to perform it. The organization chart is the drawing that records the result.

* * *

12.3 FORMAL AND INFORMAL ORGANIZATION

Walk into a bookshop at noon and watch for a moment. The off

Organization chart. A diagram showing the jobs in an organisation and who reports to whom.

The organization chart is a drawing of the result of the four steps.

Chart. Word in use. In daily life, a table or picture that shows numbers, such as a chart of rainfall. In management, a diagram of jobs and reporting lines.

A chart shows the official lines only, and it can become out of date.

Common slips in the process: skipping a step, or building the chart around people.

The recap fixes the four steps and the chart.

A shop has an official pattern

Meghna, the store manager, agreed them on 5 January 2026. Each objective lies in one key result area: sales, profit, returns, customer service and stock.

Before moving on. An objective is a specific result to be reached within a stated time. It differs from an activity, which is only something done. Objectives give direction, join effort, motivate and supply a measure. A key result area is a part of the work where results decide success, and the five objectives in the case file lie in five such areas.

PAUSE AND CHECK 9.P1 Name two key result areas for a small school, and say why a result in each area decides whether the school succeeds.

* * *

9.2 SETTING OBJECTIVES

Start with a wish that many organisations write: "Improve customer service." Can anyone check this statement in December? No one can. Now sharpen it in four moves. First, choose the key result area: customer service. Second, choose a measure for that area: the number of hours of training each employee receives. Third, add the number: 14 hours. Fourth, add the period: the year 2026. The sharpened statement reads: "14 hours of customer-service training per employee in 2026." It is one of the five objectives in the case file, and a person can check it by counting hours.

The example contains the whole technique, which has six steps. Step one: choose the key result area. Step two: state the result you want, in words that name a result and not an activity. Step three: attach a measure, which is the unit used to count progress, such as taka, hours or percentage. Step four: fix the number and the date. Step five: test the objective. Can it be reached with the people, money and time available, and does it ask for more than last year, so that effort is needed? Step six: agree the objective with the person who must deliver it. An objective that the responsible person has never discussed is only an instruction.

The words measure, number and target need care. The measure is the unit, such as taka of sales. The target is the number set for that unit, such as Tk 15,840,000. Together they make an aim that can be measured. The table below shows four vague wishes beside the measurable objectives from the case file. Read each row from left to right: the first column gives a wish, the second gives the objective that sharpens it, and the third names the measure.

FOUR VAGUE WISHES SHARPENED INTO THE OBJECTIVES IN THE CASE FILE

VAGUE WISH	MEASURABLE OBJECTIVE	MEASURE
Sell more	Sales of Tk 15,840,000 in 2026	Taka of sales
Keep returns low	Returns below 2% of sales	Share of sales

Before moving on: objectives are results with a time, and key result areas say where to place them.

Take a vague wish and sharpen it, one step at a time, until anyone can check it.

Good objective-setting follows six steps, from the key result area to agreement with the person responsible.

A target is the number in an objective, and the measure tells you what the number counts.

Target. Word in use. In daily life, a mark that you aim at, such as the centre of a board. In management, the number or level written into an objective, such as 3.2 times for stock turnover.

Organizing is an activity, and it leaves a result behind. We call the result the **organizational structure**, which means the pattern of jobs, groups and reporting lines that divides and links the work. A reporting line is the path that shows which person answers to which other person. In a bookshop the structure answers questions such as these: Who orders books? Who may refuse a refund? To whom does the cashier bring a problem? The structure exists whether or not anyone has drawn it.

Why do organisations organize at all? The first purpose is to turn a large goal into jobs of a size that one person can perform. No one can sell the books, count the stock and keep the accounts at the same moment. The second purpose is to avoid waste. When each task has an owner, it is neither forgotten nor done twice. The third purpose is to link the parts, so that the work of the buyer reaches the seller in good order. These three purposes serve the aim of the whole management process, which is to reach goals with productivity and effectiveness.

The same arrangement looks different from each viewpoint. The owner sees a way to reach the shop's goals without doing every task personally. The manager sees a tool for checking whether any task has been left without an owner. The employee sees a fixed answer to the question *what is my job, and to whom do I turn?* The customer, who never sees the structure, feels its results when the right person answers a question without delay.

Organizing also has a nature that is worth knowing, because it prevents three errors. First, organizing is continuous. A structure that suited a shop with three employees may fail when the shop has seven, so managers return to it whenever the work changes. Second, organizing serves goals. A structure is judged by whether it helps the organisation reach its goals, and not by whether it looks neat on paper. Third, organizing deals with people as well as tasks. A perfect plan of duties can fail if the people who must perform it do not understand or accept it.

There are limits to what organizing can do. A structure cannot repair a poor plan, because it only arranges the work that the plan has set. Too much structure can also slow an organisation: if every small question must travel through several reporting lines, customers wait and employees lose interest. A good structure is therefore the lightest one that still makes every task and every reporting line clear.

IN SHORT

Organizing is the function that arranges tasks, people and resources. Its result is the organizational structure. Its purpose is to make every task owned, every owner known and every part linked to the others.

Before moving on. Organizing is the function of arranging tasks, people and resources into a working structure. The structure is the pattern of jobs, groups and reporting lines. Its purpose is to give every task an owner and to link the parts. It is continuous, it serves

Organizational structure.

The pattern of jobs, groups and reporting lines that divides and links the work.

The result of organizing is an organizational structure.

Structure. Word in use. In daily life, a building or other object that has been put together. In management, the pattern of jobs and reporting lines inside an organisation.

The purpose of organizing is to turn a goal into jobs that people can perform.

From each viewpoint, organizing gives something different.

Organizing has a nature: it is continuous, it serves goals, and it follows people and work.

Exceptions: a structure can help in one setting and harm in another.

The recap fixes the function, its result and its purpose.

IN SHORT

Management by objectives is a cycle. Agree objectives together, work towards them with freedom to choose the means, review progress during the year, appraise the results at the end, and use what is learned to agree the objectives for the next year.

The case file shows the first stage of the cycle: two people agreeing objectives.

The process has six stages, and the last stage leads back to the first.

Recall: kinds of plan, Chapter 8, 'Kinds of Plans and Making Planning Work', section 8.1 'Kinds of Plan'.

In stage three, each person chooses plans and means that fit the agreed objective.

Performance appraisal is the stage where results are compared with the agreed objectives.

Performance appraisal. A review of how well an employee has done against agreed standards.

To read the ring, start at the top and follow the arrows clockwise; the last arrow returns to the start.

Case file, illustrative. On 5 January 2026, Nurul Meghna and Rafiq Meghna agreed the objectives for the year: sales of Tk 15,840,000; profit of Tk 900,000; returns below 2% of sales; 14 hours of customer-service training per employee; and stock turnover raised from 2.8 to 3.2 times. The file records the agreement between these two people. It does not record later stages of the cycle.

THE PROCESS OF MBO

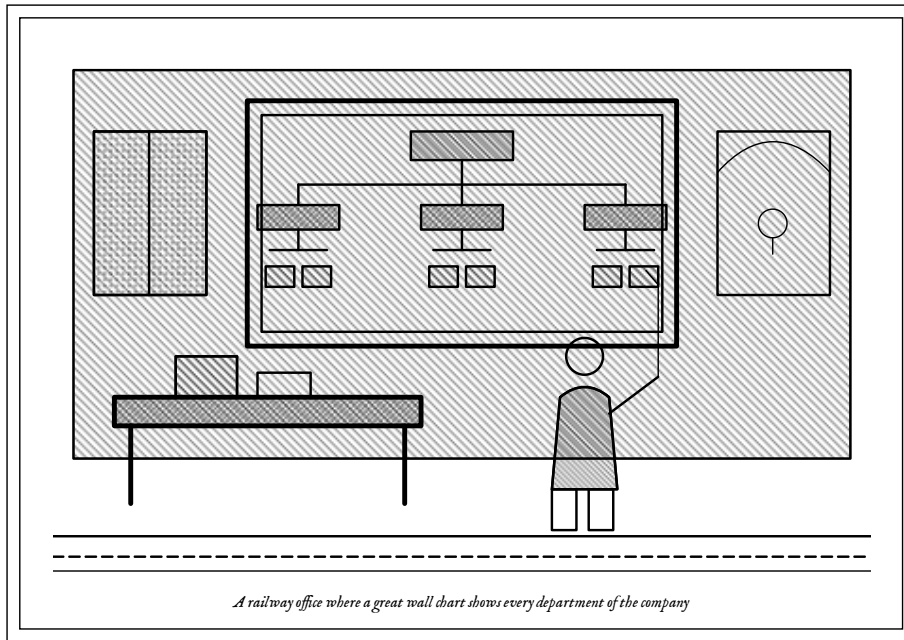
The process of management by objectives has six stages. In stage one, the organisation states its own objectives for the period. In stage two, each manager and employee agree the objectives for his or her own work, so that these support the organisation's objectives. In stage three, each person plans the means and acts. During this stage the employee has freedom to choose how to work, and checks his or her own progress; this is the self-control that Drucker named. In stage four, manager and employee review progress during the period, and they change an objective if conditions have changed. In stage five, performance appraisal takes place at the end of the period. In stage six, the results guide the objectives for the next period, and the cycle begins again.

In stage three a person chooses the means. The kinds of plan from Chapter 8 are tools here. A standing plan, such as a policy on how to handle a customer's complaint, repeats from year to year. A single-use plan, such as a project to fit security tags in one month, serves one objective once. Every plan rests on planning premises, which are the assumptions about the future that the plan accepts. An objective of Tk 15,840,000 in sales assumes that demand from schools and walk-in customers will stay about as strong as before. If that assumption fails, the review in stage four is the moment to say so.

A **performance appraisal** means a review of how well an employee has done against agreed standards. In management by objectives, the standard is the objective agreed at the start, so the appraisal asks one question: what was the result, compared with the number we agreed?⁴ For example, an employee who agreed to receive 14 hours of training and received 10 can be told the gap and asked why it arose. The appraisal is also a time to plan help, such as more training.

Figure 9.1 shows the six stages as a ring. To read it, begin at the box at the top and follow the arrows in the direction of the clock. Each box is one stage. Each arrow shows which stage comes next.

After the last box, an arrow returns to the first box, which shows ⁴ A standard is a level of performance fixed in advance as the measure of what is acceptable. Chapter 18, 'Controlling', teaches it in full. Here it means only the agreed objective.



CHAPTER XII

Organizing and Types of Organization

Who does what, and who answers to whom?



A goal cannot be reached by good intentions alone; someone must decide who does which task. This chapter explains organizing, the steps by which managers divide and link work, and the difference between the official structure and the friendships that grow beside it.

FOUR BENEFITS OF MBO AND THE MATCHING RISK

BENEFIT	MATCHING RISK
Clarity of expectations	Objectives chosen because they are easy to count
Commitment from agreement	Objectives imposed, so agreement is only on paper
Fair basis for appraisal	Results judged that the employee cannot control
Coordination across levels	Heavy paperwork that takes time from the work

Before moving on: MBO is a cycle of agreed objectives, self-control, review and appraisal, with real benefits and real risks.

Before moving on. Management by objectives is a system in which managers and employees agree objectives together and judge performance against them. Its six stages are: state the organisation's objectives, agree individual objectives, act with self-control, review progress, appraise performance, and begin the cycle again. Its benefits are clarity, commitment, coordination, fair appraisal and development. Its weaknesses are paperwork, a bias towards what is easy to count, a short-term focus, rigid objectives and pressure when it is used badly.

* * *

WORKED CASE: MAKING THE STOCK OBJECTIVE MEASURABLE

The worked case tests the stock objective in the case file from wish to number.

The situation states the objective and the figures that give it meaning.

The question asks whether the objective can be measured and what stock it implies.

The 2025 figures show where the shop starts, and the formula shows what the number means.

The conclusion and the check show that the objective is measurable and that its arithmetic holds.

Case file, illustrative. We now apply the ideas of this chapter to one objective. We take it in the order *Situation, Question, Reasoning, Conclusion, Check*, so that each step is announced and explained.

Situation. The case file records the wish that Meghna and Sons should hold stock more efficiently. The objective agreed on 5 January 2026 is stock turnover raised from 2.8 to 3.2 times. Here stock means the books that the shop holds for sale. Stock turnover means the number of times in a year that the shop sells its average stock, found by dividing cost of sales by average stock. Cost of sales is Tk 9,800,000, and average stock is Tk 3,500,000.

Question. Is the objective measurable? And if cost of sales stays at Tk 9,800,000, how much average stock does the shop need to reach 3.2 times?

Reasoning. First we confirm the starting point. The formula in words is: stock turnover equals cost of sales divided by average stock. With the numbers: $9,800,000 \div 3,500,000 = 2.8$. So 2.8 is the 2025 starting point, and the objective is measurable, because it names a measure (times per year), a target (3.2) and a period (2026). Next we find the stock that gives 3.2. Rearrange the formula: average stock equals cost of sales divided by turnover. Then $9,800,000 \div 3.2 = 3,062,500$. The shop therefore needs average stock of Tk 3,062,500. The reduction is $3,500,000 - 3,062,500 = \text{Tk } 437,500$.

Conclusion. The objective is measurable. If cost of sales stays at Tk 9,800,000, reaching 3.2 times means holding average stock of Tk 3,062,500, which is Tk 437,500 less than in 2025. **Check.**

In 1978 Simon received the Nobel prize in economics, and later in his career he turned to artificial intelligence by computer. His idea fits this chapter. A tree and a decision support system widen what a manager can examine. A group widens what is known. But each costs time, so managers still stop when the choice is good enough.

His idea explains why tools and groups help, but cannot make the search endless.

REVIEW QUESTIONS

- 11.1 A probability is 0.25. What does this number mean, and what is the probability that the event does not happen?
- 11.2 Find the expected value: probability 0.7 of a profit of Tk 50,000, and probability 0.3 of a loss of Tk 80,000.
- 11.3 Find the expected value: probability 0.2 of Tk 100,000, probability 0.5 of Tk 40,000, and probability 0.3 of a loss of Tk 60,000.
- 11.4 Choice X gives Tk 200,000 with probability 0.6 and -Tk 100,000 with probability 0.4. Choice Y gives Tk 70,000 for certain. Which has the higher expected value, and what else might a manager weigh?
- 11.5 In a decision tree, what do a square and a circle show, and in which direction do we draw the tree and in which do we calculate it?
- 11.6 Describe two ways in which a decision support system helps a manager, and one limit.
- 11.7 Give two advantages and two disadvantages of a group decision.
- 11.8 Name three factors, other than money and probability, that shape a decision.
- 11.9 Using Chapter 10, say why a decision tree suits a decision under risk, and what changes when the manager has certainty.

towards what is easy to count, and misuse as a tool of pressure. These faults are in section 9.3. Drucker's central point remains useful: people work better when they know the result expected and have helped to define it.

REVIEW QUESTIONS

The review questions test each main idea of the chapter.

- 9.1 Define objective in your own words, and say how an objective differs from an activity.
- 9.2 Rewrite each wish as a measurable objective, adding a measure, a target and a period: "Sell more books to schools", "Keep stock low", "Make staff more helpful". Use figures of your own.
- 9.3 In the case file, stock turnover is cost of sales divided by average stock. If cost of sales is Tk 9,800,000, find the average stock that gives a turnover of 3.2 times.
- 9.4 List the six stages of the MBO process in order, and name the stage in which performance appraisal takes place.
- 9.5 Give two benefits and two weaknesses of management by objectives.

A GROUP COMPARED WITH ONE MANAGER

POINT	GROUP DECISION	ONE MANAGER DECIDING
Speed	Slower	Faster
Range of ideas	Wider	Narrower
Support from those affected	Stronger	Weaker, unless explained
Responsibility	Shared and less clear	Clear
Cost in time	Higher	Lower

The table suggests a rule. Choose a group when the problem is important, unclear, and affects many people, and when time allows. Choose one manager when the problem is urgent or simple, or when one person holds the authority and the knowledge. The exception is that no rule fits every case, and a manager may still choose to consult before deciding alone.

OTHER FACTORS IN DECISIONS

A decision tree uses taka and probabilities. Real decisions also depend on other factors. The first is time: some decisions cannot wait for a long study. The second is cost: a study can cost more than the decision is worth. The third is importance: a small choice deserves a small effort. The fourth is the manager's own values and attitude to risk, for two managers may read the same expected value and choose differently. The fifth is authority: who is allowed to decide, and whose agreement is needed.

One last fact explains much of this. Managers rarely have time or information to compare every alternative. They usually stop when they find one that is good enough. The section *In History*, later in this chapter, tells who named this habit.

Before moving on. A group decision is a choice reached by several people discussing and agreeing. It brings more ideas and support, and it costs time and can blur responsibility. Time, cost, importance, values and authority also shape a decision.

* * *

WORKED CASE: CHOOSING BETWEEN TWO SITES

Situation. Selim runs a stationery stall and may rent one of two sites. At Site A, success (probability 0.5) gives a profit of Tk 80,000, and failure (probability 0.5) gives a loss of Tk 20,000. At Site B, success (probability 0.8) gives a profit of Tk 30,000, and failure (probability 0.2) gives a loss of Tk 10,000. Waiting gives Tk 0. *Question.* Which choice has the highest expected value, and is that choice automatically the right one?

Reasoning. We fold back the tree from the right. Site A: $0.5 \times 80,000 = 40,000$, and $0.5 \times (-20,000) = -10,000$, so the expected value is $40,000 - 10,000 = \text{Tk } 30,000$. Site B: $0.8 \times 30,000 =$

A group suits important, unclear problems; one manager suits urgent, clear ones.

Money values and probabilities are not the only things that shape a decision.

Managers rarely find the best answer; they stop at a good one.

Before moving on: a group decision gains knowledge and support but costs time, and several factors besides money shape any choice.

Selim must compare two sites and waiting, using expected value.

We fold back the tree: first each circle, then the square.

You will need: Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'; Chapter 9, 'Objectives and Management by Objectives', section 9.2 'Setting Objectives'.

A simple choice about travel hides a method that managers use every day.

A short choice already holds several separate acts.

A manager makes such choices with more at stake and with others affected.

The chapter teaches the method first and the conditions second.

GO SLOWLY HERE.

The idea of a decision begins with an everyday choice of one thing from several.

Recall: Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is' — planning.

Rina's choice has the shape of every decision: several options and one choice.

Decision making. Choosing one course of action from several, in order to solve a problem or use an opportunity.

Decision. Word in use. In daily life, any choice that a person has made, or the act of settling a question. In management, a choice of one course of action from several, made to solve a problem or use an opportunity.

Every plan is made of

It is seven in the morning, and your class begins at nine. You can walk, take a bus or hire a rickshaw. You pause for a moment, think of the cost and the time, and choose. By the time you reach the door you have made a decision. Did you do anything that could be written down as a method?

Look again at what you did. You noticed that you had to be somewhere by nine. You asked what mattered to you: the cost, the time, or both. You listed the ways to go, compared them and picked one. Then you went, and you learned something from the journey. Most people perform all of these acts in a few seconds and never name them.

A manager performs the same acts under heavier conditions. The choice may cost thousands of taka, or change the work of seven employees. Other people will live with the result, and the manager cannot test it in advance. A fast habit that works for a bus ride can fail badly when the stakes are high.

This chapter teaches the method first, in clear steps that a manager can follow and check. It then describes the three conditions in which every decision is made. We follow one real choice, made by a bookshop in Dhaka on 20 August 2026. We also visit a launch pad in Florida, where in January 1986 a different choice ended in disaster.

* * *

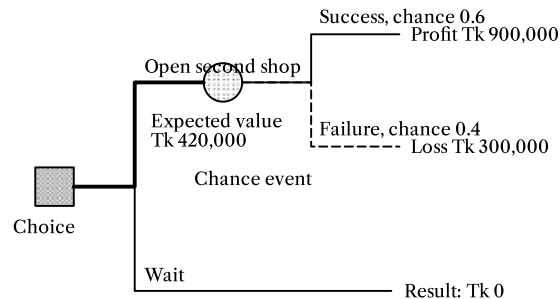
10.1 WHAT A DECISION IS

Begin with something small. Rina is a student, and on Friday she has Tk 500 to spend. She can buy a book, buy a cinema ticket or keep the money. She cannot do all three with Tk 500. She must choose.

Her choice has a name in management. **Decision making** means choosing one course of action from several, in order to solve a problem or use an opportunity. Rina has three courses of action, and she picks one. A shop owner who chooses between two sites for a new shop does the same thing with larger sums. Notice the word *several*: with only one course of action open, nothing is chosen, and no decision exists.

Chapter 7 taught that planning means deciding in advance what

waiting, and it ends at Tk 0. The thick line marks the choice with the higher value.



Square: a choice. Circle: a chance event.
Thick line: the choice with the higher value.
Dashed line: the failure result.

FIG. 11.1. *Decision tree for the second shop*

Now the exceptions. A tree is only as good as its numbers, and the numbers 0.6 and 0.4 are estimates. A tree also needs every result to be written in taka, so it ignores matters that have no price, such as family strain or the shop's name among its customers. Finally, a high expected value does not remove the chance of the loss. The result of Tk 420,000 will never happen: the shop will earn Tk 900,000 or lose Tk 300,000.

Before moving on. A decision tree shows choices as squares, chance events as circles and results at the branch ends. We fold it back from right to left, finding an expected value at each circle, and then choose at the square. Its answer is only as good as its estimates.

* * *

11.2 DECISION SUPPORT SYSTEMS

THE SYSTEM ITSELF

A shop owner wants to know which books sell best each week. Counting through paper receipts takes days, and by then the answer is old. A computer can give the same answer in seconds, and can answer follow-up questions as well. Managers have used this idea in every kind of organisation, from banks to hospitals.

A **decision support system** is a computer-based tool that helps managers study data and compare alternatives. A program that stores a firm's sales records and answers questions about them is one example. Such a system usually has three parts: stored data, tools that analyse the data, and a way to ask *what if* questions, such as how profit would change if a price rose by ten per cent.

Case file, illustrative. The sales software of Meghna and Sons lists the top 20 titles each week.

The tree's answer depends on its numbers and ignores what has no price.

Before moving on: a decision tree is read left to right and calculated right to left.

Hand counting takes days, and by then the answer is old.

Decision support system. A computer-based tool that helps managers study data and compare alternatives.

A decision support system is a computer tool for studying data and comparing alternatives.

The case file shows a small system at work.

10.2 THE DECISION PROCESS

GO SLOWLY HERE.

A bus ride shows each step of the method before we use it on a business.

Step 1 is to state the problem or opportunity in plain words.

Step 2 is to decide how the best choice will be judged.

Decision criterion. A standard used to judge which alternative is best.

Criteria should be set before the options are examined.

Step 3 is to list the possible courses of action.

Step 4 is to test each alternative against each criterion.

A table shows which alternatives pass which tests.

Step 5 is to choose the alternative that best meets the criteria.

A method is easier to learn on a small case. Return to the morning at the start of the chapter, and give it numbers. Rina must reach class by 8:55, and she lives 40 minutes away by bus. She has little money. We will follow her through each step, and then apply the same steps to a shop.

Step 1: name the problem or opportunity. Rina writes it plainly: "I must be in class by 8:55 and I must not spend much." A vague statement leads to a vague decision. Many students find this step easy to skip, because the problem feels obvious. A decision made on the wrong problem can be perfect in every other way and still fail.

Step 2: set the criteria. A **decision criterion** means a standard used to judge which alternative is best. Rina has two criteria: arrive by 8:55, and spend no more than Tk 50. Each criterion is a test that an option must pass. Criteria come from the goals that the decision serves. Chapter 9 showed how a goal becomes a measurable objective, and a good criterion often comes straight from an objective.

The order matters. Set the criteria before you look at the options. If you see the options first, you may choose the criteria that favour the option you already like. A manager who sets the criteria first can defend the choice later, because the standard was fixed before the result was known.

Step 3: develop the alternatives. An alternative is one possible course of action. Rina lists three: walk, take the bus and hire a rickshaw. A manager should seek more than two or three, and should look beyond the first idea that comes to mind. A common error is a list of one real option and two weak ones, which only appears to give a choice. Include the option of doing nothing or waiting, because it too has a cost.

Step 4: evaluate the alternatives. To evaluate means to test each alternative against each criterion and record the result. Rina's walk takes 70 minutes and costs nothing, so she arrives at 9:10 and fails the first test. The bus takes 40 minutes and costs Tk 20. The rickshaw takes 25 minutes and costs Tk 100. The table below sets these results side by side.

Read the table one row at a time. The first cell names an alternative, the next two cells give its results, and the last cell shows whether it passes both criteria. The criteria are on the top row.

RINA'S THREE ALTERNATIVES TESTED AGAINST TWO CRITERIA

ALTERNATIVE	ARRIVAL TIME	COST	PASSES BOTH CRITERIA?
Walk	9:10	Tk 0	No: arrives after 8:55
Bus	8:40	Tk 20	Yes
Rickshaw	8:25	Tk 100	No: costs more than Tk 50

Step 5: choose. Only the bus passes both tests, so Rina chooses it. Choosing is often the shortest step. The hard work has already been done in the earlier steps. When no alternative passes all the criteria, or when several pass, the manager must rank the criteria

EXPECTED VALUE

Suppose a friend offers you a game. A fair coin is tossed once. If it lands heads, your friend pays you Tk 100. If it lands tails, you pay your friend Tk 20. Would you play? Most people say yes, but cannot say why. The reason is that the game is worth something to you, and we can find out how much. Think about playing ten times. You would expect about five heads and five tails.

Five heads bring $5 \times 100 = \text{Tk } 500$. Five tails cost $5 \times 20 = \text{Tk } 100$. The net gain over ten plays is $500 - 100 = \text{Tk } 400$. Divided by ten plays, this is Tk 40 for each play. No single play pays Tk 40, because each play pays Tk 100 or costs Tk 20. But Tk 40 is the average, and it tells you what the game is worth.

That average has a name. **Expected value** means the average result of a choice, found by weighting each possible result by its chance. In the game above, the expected value is Tk 40. The word *expected* does not mean that the result will happen. It means the result you would see on average, if the same choice were made many times.²

The method has one rule. In words: expected value equals the sum, over every possible result, of the chance of that result multiplied by the result. In symbols: $EV = (p_1 \times x_1) + (p_2 \times x_2) + \dots$ Here EV stands for expected value, p stands for the probability of a result, and x stands for the result in taka. The small numbers 1 and 2 label the first and second results. A gain is written as a positive number and a loss as a negative number. The probabilities must add up to 1.

Given. Heads (probability 0.5) pays Tk 100. Tails (probability 0.5) costs Tk 20, written as -20. *Find.* The expected value of one play. *Formula.* $EV = (p_1 \times x_1) + (p_2 \times x_2)$. *Steps.* Heads: $0.5 \times 100 = 50$. Tails: $0.5 \times (-20) = -10$. Add: $50 + (-10) = 40$. *Answer.* The expected value is Tk 40 for each play. *Check.* The probabilities add up ($0.5 + 0.5 = 1$), and Tk 40 lies between the worst result (-20) and the best (100). It also equals the Tk 400 we found over ten plays, divided by ten.

Now a messier case. A trader plans a stall at a three-day fair. A busy fair (probability 0.5) gives a profit of Tk 4,000. A normal fair (probability 0.3) gives Tk 1,500. A wet fair (probability 0.2) gives a loss of Tk 1,000. *Find* the expected value. *Formula.* $EV = (p_1 \times x_1) + (p_2 \times x_2) + (p_3 \times x_3)$. *Steps.* Busy: $0.5 \times 4,000 = 2,000$. Normal: $0.3 \times 1,500 = 450$. Wet: $0.2 \times (-1,000) = -200$. Add: $2,000 + 450 - 200 = 2,250$. *Answer.* The expected value is Tk 2,250. *Check.* $0.5 + 0.3 + 0.2 = 1$, and 2,250 lies between -1,000 and 4,000.

Three slips are common. The first is to forget the minus sign on a loss, which makes a bad choice look good. The second is to use probabilities that do not add up to 1. The third is to mix units, for example profit in thousands of taka in one line and in taka in the next. Write every result in the same unit before you multiply.

² Some books use the longer name *expected monetary value* when all results are sums of money. This book uses *expected value*.

A fair way to describe an uncertain choice is the average result over many tries.

GO SLOWLY HERE.

Over ten plays the game pays Tk 400, which is Tk 40 for each play.

Expected value. The average result of a choice, found by weighting each possible result by its chance.

Value. Word in use. In daily life, how much something is worth or how important it is. Here, a number in taka: the average result of a choice.

Expected value is the average result of a choice, weighted by chance.

The formula multiplies each result by its chance and adds the products.

The first example uses small numbers so that the method is the only new thing.

A second example has three results and is more realistic.

The common slips are wrong signs, probabilities that miss 1, and mixed units.

10.3 CONDITIONS OF DECISION MAKING

GO SLOWLY HERE.

Every decision faces a different amount of knowledge about the future.

Certainty means the manager knows the exact result of each alternative.

Certainty. A condition in which the result of each alternative is known in advance.

Risk means results are unknown but their chances can be estimated.

Risk. A condition in which the result is not known but its chances can be estimated.

Risk. Word in use. In daily life, a danger of harm. In management, a condition in which the result is not known but its chances can be estimated, and the result may be good or bad.

Risk depends on records or experience that allow a chance to be estimated.

Uncertainty means that neither the results nor their chances can be estimated.

Uncertainty. A condition in which neither the results nor their chances can be estimated.

A table sets the three conditions side by side with their tests.

Why do two decisions of the same size feel so different? One may be calm and the other frightening. The difference lies in how much the manager knows about what will happen after the choice. Management names three conditions, from the most knowledge to the least. We take them one at a time, with a small example for each.

The first condition is **certainty**, which means a condition in which the result of each alternative is known in advance. Rina knows that the bus fare is Tk 20, and she knows that the bus takes 40 minutes. A shop that signs a rent of Tk 38,000 a month knows what it will pay each month. Certainty is rare in business, but it does exist for costs that a contract fixes.

The second condition is **risk**. Risk means a condition in which the result is not known but its chances can be estimated. A tea stall owner has kept records for 10 market days. It rained on 4 of them. The owner cannot know whether tomorrow will be wet, but can estimate that the chance of rain is 4 in 10, or 0.4. The word *risk* has an everyday meaning of danger. In management it means that the result is unknown and its chances can be estimated, and the result may be good or bad.

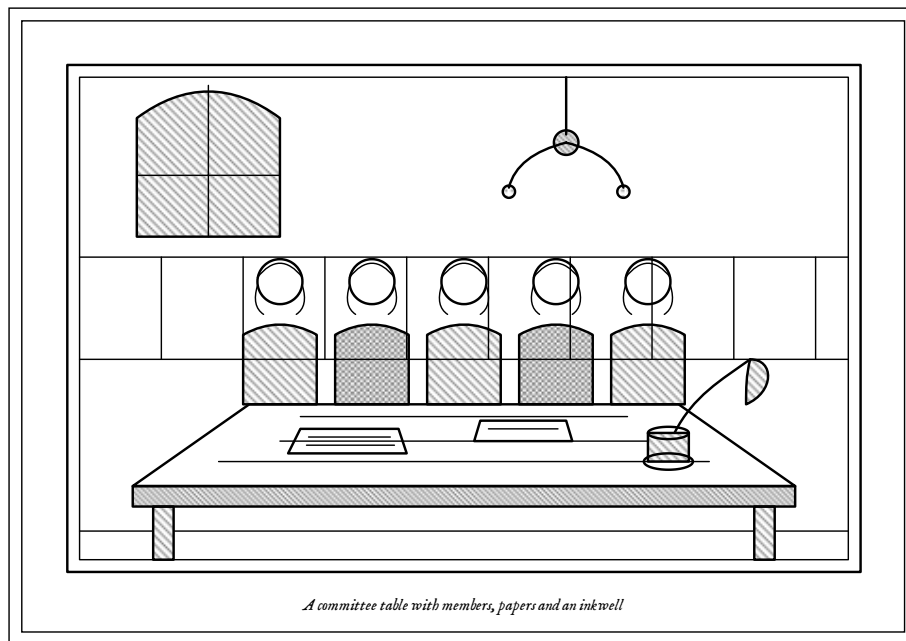
How does a manager estimate a chance? The usual source is past records or experience of similar cases. The owner's 10 days of records give a rough chance of 0.4. More records would give a better estimate. When the records are few, the estimate is weak, and a manager should say so honestly.

The third condition is **uncertainty**, which means a condition in which neither the results nor their chances can be estimated. Suppose the tea stall owner wants to open in a town where she has no records and no comparison. She cannot say what the sales might be, and she cannot say how likely each result is. Many students find the difference between risk and uncertainty confusing, because both involve not knowing. The test is whether a chance can be estimated: if yes, it is risk; if no, it is uncertainty.

The table below compares the three conditions. Read each row from the left. The first cell names the condition, the second says what the manager knows, and the third gives the tea stall example.

THE THREE CONDITIONS OF DECISION MAKING

CONDITION	WHAT THE MANAGER KNOWS	TEA STALL EXAMPLE
Certainty	The result of each alternative	The rent for the month, fixed by a contract
Risk	The possible results and their chances	Rain on 4 of the last 10 market days
Uncertainty	Neither the results nor their chances	A town where the owner has no records



CHAPTER XI

Decision Tools and Group Decisions

How can a manager make a doubtful choice less doubtful?



Managers seldom know what will happen after they decide; this chapter measures that doubt with numbers, compares choices on one diagram, uses computers for facts, and shows when groups should decide.

A table collects the results so that the options can be compared.

Steps 5 and 6: Mirpur wins on both criteria, and the follow-up compares actual with expected sales.

The conclusion states what the figures support and what they leave open.

Rung 2 reshapes Rina's case so that no option passes both tests.

Rung 3 moves the method to a school with a very different setting.

Rung 4 combines the process with the three conditions and with planning.

A short summary of the chapter's five defined terms and the process.

expected sales for each taka spent. Criterion B: $5,100,000 - 624,000 = \text{Tk } 4,476,000$.

The table collects the results. Read each row from the left: the option, the first-year cost, the two criteria. The last row shows that waiting cannot be scored.

MIRPUR AND UTTARA ON THE TWO CRITERIA

OPTION	FIRST-YEAR COST	A: SALES PER TAKA SPENT	B: SALES LESS RENT
Mirpur	Tk 1,106,000	4.88	Tk 4,944,000
Uttara	Tk 1,344,000	3.79	Tk 4,476,000
Wait a year	Tk 0	Not scored	Not scored

Reasoning, steps 5 and 6. Mirpur is higher on both criteria, so on these figures it is the better site. If the owner chose Mirpur, the follow-up would record the actual sales and costs in the first year and compare them with Tk 5,400,000 and Tk 1,106,000. The size of any gap would show how far the estimates could be trusted. The condition matters here: rent and fitting are certain, but expected sales hold no estimated chance, and so belong nearer uncertainty.

Conclusion. On the assumed goal, Mirpur is the better site, because it costs less and expects more sales. The figures do not say whether waiting is better, since they do not measure it. The case file records the options and does not record the final choice. *Check.* Mirpur has the lower rent, the lower fitting cost and the higher expected sales, so every figure points the same way. It follows that both criteria must favour Mirpur, and they do.

CHANGING THE SITUATION

The same method works in new situations. Rung 1 is Rina's bus ride, solved above. In rung 2, suppose the bus fare rises to Tk 60. Now no alternative passes both criteria. Rina must rank her criteria and decide which counts more, and the process returns to step 2.

In rung 3, a school head must choose a place for a class visit. The head sets criteria for cost, distance and safety, lists four places, tests each one and chooses. The steps are the same, though the people and the sums are different. Chapter 7 showed that planning places the choice inside a larger plan, and here the visit is that plan.

In rung 4, a small clinic must decide whether to buy a new machine. The price is certain. The number of patients who will need it can be estimated from last year's records, so that part is a risk. The reaction of a rival clinic cannot be estimated, so that part is uncertainty. The owner uses all six steps and treats each part according to its condition.

* * *

SUMMARY

Decision making is choosing one course of action from several, in order to solve a problem or use an opportunity. The process has

The record shows the gap described above. Fourteen suggestions are creativity at work. Three adopted suggestions mark the step into innovation. The reading corner is an idea put into use, and the weekend sales figure records what followed. We now apply these figures in the worked case below.

Before moving on. Creativity is the ability to produce new and useful ideas. Innovation is turning a new idea into something that is put to use. Managers encourage ideas by listening, by giving time and freedom, by making failure safe, and by providing a path from idea to action.

* * *

WORKED CASE: SUGGESTIONS AND THE READING CORNER

This section applies the ideas of the chapter to the shop's record. It does not give a model answer. It shows one way of reasoning, step by step, so that you can follow each decision and then repeat it on another case.

Situation. In January to June 2026, Meghna and Sons recorded 14 suggestions from its staff, and 3 were adopted. A reading corner was built in April 2026 at a cost of Tk 45,000. Weekend sales rose from Tk 180,000 to Tk 207,000 a month.

Question. Which of the figures show creativity and which show innovation? What share of suggestions was adopted? What can be said about the reading corner and the rise in weekend sales?

Reasoning. First we separate the two ideas. The 14 suggestions measure creativity, because each is an idea offered by a person. The 3 adopted suggestions measure innovation, because they were put to use. The share adopted is 3 divided by 14, which is about 0.214, or 21%. So about one suggestion in five became part of the shop's work. This is not a poor result. It shows that the shop listens and also chooses.

Second, the reading corner. The record shows that it was built in April 2026 at Tk 45,000, so it is an idea put into use. The record does not say whether it came from one of the 3 adopted suggestions, and we shall not assume so. The weekend sales rose by Tk 207,000 - Tk 180,000 = Tk 27,000 a month. As a share of the earlier figure this is 27,000 divided by 180,000 = 0.15, or 15%, which agrees with the record. The corner may have helped, but weekend sales can rise for other reasons, such as a change in the weather or in the school year.

Third, a rough comparison. If the rise of Tk 27,000 a month continued, then in about 45,000 divided by 27,000 = 1.7 months the extra sales would equal the cost of the corner. This is only an illustration, because sales are not profit: the shop must still pay for the books it sells. The owner would need the profit on those extra sales to judge whether the corner paid for itself.

Conclusion. The record shows a shop that collects ideas and adopts one in about five. It shows one idea, the reading corner, put into use, followed by a rise of 15% in weekend sales. To do more

The record shows the gap between ideas and use, and one idea that crossed it.

The recap fixes creativity, innovation and the encouraging of ideas.

The worked case applies the chapter's ideas to the shop's record.

The situation lists the suggestions, the reading corner and the sales.

The question asks what the figures show about creativity and innovation, and what they cannot show.

Reasoning begins by separating ideas from ideas put to use.

The reading corner shows an idea put into use, but its link to the sales rise needs care.

A rough comparison of cost and extra sales helps the owner, but sales are not profit.

The conclusion states what the figures show and what they leave open.

Formal organization. The official structure of jobs, rules and lines of authority that management sets up.

The formal organization is the official structure that management sets up.

The formal organization gives order, stability and clear authority, but it can be slow.

Informal organization. The network of friendships and social groups that forms among employees on its own.

The informal organization is the network of friendships that forms without management's instruction.

The case file records an informal group at Meghna and Sons.

The lunch group is informal, because no manager created it and it appears on no chart.

People join informal groups to meet needs that the official structure does not meet.

the store manager walks between them. Now watch where people stand during a quiet hour. Some of them gather in small groups, talk about books and families, and pass along advice that no one has ordered them to give. A second pattern has formed beside the first. Both patterns shape how the shop works, and a manager who sees only the first will often be surprised by the second.

The first pattern is the **formal organization**. It means the official structure of jobs, rules and lines of authority that management sets up. It is deliberate, which means that managers choose it on purpose. It is usually written down, in job descriptions, rules and the organization chart. Everything we studied in section 12.2 produces a formal organization, because the four steps are decisions that management takes and records.

The formal organization has real strengths. It gives each employee a known job and a known person to ask. It lasts when individuals come and go, because the jobs remain even when the people change. It supports the control of work, since someone is answerable for each task. Its weakness is that it cannot describe everything people do. Official rules cannot foresee each small situation, and an organisation that relies only on its rules can become slow and stiff.

PAUSE AND CHECK 12.P2 A shop's job description says that the stock clerk reports to the store manager. Is this part of the formal or the informal organization? Give one reason.

The second pattern is the **informal organization**. It means the network of friendships and social groups that forms among employees on its own. No manager designs it, no document records it, and it has no official leader. It appears wherever people work together over time. Employees who share a bus route, a hometown or an interest in poetry will talk to one another more than the chart would predict, and a group of friends is the usual result.

Case file, illustrative. In 2026, 3 sales assistants at Meghna and Sons form a lunch group and exchange book tips informally. The record places the group beside the formal structure and does not show it on the organization chart.

The lunch group is part of the informal organization. No manager created it. No chart records it. Yet it does useful work: when one sales assistant learns which new title a school teacher has asked about, the others hear it at lunch. The knowledge reaches the counter faster than any written notice would. This example shows an important point. The informal organization is neither good nor bad in itself. Its effect depends on what the group does and on how it relates to the goals of the shop.

Why do people join informal groups? The reasons are ordinary ones. Employees want company, because work fills much of the day and people dislike spending it alone. They want help, because a colleague can often explain a task faster than a manual. They want information, because the official channels are slow or incomplete. They want to feel that they belong and that their opinion counts. Each of these is a need that the formal structure rarely meets, and

Not every group is a team. A **team** means a small group of people with joined skills, working toward a common goal for which they are all accountable. To be accountable is to be answerable for the result. Three features separate a team from an ordinary group. The members have different skills that fit together. They share one goal. They are answerable together for whether the goal is reached. A stock clerk, a buyer and a cashier who work together to prepare a school order form a team. The four craftsmen eating lunch are a group but not a team.

The table below sets three kinds of gatherings side by side. Each row names a feature, and the columns show how the feature appears in a crowd, a group and a team. Read across a row to compare the three on one feature.

CROWD, GROUP AND TEAM COMPARED

FEATURE	CROWD	GROUP	TEAM
Interaction	None or little	Regular	Close and planned
Goal	None shared	Often personal	One common goal
Accountability	None	Each for self	All for the result
Example	Passengers at a bus stop	Colleagues at lunch	Staff preparing a school order

Every group, whether it is a team or not, develops rules of behaviour. The rules are rarely written, yet members know them well. A newcomer to the workshop learns, without being told, that one does not leave a mess on the common bench and that one does not criticise a colleague in front of the master. Such rules have a name in management.

A **group norm** means an unwritten rule that members of a group follow and expect others to follow. A norm may say how fast to work, how to speak to customers, or when to take a break. A norm is shown by what happens to someone who breaks it. The group may laugh, give a look, or leave the person out of conversation. Norms hold a group together, and they are an important part of the informal organization.

Norms can serve the organisation or oppose it. A norm that says *we help new employees learn the shelves* serves the shop. A norm that says *no one should work faster than the slowest member* works against it, because it limits what the group produces. Managers cannot abolish norms, but they can influence them. They can reward helpful norms, discuss harmful norms openly, and set an example in their own behaviour.

PAUSE AND CHECK 15.P2 The three sales assistants of a shop have an unwritten rule that no one sells a book to a customer who is already talking with another assistant. Is this a group norm? Say

Team. Word in use. In daily life, the players on one side in a sport. In management, a small group of people with joined skills who work toward a common goal.

Team. A small group of people with joined skills, working toward a common goal for which they are all accountable.

A team is a small group with joined skills, a common goal and shared accountability.

A table compares a crowd, a group and a team.

Groups develop rules that nobody writes, and these rules are called norms.

Norm. Word in use. In daily life, what is normal or average. In management, an unwritten rule of behaviour that a group expects its members to follow.

Group norm. An unwritten rule that members of a group follow and expect others to follow.

A group norm is an unwritten rule that members follow and expect others to follow.

Norms can help the organisation or work against it.

form a counter group of 2 duties. Accounts, online orders, school accounts and marketing all deal with records or with customers who are not at the counter, so they form a records-and-outside-customers group of 4 duties. The grouping is one reasonable choice, and the file does not say which choice the shop made.

For step three, the base record states that Tariq Meghna handles accounts and online orders, and both duties sit in the third group. That agrees with our grouping, which is a useful sign. The file does not record who holds the other duties, so we do not invent assignments. We can, however, work out how heavy the load is. Ten duties among seven people give an average of 10 divided by 7, which is about 1.4 duties per person. For step four, the store manager would take the reporting of the stock group and the counter group, and the owner would receive reports from the store manager and from Tariq Meghna.

The lunch group of 3 sales assistants is informal. It has no place on the organization chart, because no manager created it and it carries no authority. The store manager should still know of it, because the group already carries useful knowledge about which books customers ask for.

Conclusion. The shop's ten duties fall into three groups of 4, 2 and 4 duties. The load is about 1.4 duties per person on average, so some people hold more than one duty. The lunch group belongs to the informal organization, and managers should use it, not suppress it.

Check. We test the counting. The groups give 4 plus 2 plus 4, which equals 10, so no duty is lost or counted twice. The load check: 7 people times 1 duty each gives only 7 duties, so $10 - 7 = 3$ duties must be additional ones. This confirms that at least some people hold two duties. The shop should also confirm each person's duties in writing, because a task that is not assigned is the first step toward the puzzle that opened this chapter.

* * *

SUMMARY

Organizing is the function that arranges tasks, people and resources into a working structure. Its result is the organizational structure, the pattern of jobs, groups and reporting lines. The process has four steps: identify the tasks, group the tasks, assign the duties and set the relationships. The organization chart records the result. Beside the formal organization, which management sets up, an informal organization of friendships forms on its own. Effective managers design the formal structure well and work with the informal groups.

THINK IT THROUGH

Each question below changes the situation a little more than the one before. Answer them in order.

For Step three, the base record already gives one assignment, and the other assignments are not recorded.

The lunch group is informal, so it stays off the chart but not out of the manager's view.

The conclusion states the structure and the status of the lunch group in plain sentences.

The check tests the counting and guards against a hidden gap.

The summary restates the chapter in a few sentences.

Four situations test the chapter, from familiar to combined.

belongs to the organisation, but it is felt first in the groups and then by each person.

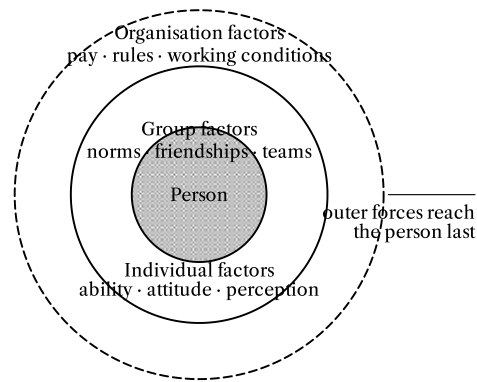


FIG. 15.1. *Individual, group and organisation factors as nested rings*

The innermost ring contains the individual factors, and these are the subject of the rest of this section. The group factors are studied in section 15.2. Organisation factors, such as pay, rules and working conditions, are treated in later chapters on motivation and leadership. We begin with the person because every group is made of persons, and a manager who understands one person at a time is better prepared to understand a group.

The first individual factor is the set of individual differences, which means the ways in which people vary from one another. People differ in ability: one can add a column of numbers quickly, and another can calm an angry customer. They differ in attitude, which is a person's settled way of judging something as good or bad. They differ in needs, in experience and in background. No two employees are alike, and a rule that suits one employee may confuse another.

Individual differences have a limit that managers must respect. They help to explain why people act differently, but they must not be used to label a person unfairly. A manager who decides, after one slow morning, that an employee is lazy has drawn a conclusion from too little evidence. The correction is to observe over time, to ask the employee, and to look for causes outside the person, such as unclear instructions or a broken tool.

The second individual factor is perception. Perception means the way a person selects, arranges and understands what he or she sees and hears. Two people can look at the same event and report different things, because each notices some parts and ignores others. The first assistant saw a pile of books as a simple task. The second assistant saw a pile that might block customers. Each assistant described what was real to him or her.

Perception depends on a person's experience, mood and

³ Many students find perception difficult to separate from opinion. A useful test is to ask: what did I observe, and what did I add from my own mind? Only the first is fact.

Individual factors come first, because every group is made of single persons.

Individual differences are the ways in which people vary in ability, attitude and background.

Individual differences have limits: they explain much, but they must not be used to label people.

Perception is the way a person selects and understands what he or she sees and hears.

Perception is shaped by experience, mood and

REVIEW QUESTIONS

- 12.1 Separate these into formal or informal: the job description of a cashier; a lunch group of friends; the reporting line from stock clerk to store manager; a habit of sharing book tips at noon.
- 12.2 State the four steps of the organizing process in order, and say which question each step answers.
- 12.3 In your own words, give the difference between the formal organization and the informal organization. Give one strength of each.
- 12.4 Explain why people join informal groups, and say how a manager can work with such groups.

Two ideas from Chapter 12 support this chapter.

A shopkeeper can watch three assistants, but no one can watch three thousand soldiers.

The puzzle has two parts: how many people to lead directly, and how to group the rest.

The chapter answers both parts and then names the forces that shape the answer.

A teacher with a class shows that the number of people led changes how well they are led.

Span of management. The number of employees who report directly to one manager.

The span of management is the number of employees reporting directly to one manager.

You will need two ideas from the previous chapter. The first is organizing, taught in Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is'.¹ The second is the formal structure, taught in Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.² Read those sections again if either idea feels uncertain.

Think of a shopkeeper who has three assistants. She sees every sale, hears every conversation and corrects a mistake within a minute. Now think of a general who has three thousand soldiers. He cannot see them all, and he cannot speak to each one in a single day. Somewhere between three and three thousand there lies a point where one person can no longer lead everyone directly. Where is that point, and what should the leader do when the point has been passed?

The puzzle has a second part. Suppose a Victorian customs department, with desks in tiers and a chief clerk above them, grows from twenty clerks to two hundred. The chief clerk must now decide how many officers will report to him, and how the clerks will be grouped beneath those officers. Some departments group clerks by the kind of task. Others group them by the ships they handle, or by the port where they work. Each choice changes how fast the work moves and how well the clerks understand their duties.

In this chapter we answer both parts of the puzzle. We begin with the number of people a manager leads directly. Then we study four ways of grouping jobs. Finally we examine the forces that push a manager toward one design or another, and we study the old idea that the work itself should be divided into small tasks. Throughout, we shall use the small Dhaka bookshop Meghna and Sons, whose records give us numbers to work with.

* * *

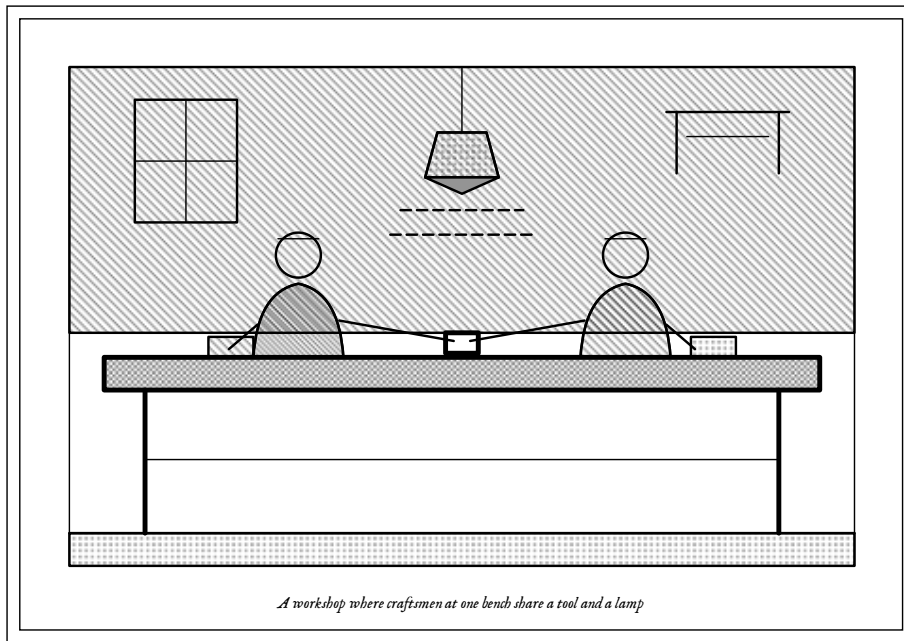
13.1 SPAN OF MANAGEMENT

Consider a teacher who gives private lessons to two students. She knows each student's weak points and can reply to every question at once. When the same teacher faces a class of sixty, she can no longer follow each student. Her skill has not changed. The number of people she leads has changed, and that number alters the quality of her leadership. Managers meet the same fact in every organisation.

Management gives this number a name. The **span of management** means the number of employees who report directly to one manager. The word *directly* matters. A manager with five employees and fifty more below them has a span of five, not fifty-five. A store manager who has three sales assistants, a cashier and a stock clerk reporting to him has a span of five.

¹ Recall: organizing, Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is'.

² Recall: formal organization, Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.



CHAPTER XV

Human Factors, Groups and Creativity

If two people have the same tools, why does the work differ?



Machines behave the same way each day, but people do not. This chapter explains the human factors that shape work, shows how groups and teams form and set their own rules, and ends with the ways in which new ideas are born and put to use.

CHANGING THE SITUATION

The same ideas apply in new situations. Rung 1 is the case above. In rung 2, suppose a return is worth exactly Tk 5,000 and Rafiq is absent. Rafiq's right does not pass to anyone else, so the return waits for him, or goes to Nurul. This shows that authority belongs to the post and to its holder, and it does not pass without a decision.

In rung 3, a school head delegates the booking of class visits to a teacher, with a limit of 20,000 taka for each visit. The teacher has the task, the right to book up to the limit, and the duty to report to the head. The three parts match, and the head still answers to the school's governors.

In rung 4, an owner has 12 employees and delegates to three team leaders. Chapter 13 explained that the span of management is the number of employees a manager can direct well. With three leaders, each has a span of four employees, which is easy to manage. The owner holds a weekly meeting of the leaders to coordinate the three teams.

* * *

SUMMARY

Authority is the right to give orders and to expect them to be obeyed, given by position. Power is the ability to influence others, with or without a position. Line authority directs work along the chain of command, and staff authority advises without commanding. Delegation gives an employee the right to decide and act for a stated task, and accountability is the duty to answer for results to the person who delegated. Centralization keeps decisions at the top, and coordination links separate work toward one goal.

THINK IT THROUGH

14.T1 Rafiq is away on a day when a customer returns books worth Tk 5,400. Explain who may approve the return under the case file, what Rafiq's authority does and does not cover, and one way Nurul could change the arrangement so that returns do not wait.

14.T2 Tariq has been advising Nurul to refuse credit to a school. A sales assistant says Tariq has told her to refuse the order herself. Which kind of authority has Tariq used, and is it within his right? Explain your reasoning.

WHERE PEOPLE GO WRONG

Mistake: "Authority and power are the same thing." Authority comes from a position and stays with the post. Power comes from the person and leaves with the person. A manager may have one without the other.

Mistake: "If I delegate, the result is no longer my concern." The employee is accountable to the manager who delegated. The man-

Rung 2 reshapes the case by changing the amount.

Rung 3 moves the idea to a new setting.

Rung 4 combines delegation with the ideas of the span of management and coordination.

A short summary of the chapter's eight defined terms.

Mistake one: believing that authority and power are the same.

Mistake two: believing that delegating gives away

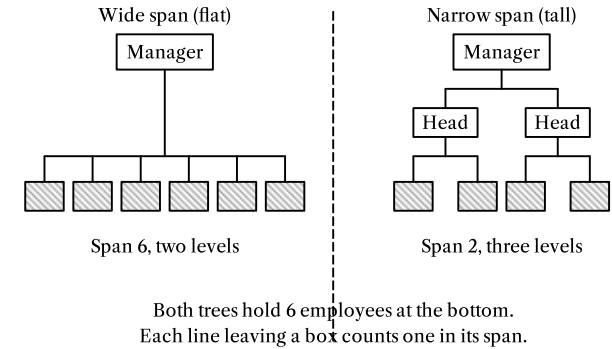


FIG. 13.1. Wide and narrow spans shown as two trees

Each design has benefits and costs, and neither is best in every case.

The recap fixes span, the factors, and tall and flat designs.

Each design has benefits and costs. A tall structure allows close supervision and gives employees many steps to climb, but messages must pass through many levels, so decisions are slow, and the many managers add to the cost of wages. A flat structure sends messages quickly and gives employees more freedom, but each manager is overloaded if the span is too wide. A manager who has too wide a span may miss mistakes. A manager who has too narrow a span may watch too closely and discourage initiative.

Before moving on. The span of management is the number of employees who report directly to one manager. The suitable span depends on factors such as skill, similarity of tasks, clarity of plans, nearness and the manager's other duties. A tall structure has many levels and small spans. A flat structure has few levels and wide spans.

* * *

13.2 DEPARTMENTALIZATION

A bookshop with three employees needs no groups. Everyone knows everyone, and the owner speaks to all of them daily. When the shop grows to seven employees, the owner must decide how to divide them. One idea is to put the people who buy and shelve books together, and the people who serve customers together. That is a division by the kind of work. Another idea is to place together everyone who deals with schools. That is a division by customer. The choice affects how the shop works every day.

Management has a name for this choice. **Departmentalization** means grouping jobs into departments according to a common basis, such as function or customer. A department is a group of jobs placed under one manager. The basis is the feature that the jobs in one group share. Four bases are common: function, product, customer and territory. We take each in turn.

Departmentalization. Grouping jobs into departments according to a common basis, such as function or customer.

Departmentalization is the grouping of jobs into departments on a common basis.

answer for results to the person who delegated the work. Responsibility, authority and accountability must be kept in balance. A manager who delegates still answers upward for the final result.

You should now be able to say: delegation hands over a stated task and the right to act, and accountability stays with the person who must answer for the result.

* * *

14.4 CENTRALIZATION AND DECENTRALIZATION

Delegation raises a larger question. Across the whole organisation, how many decisions should the top keep, and how many should it hand down? The answer is a matter of degree, and it ranges from almost all at the top to almost none.

Centralization means keeping most decision-making power at the top of the organisation. In a very centralized shop, the owner approves every return and every order. The opposite is decentralization, which means spreading decision-making power to lower levels. It is the sum of many acts of delegation. The two ideas mark the ends of a line, and real organisations sit between them.

Which should an organisation choose? Several things point the way. A large organisation cannot decide everything at the top and tends to spread decisions downward. Skilled and trusted employees make decentralization safer. Where speed matters, a decision made close to the customer is faster. Where a mistake is very costly, or where all parts must act in the same way, the top may wish to keep control. Tall structures, with many levels, slow decisions that must climb to the top.

Before moving on. Centralization keeps most decision-making power at the top, and decentralization spreads it downward. Neither is always best. Size, skill, speed and the cost of mistakes guide the choice.

* * *

14.5 COORDINATION

Division of work brings a problem. When a task is split among departments, each does its part, and the parts must still fit together. A shop whose buyer orders books the sales assistants cannot sell, or whose sales assistants promise books that the stock clerk has not received, is a shop whose parts do not fit.

The remedy is **coordination**, which means linking the work of different people and departments so that they pull toward the same goal. Coordination is not a separate task done once. It is a quality that good management gives to all the other tasks. The need grows with the number of people, since each added person adds more links to keep in order.

Two groups of methods help. The basic activities are shared

A sentence to test the reader's grasp of the section.

Every organisation must decide how many decisions to keep at the top.

Centralization keeps most decision-making power at the top.

Centralization. Keeping most decision-making power at the top of the organisation.

Which to choose depends on size, skill, speed and the cost of mistakes.

The recap of section 14.4.

When work is divided among many people, something must join it again.

Coordination links the work of separate people toward one goal.

Coordination. Linking the work of different people and departments so that they pull toward the same goal.

Basic activities and

FOUR BASES OF DEPARTMENTALIZATION

BASIS	PUTS TOGETHER	STRENGTH	WEAKNESS
Function	Jobs needing the same skill	Builds skill, saves effort	Departments may drift apart
Product	Jobs for one good	Focus on each product	Repeated skills, higher cost
Customer	Jobs for one kind of buyer	Knows the buyer well	Repeated skills, needs more people
Territory	Jobs in one place	Responds to local needs	Branches may differ in rules

Few organisations use only one basis. A shop may group by function at the top and by customer inside the sales department. A large company may group by territory, and inside each territory by product. The test is the same in every case: does the grouping help the organisation reach its goals, with a reasonable cost? A grouping that was right for a shop with five employees may fail for a shop with fifty.

Before moving on. Departmentalization is grouping jobs into departments on a common basis. The four common bases are function, product, customer and territory. Each has strengths and weaknesses, and most organisations mix them.

* * *

13.3 FORCES BEHIND STRUCTURE

A manager who has studied spans and departments may expect a single best design to exist. It does not. Two organisations of similar size can use different structures and both succeed. The reason is that managers work under forces, and the forces differ from place to place. We now name the main forces, beginning with two old ideas about what a structure should be.

The first idea is the bureaucratic model. Max Weber, a German sociologist who lived from 1864 to 1920, described an organisation built on written rules, fixed duties, a clear chain of command from top to bottom, and promotion by ability.³ Its aim was fairness and reliability: the same rule applies to everyone, and the work continues when a person leaves. Its weakness is stiffness. An organisation that follows rules without exception may respond slowly to new situations.

The second idea is the behavioural model. Writers of this school noticed that employees are not machines. People work better when they feel valued, when they have some say in decisions, and when the groups they belong to are respected. A structure built on this idea gives employees wider tasks, more freedom and more contact with managers. This model is linked to the informal organization that we studied in the previous chapter. Its weakness is that

³ Weber described this model in writings that were published after his death. The word *bureaucracy* has a neutral meaning in management, although in daily speech it is often used as a complaint about slow offices.

Most organisations mix the bases, and the right mix depends on the size and the work.

The recap fixes departmentalization and the four bases.

Managers do not choose a structure freely; several forces push them toward one design.

The bureaucratic model asks for fixed rules, clear levels and impersonal treatment.

The behavioural model asks for a structure that fits the needs and habits of people.

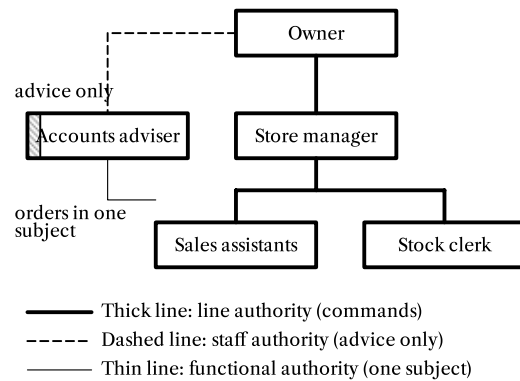


FIG. 14.1. Line, staff and functional authority

Read the figure from the owner at the top downwards. The thick lines carry line authority down the chain of command. The dashed line carries staff authority, which is advice only, and it reaches the line manager but not the employees below. The thin line carries functional authority, which is a right to direct in one subject. The key at the bottom repeats these meanings in words, so the figure reads the same in grey.

The three kinds can cause trouble. A staff person who begins to give orders to employees weakens the line manager, and the employee does not know whom to obey. An employee who receives two different orders is in a hard position. The remedy is to say plainly which kind of authority each person holds and in which matters. Where authority is stated in writing, such conflicts are fewer.

Before moving on. Line authority is the right to direct employees along the chain of command. Staff authority is the right to advise and support line managers without commanding their employees. Functional authority is a staff specialist's right to direct others in one subject only. In the case file, Tariq holds staff authority.

You should now be able to say: line authority commands, staff authority advises, and functional authority commands in one subject only.

* * *

14.3 DELEGATION

Suppose Nurul, the owner of a bookshop, personally approved every return and every purchase. Each task would wait for Nurul to be free, and customers and suppliers would wait too. A manager has a limit on time and attention. Some tasks must therefore be handed to others, and the handing must be done with care.

The word for this is **delegation**, which means giving an employee the right to make decisions and take action for a stated

The figure shows the three kinds of authority as three kinds of line.

The figure is read from the owner at the top, with the line types in a key below.

The three kinds cause trouble when they are not kept apart.

The recap of section 14.2.

A sentence to test the reader's grasp of the section.

GO SLOWLY HERE.

A manager cannot do all the work personally, so tasks must be handed on.

Delegation hands a stated task to an employee together with the right to act.

The situation gives the spans, the present grouping and the alternative.

The question asks whether the spans are reasonable and whether the shop should change its grouping.

Reasoning begins with the spans and the factors that affect them.

Next, the sales figures show how large each customer department would be.

The weakness of customer grouping is repeated skills, which a small shop can ill afford.

The conclusion states what the records support.

The check tests the arithmetic and counts the reports.

The summary restates the chapter in a few sentences.

Situation. In 2026 Nurul Meghna has 2 direct reports and Rafiq Meghna has 5. The shop groups its jobs by function: sales, stock and accounts. The alternative is to group by customer type, using the 2025 sales: walk-in Tk 9,000,000; schools Tk 3,600,000; online Tk 1,800,000.

Question. Are the two spans reasonable for a shop of this size? Would a grouping by customer type be a better choice than the present grouping by function?

Reasoning. First, the spans. Rafiq's span of 5 is the larger. The factors from section 13.1 help us judge it. His employees work in one shop, so nearness is favourable. Their tasks, which are selling, the counter and stock handling, differ somewhat but are routine. The record gives no sign of difficulty. Nurul's span of 2 is small, but the owner has other duties, such as dealing with the world outside the shop, so a small span is reasonable. Both spans seem suitable.

Second, the grouping. A customer grouping would create three departments. We first find how large each customer type is, by dividing each customer's sales by the total of Tk 14,400,000. Walk-in sales are Tk 9,000,000, which is 62.5% of the total. School sales are Tk 3,600,000, which is 25%. Online sales are Tk 1,800,000, which is 12.5%. A walk-in department would be the main part of the shop. An online department would serve only one-eighth of the sales, yet it would still need its own skills in stock handling and accounts.

With only 7 employees, three customer departments would leave very few people in each. Each would still need stock handling and accounts, so the work would be repeated, and the cost per sale would rise. The present grouping by function avoids repetition: one stock group and one accounts group serve all three kinds of customer. Its weakness, as section 13.2 explained, is that sales and stock may drift apart, and the shop's small size makes that risk low.

Conclusion. The two spans appear suitable, and the present grouping by function is reasonable for a shop of this size. A customer grouping becomes more attractive if school or online sales grow so that a customer department could keep several people busy.

Check. We test the numbers. The three sales figures add to Tk 9,000,000 + Tk 3,600,000 + Tk 1,800,000 = Tk 14,400,000, which equals the total sales of 2025. The three shares add to 62.5% + 25% + 12.5% = 100%. The spans add to 2 + 5 = 7, which equals the 7 employees on the staff list, so every employee reports to exactly one manager in our reading of the record.

* * *

SUMMARY

The span of management is the number of employees who report directly to one manager, and the suitable span depends on factors such as skill, similarity of tasks, clarity of plans, nearness and the

Authority has a size. A store manager has more authority than a sales assistant, and an owner has more than a store manager. It has a source, which is the organisation. It also has a limit: a store manager may direct the shop's work but may not order a customer to buy. Authority reaches only as far as the organisation has given it.

Here is a simple test. If the manager leaves and a new manager is appointed, who holds the authority? The new manager does. Authority stays with the post and passes to each new holder. This is why we say it is given by position.

Now consider the friendly colleague. The colleague holds no higher position than the cashier, yet the cashier does the favour. **Power** means the ability to influence other people's behaviour, whether or not one holds a position. The colleague has power because the cashier likes and trusts the colleague. If the colleague moved to another shop, the power would move too, since it belongs to the person.

Where does power come from? Writers on management name five common sources. The first is position, which is authority itself. The second is the ability to give rewards, such as a bonus or a better shift. The third is the ability to punish, such as a warning. The fourth is knowledge and skill: an employee who knows the stock system better than anyone else is listened to. The fifth is personal liking and respect. Chapter 17, 'Leadership', section 17.1 'Leader and Manager', studies the power of leaders in more detail.

The table below compares authority and power. Read each row from the left. The first cell gives a question, the second answers it for authority, and the third answers it for power.

AUTHORITY COMPARED WITH POWER

QUESTION	AUTHORITY	POWER
Where does it come from?	The position, given by the organisation	Knowledge, rewards, punishment, liking or position
Who holds it?	Whoever holds the post	A person, with or without a post
What happens when the holder leaves?	It stays with the post	It leaves with the person
Example	A manager orders a till count	A trusted colleague asks for a favour

A manager may hold both, and the strongest managers do. A manager may also hold one without the other. A newly appointed manager has full authority and may have little power, because the employees do not yet know or trust the new person. A senior employee may have much power and no authority. Employees often follow such a person, and a wise manager notices it.

Authority has a limit that is easy to miss. A right to give orders means little if the orders are not obeyed. In practice, authority

Authority can be small or large, but it always belongs to a post.

The test of authority is whether it would pass to a new holder of the post.

Power is a different idea: the ability to influence, with or without a position.

Power. The ability to influence other people's behaviour, whether or not one holds a position.

Power. Word in use. In daily life, physical strength, or electricity ("a power cut"). In management, the ability to influence what other people do.

Power has several sources that managers can name.

The two ideas are compared in a table with four questions.

A manager usually has both, and the two can work apart.

Authority works only when the people below accept it.

The central office shows that grouping by product does not remove the need for coordination. Writers still debate how far central control should reach, and the debate has no final answer. The design became a pattern for many large companies in later years.

REVIEW QUESTIONS

13.1 A manager has 4 employees who report to him directly, and each of them leads 3 others. State his span and the total number of people below him.

13.2 Explain why a store manager who must also keep accounts should have a smaller span than one who has no other duties.

13.3 Name the four bases of departmentalization and match each to an example from a bookshop.

13.4 Explain, in your own words, the difference between a tall structure and a flat structure, and give one cost of each.

mixture: the year-end count is feedback, the tags act during the event, and the weekly count of a sample gives feedback often enough to behave almost like a live check.

PAUSE AND CHECK 18.P2 A manager reads last month's complaints and changes the rota. Is this feedforward, concurrent or feedback control? Give one reason.

MONITORING

Monitoring means watching progress regularly through reports, visits and counts, so that deviations appear early. It is not a fourth type of control but a habit that supports all three. A weekly report on sales, or a short daily meeting at the shop door, are forms of monitoring.

Before moving on. Feedforward control acts before work begins, to prevent problems. Concurrent control acts while the work is being done. Feedback control acts after the work is done, using the results to correct later work. Organisations use all three, with monitoring to keep watch.

* * *

18.4 MAKING CONTROL EFFECTIVE

REQUIREMENTS OF EFFECTIVE CONTROL

For control to help, it must be timely, so that the information arrives while action is still useful. It must be accurate, because wrong figures lead to wrong actions. It must be understood by the people who are checked, and it must be economical, costing less than it saves. It should also focus on key points, and it should suit the plan and the people it serves.

BARRIERS TO CONTROL

Several barriers stand in the way. Employees may resist, because they fear blame or feel distrusted. Some results are hard to measure, such as the quality of service or the spirit of a team. Measuring everything is costly and can slow the work. And too much control can harm: when every step is checked, employees may stop thinking for themselves and wait for orders.

POWER AND CONTROL

A person who checks the work of others holds power: the power to praise, to blame and to correct. Power is a separate term, studied elsewhere in this book. The danger is that one person controls too much alone. If the same person does the work and also checks it, no one else can see a mistake or a deceit. For this reason good organisations separate the doing from the checking.

Timing deserves a closer look. A shop that learns of a shortage in the first week can search the till records, ask the staff and check the door while memories are fresh. A shop that learns in the twelfth month can do none of these. The same figure has very different value on different days, and so the speed of reporting is part of the

Monitoring is regular watching of progress, and it supports all three types.

Before moving on: control acts before work (feedforward), during work (concurrent) or after work (feedback), and monitoring supports all three.

Effective control meets a short list of requirements.

People and costs can stop control from working.

Control gives power to those who check, and that power needs its own limits.

Timing matters because information loses value as it ages.

* * *

SUMMARY

The summary restates the chapter in a few sentences.

Human factors are the traits, feelings and relationships of people that affect how they work, and they act at the levels of the person, the group and the organisation. Individual differences and perception explain why the same instruction can have different results. People form groups for security, belonging, status, help and shared interests. A team is a small group with joined skills, a common goal and shared accountability, and every group develops group norms. Creativity produces new and useful ideas; innovation puts them to use; and managers encourage both by listening and by making failure safe.

THINK IT THROUGH

Four situations test the chapter, from familiar to combined.

Each question below changes the situation a little more than the one before. Answer them in order.

15.T1 A new employee in a shop smiles at every customer. A second new employee rarely smiles but answers every question correctly. Explain, using individual differences, why a manager should not decide at once which is the better employee.

15.T2 The three sales assistants have an unwritten rule that they eat lunch at the same time every day. A manager asks them to eat at different times so that the counter is always covered. Name the norm, and explain one way the manager could win their agreement.

15.T3 Employees stop giving suggestions after a manager repeatedly answers *we tried that before*. Name the habit, explain its effect on creativity and innovation, and suggest one change.

15.T4 A clinic wants to find ways to reduce waiting time. It asks a nurse, a receptionist and a doctor to work together and report on the result. Is this group a team? Name the three features and say which of them the clinic must make clear for the group to be accountable.

WHERE PEOPLE GO WRONG

Some managers treat employees as if they were machines that respond in the same way to the same instruction. People differ in ability, attitude and background, and a method that works for one employee may fail for another.

Some students believe that a group and a team are the same thing. A team is a small group with joined skills, a common goal and shared accountability. A group of friends at lunch may have none of these.

Some students believe that norms are always a problem for managers. A norm that helps new employees learn the work is useful.

Mistake one: treating employees as if they were machines.

Mistake two: believing that a group and a team are the same.

Mistake three: believing that norms are always harmful or always helpful.

percentage points. *Answer.* The loss is Tk 21,000 above the standard, which is 1.6% against 1%. *Check.* $0.6\% \text{ of } 3,500,000 = 0.006 \times 3,500,000 = \text{Tk } 21,000$, which agrees.

Three slips are common. The first is to subtract the wrong way and so reverse the meaning of the sign. The second is to confuse a percentage with a percentage point: from 1% to 1.6% is a rise of 0.6 percentage points, but it is also a rise of 60 per cent of the standard rate. The third is to compare figures for different periods, such as a week's loss with a year's standard.

Three slips are common when comparing.

CORRECTING

A deviation calls for a question: why? The manager looks for the cause before choosing the action. Possible actions are to improve performance (stop the cause), to change the plan, or to revise the standard if it was unrealistic. If the deviation is small and temporary, the manager may do nothing, and this too is a decision.

The fourth step is to act on the difference.

Case file, illustrative. As the correction, Meghna and Sons fitted security tags at Tk 30,000 in 2026.

The tags are the correction chosen for the stock loss.

A correction also has a cost, so it must be weighed against the benefit. Suppose, as an illustration, that the tags remove the whole excess loss of Tk 21,000 a year. Then $\text{Tk } 30,000 \div \text{Tk } 21,000$ is about 1.4, so the tags repay their cost in a little under a year and a half. If they remove only half the excess, the repayment takes about three years. The cost and the benefit must both be judged.

A correction has a cost, so it must be weighed against its benefit.

The figure shows the process as a loop. To read it, start at the box *Set standard* and follow the arrows: measure, compare, correct. The dashed arrow shows that what is learned returns to the first box and to planning.

The figure shows the four steps as a loop that returns to planning.

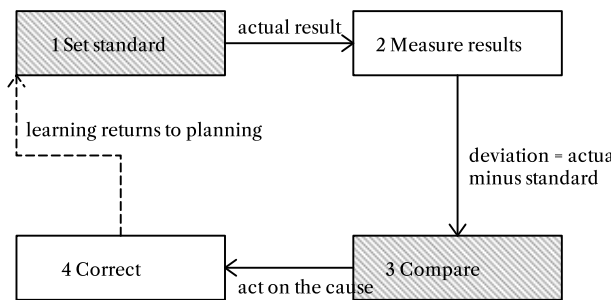


FIG. 18.1. *The control process as a loop*

PAUSE AND CHECK 18.P1 A cafe sets a standard of 5 spoiled cups in a week. In week 6 it has 9. What is the deviation, and what should the manager do first?

IN SHORT

The four steps are: set a standard, measure performance, compare

Viewpoints differ. The owner sees control as protection of money and name. The manager sees it as a way to learn early where help is needed. The employee may see it as distrust, unless the purpose is explained and the checks are fair. The customer meets control as reliable quality, such as the right book on the shelf at the right price. A good manager keeps all four views in mind when designing a check.

Before moving on. Controlling means checking results against plans and taking action to correct any difference. It has two halves, the check and the action. It is the last function of management, and its findings feed the next plan.

NATURE AND PRINCIPLES

Five features describe the nature of control. It is *continuous*, because results change every week, and one check at year end is not enough. It is *forward-looking*, because its purpose is to improve future results, not to blame the past. It is *universal*, because every manager at every level controls something, from the owner who watches profit to the cashier who counts the till. It is *tied to plans*, because without a plan there is nothing to check against. And it is *connected to people*, because people carry out the checks and people are checked.

Five principles guide the work. First, control should follow the plan, so that it checks what matters to the plan. Second, it should watch key points. A manager cannot watch everything, so the manager watches the key result areas, the few areas where results decide success. Third, it should act quickly, because a late correction is costly. Fourth, it should be economical: its cost must stay below the loss it prevents. Fifth, it should be flexible, so that it still works when the plan changes.

The economical principle deserves an example. Suppose a stall loses about Tk 500 a year to small errors. A system that costs Tk 2,000 a year to prevent this is poor value, since it spends four times what it saves. Controlling every item with equal care is therefore wasteful. Wise managers spend most effort where the possible loss is largest.

Before moving on. Control is continuous, forward-looking, universal, tied to plans, and connected to people. It should follow the plan, watch key points, act quickly, cost less than the loss it prevents, and remain flexible.

THE LINK WITH PLANNING

Planning and controlling are like two sides of a coin. Planning sets the objectives and chooses the course of action. Controlling checks whether the objectives are being reached. Without plans, control has nothing to measure against, so it cannot begin. Without control, planning is only hope, because no one learns whether the plan worked. Control also supplies facts for the next plan: a premise that proved wrong in this year can be corrected in the next.

Before moving on. Plans give controlling its measures, and controlling shows whether the plan works. The findings return to

Different people see control from different sides.

Before moving on: controlling is checking results against plans and correcting differences.

Control has certain features that hold in every organisation.

Good control follows a few simple principles.

A check costs money, so its cost must be compared with the loss it prevents.

Before moving on: control is continuous, forward-looking and universal, and it follows a few clear principles.

Planning and controlling depend on each other.

Recall: planning, Chapter 7, section 7.1 'What Planning Is'; objectives, Chapter 9, section 9.1 'Objectives'.

Plans give control something to measure, and control tells planners if the plan works.

Before moving on: planning and control are a pair, and each makes the other useful.

This chapter builds on two earlier sections.

Two employees with the same pay behave very differently.

Pay alone cannot explain the difference, so something else must be at work.

A person who works hard in one place and not in another has a reason that can be studied.

GO SLOWLY HERE.

Before a word is defined, an everyday case shows the idea.

Motivation. The inner push that makes a person choose to act and keep acting toward a goal.

Motivation is the inner push that starts and sustains a person's action toward a goal.

Motivation lives inside a person, so a manager sees only its results.

You will need: Chapter 15 'Human Factors, Groups and Creativity', section 15.1 'Human Factors in Management'; and Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

Imagine a shop with two sales assistants. They were hired on the same day. They receive the same pay, work the same hours, and sell the same kinds of goods. Yet one of them arrives early, learns the names of regular customers, and tidies shelves without being asked. The other watches the clock, serves each customer correctly, and leaves at the minute the day ends. Neither person breaks a rule. Both are paid fairly. Why is the work so different?

The first answer that comes to mind is pay, but pay is equal here, so pay cannot be the whole explanation. A second answer is skill, yet both assistants are equally skilled. A third answer is character: one is hardworking, the other lazy. That answer is attractive, because it is short. But it explains nothing, and it gives the owner no action to take. Moreover, the "lazy" assistant may behave very differently at a cricket match or a family wedding, where the same person works hard for hours without pay.

This fact is the key. The same person can be eager in one setting and careless in another. The difference lies in what that setting offers: what the person wants, what the person hopes to gain, and what the person believes is possible. Managers cannot see these inner matters, but they can study them, and they can arrange work so that the right wants are met. Whole theories have been built on this idea, and this chapter introduces the best-known ones.

* * *

16.1 WHAT MOTIVATION IS

MOTIVATION

Start with an everyday case. A student has an examination in ten days. One student studies for two hours every evening. Another student, with the same books and the same teacher, leaves the books closed until the last night. The first student has a reason that is strong enough to start the work and to keep it going each evening. The second student has a weaker reason, or none. Both students are able to study. What differs is the push.

That push has a name. **Motivation** means the inner push that makes a person choose to act and keep acting toward a goal. The goal may be a pass in the examination, a higher income, or the respect of friends. The word *keep* matters. A short burst of effort that stops after one day is weak motivation. Strong motivation continues when work becomes tiring or dull.

Three points follow. First, motivation is inside the person. A manager cannot look at an employee and measure it. A manager can only observe what the employee does: the effort, the care, the time. Second, motivation is not the same as ability. A highly skilled employee may have little motivation, and a less skilled employee may have a great deal. Third, motivation is not force. A person who

Before moving on: managers need motivation because it affects output, results and staff retention, and an incentive works only if it meets a need.

Every motivation theory starts from some assumptions about people.

The first two assumptions say that behaviour has causes and that people differ.

The last two assumptions say that managers shape conditions and that motives change.

Before moving on: the theories assume that behaviour has causes, that people differ, that managers shape conditions and that motives change.

Need theories explain motivation by asking which needs a person feels.

Recall: human factors, Chapter 15, section 15.1 'Human Factors in Management'.

If motivation grows from needs, a good theory should list the needs and show how they act.

GO SLOWLY HERE.

An everyday case begins the idea: a hungry person cares little about anything else.

Before moving on. Managers get results through people, so motivation affects productivity, effectiveness and the number of employees who stay. An incentive is a reward offered to encourage a person to act in a wanted way. It works only when it meets a need that the person feels.

ASSUMPTIONS ABOUT MOTIVATION

Every manager holds beliefs about what moves people, whether or not the manager has thought about them. These beliefs guide daily choices. The theories in this chapter make the beliefs explicit, so that we can examine them. Four assumptions run through most of the theories.

The first assumption is that behaviour has causes. People do not act for no reason, even when the reason is hidden. The second is that people differ. An owner may be moved by the wish to leave a legacy, a sales assistant by the wish to support a family, a customer by price. A manager must therefore look at each person, and not only at the group.

The third assumption is that managers cannot place motivation inside another person. They can only arrange the conditions in which a person's own motives are likely to work. The fourth is that motives change. An employee who first seeks a secure income may later seek responsibility. A manager who studied the person last year may need to study again.

Before moving on. Behaviour has causes. People differ. Managers shape conditions and do not insert motivation. Motives change over time. You should now be able to say: *motivation is an inner push from a felt need, and a manager can only arrange conditions in which that push is likely to appear.*

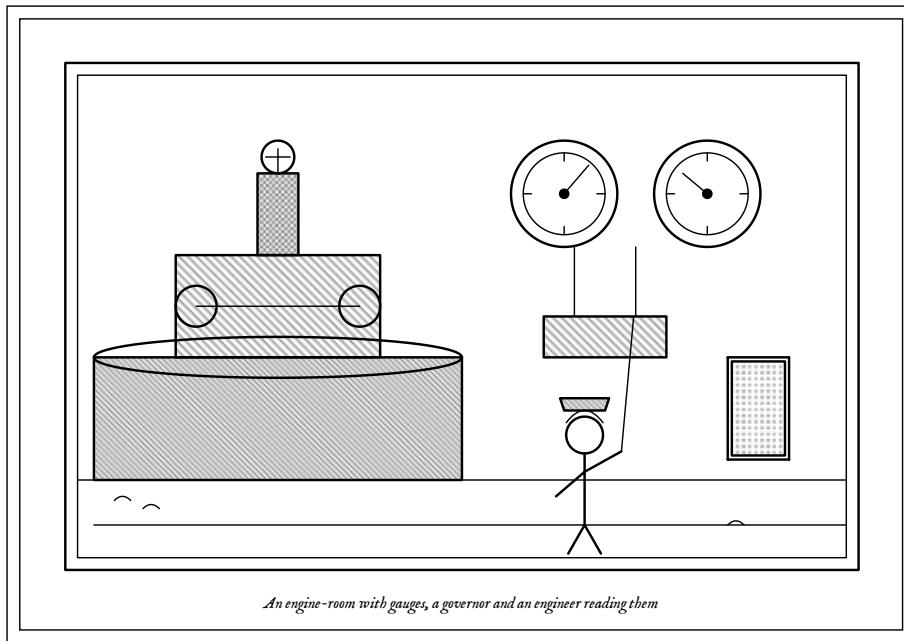
* * *

16.2 NEED THEORIES

In Chapter 15 you studied the human side of management. Now we ask a sharper question about that human side. If motivation grows from needs, which needs are there, and in what order do they matter? Three well-known answers follow. The first, and the most famous, is the work of Abraham Maslow. The second shows how any need leads to action. The third, by Frederick Herzberg, divides the features of a job into two groups.

MASLOW'S NEED HIERARCHY

Begin with a person who has eaten nothing for two days. Offer this person a place on a respected committee. Will the person care? Probably not. Hunger fills the mind, and the offer seems unimportant. Now offer a meal. The person eats, and the hunger fades. An hour later the same person starts to wonder about shelter for the night. Later again, about friends. The needs do not arrive all at once. They arrive in an order.



CHAPTER XVIII

Controlling

How does a manager know that the plan is working?



A plan is only a promise until its results are checked. This chapter explains how managers set standards, measure results, compare them with the plan and correct the difference, and it shows the three moments at which control can act.

the leader, the followers and the situation. Tannenbaum and Schmidt described seven patterns on a continuum. Autocratic, participative and free-rein are the classic styles. Emotional intelligence, transformational leadership and coaching leadership are modern ideas.

THINK IT THROUGH

A hospital ward has twelve nurses and one ward manager. The ward manager has authority from the hospital, but several nurses go to a senior nurse, who holds no position of authority, whenever they need advice or support. Use the ideas of this chapter to answer both questions.

17.T1 Is the senior nurse a leader, a manager, or both? Explain your answer using the definitions of leadership and management in section 17.1.

17.T2 A fire alarm sounds in the ward. Which point of the continuum should the ward manager use, and why? Then say which point suits a monthly meeting on improving the duty roster.

WHERE PEOPLE GO WRONG

The first misunderstanding is that leaders are born and cannot be made. Correction: some qualities help, but studies found no list that works everywhere, and people can learn many leadership skills. The second is that a leader and a manager are the same. Correction: a manager holds a position; a leader holds influence; the two overlap but are not equal.

The third misunderstanding is that one style is always best. Correction: each style fits some situations and fails in others. The fourth is that the participative style is always better because it is kinder. Correction: in an emergency, or with a very new group, slow discussion may cause harm. The fifth is that power is only the ability to punish. Correction: power has five sources, and referent power often lasts longest. The sixth is that leadership is always good. Correction: the definition describes influence toward group goals and says nothing about whether the goals are good.

IN HISTORY

Ernest Shackleton was a British explorer. On 8 August 1914, he sailed from Plymouth, in England, in the ship *Endurance*, with a crew of 28 men. His aim was to cross the continent of Antarctica. In January 1915 the ship became trapped in the ice of the Weddell Sea. The ice crushed her, and the men abandoned the ship on 27 October 1915. She sank in November 1915.

The men then lived for months on the ice, and in April 1916 they reached Elephant Island, an uninhabited island, in three small boats. On 24 April 1916, Shackleton and five others sailed one boat, the *James Caird*, about 800 miles to South Georgia to find help. He then crossed the mountains of that island on foot with two com-

The problem asks you to apply the chapter to two situations.

Six common misunderstandings about leadership are stated and corrected here.

Further misunderstandings concern the best style, the meaning of power, and the nature of a leader's goals.

Shackleton led the Endurance expedition from 1914 to 1916 and brought all 28 men home.

After the loss of the ship, Shackleton's choices kept his men together through months on the ice and in open boats.

Maslow's key claim is that a need that is met stops pushing, and a new need takes its place.

The cashier's training offer is a second kind of incentive.

The cashier's training touches esteem and growth, which sit high on the stair.

The hierarchy has limits, and Maslow himself did not claim a rigid order.

Before moving on, the need hierarchy ranks five levels of need and treats a met need as no longer pushing.

You should now be able to say what the five levels are and why an incentive may fail.

GO SLOWLY HERE.

A simple loop shows how a need becomes action and then returns.

Now the central claim, which many students misread. Maslow did not say that a person has only one need at a time. He said that an unmet lower need pushes most strongly, and that a need which is met stops pushing. When it stops pushing, the next need above it comes forward. The consequence for a manager is direct. An incentive works only when it touches the need that is pushing at that moment. A praise card means little to an employee who cannot pay the rent. A larger salary may mean less to an employee who already earns enough and wants respect.

Case file, illustrative. In the 2026 plan of Meghna and Sons, the cashier is offered training for a supervisor post.

Where does this offer sit on the stair? Training for a supervisor post offers a title and responsibility, which belong to esteem, and a chance to learn, which belongs to self-actualisation. Compare it with the bonus of one month's pay: that incentive speaks mainly to the lower levels, because income buys food, shelter and security. A shop owner who wants to keep a particular employee might offer either, but the theory tells the owner to ask which need is pushing that person now.

Now the exceptions. The order is not the same for everyone. Some people place respect above friendship, and some pursue a creative goal while poor. Needs also overlap, so a person may be partly secure and partly hungry for respect at the same time. Later researchers have found that the five levels do not always appear in the same order in every culture. Maslow himself said that the order is not fixed for every person. So the hierarchy is a useful guide to questions, not a rule that predicts one person's behaviour. **PAUSE AND CHECK 16.P2** A stock clerk has a steady job and a safe warehouse, and says the work feels pointless because nobody notices it. Which level of need is probably unmet?

IN SHORT

The need hierarchy ranks needs in five levels: physiological, safety, social, esteem and self-actualisation. An unmet lower need pushes hardest. When it is met, the next level comes forward.

Before moving on. The need hierarchy is Maslow's idea that human needs rank in five levels, and a higher need matters only after the lower ones are met. An incentive works when it touches the need now pushing. The order is a guide and not a law: it differs between people and cultures.

You should now be able to say: *needs rank in five levels, and an incentive works only when it meets the level that is pushing the person now.*

THE NEEDS-GOAL MODEL

Maslow told us which needs exist. A second idea, which we may call the needs-goal model, shows how any single need turns into action. Return to hunger. First comes the lack: the stomach is empty. Then comes tension, which is uncomfortable. The tension moves the person to act, so the person walks to the kitchen. The

the participative style lies near the middle, and the free-rein style lies near point 7.

Each style has a time and a place. The autocratic style fits an emergency, a very new group, or a task where one wrong move is costly; its risk is that people stop offering ideas. The participative style fits a problem where employees know things the leader does not, and where commitment matters; its risk is that decisions take longer. The free-rein style fits a group of skilled and motivated people; its risk is that the group drifts, if people lack skill or goals are not clear. The table sets the three side by side. Read each row from left to right: the first column names the style, the second says who decides, and the third and fourth give a fitting case and the main risk.

THREE CLASSIC LEADERSHIP STYLES

STYLE	WHO DECIDES	FITS WHEN	MAIN RISK
Autocratic	The leader alone	Time is short or danger is high	Ideas stop coming
Participative	The leader with the group	Employees know the work well	Slow decisions
Free-rein	The group	People are skilled and motivated	Drift without direction

Look again at the layout change. Nurul decided and told the staff, which is the autocratic style, and the first point of the continuum. Rafiq asked the sales assistants first and then decided, which is the participative style and lies near the fifth point of the continuum. Both are patterns that a leader can choose. A good leader knows the range, and chooses the pattern that fits the people and the problem.

Emotional intelligence means the ability to notice, understand and manage one's own and other people's feelings. For example, a manager who notices that a tired employee is losing patience, and who responds calmly, shows emotional intelligence. Two psychologists, Peter Salovey and John Mayer, introduced the term in 1990, and the writer Daniel Goleman made it widely known in 1995. Its parts include knowing one's own feelings, controlling them under pressure, understanding other people, and handling relationships with care. The idea has limits. It does not replace knowledge of the work, and researchers still debate how well it can be measured.

Transformational leadership means leading by inspiring followers to look beyond self-interest and to change themselves and the organisation. The political scientist James MacGregor Burns drew the idea in 1978 from his study of political leaders. He contrasted it with a leader who trades, giving pay or praise in return for work done. The trading leader gets the agreed work. The transformational leader hopes for more: followers who accept a larger purpose and who grow in skill and confidence. For example, an owner who explains to the whole shop why good service matters to the neighbourhood, and who asks each person to help build that reputation, is leading in this way. The risk is that an inspiring leader

Each style fits some situations and fails in others, as the table shows.

The case file shows both ends of the range of behaviour in one shop: deciding and telling, and asking and then deciding.

Emotional intelligence is the ability to understand and manage feelings, one's own and other people's.

Emotional intelligence. The ability to notice, understand and manage one's own and other people's feelings.

Transformational leadership inspires followers to look beyond self-interest and to change themselves and the organisation.

Transformational leadership. Leading by inspiring followers to look beyond self-interest and to change themselves and the organisation.

upper levels, since recognition and growth serve esteem and self-actualisation. The match is rough, not exact. Both views agree on one lesson for managers: fixing the conditions around a job removes complaints, but it does not by itself create commitment.

The exceptions deserve attention. Later studies found that the split between the two sets is less sharp than Herzberg described, and that pay can motivate some people, especially when income is short. The sensible reading is to treat the two factors as a prompt for two questions. Are the conditions around the job fair? Is the work itself worth doing?

Before moving on. The two-factor view says that conditions around the job, such as pay and rules, prevent unhappiness, while the work itself, such as achievement and responsibility, builds effort. The split is rough, but it reminds managers to ask both questions.

* * *

16.3 THEORY X AND THEORY Y

THEORY X

Suppose two shop managers each find an employee arriving late. The first manager says that employees avoid work unless watched, and installs a clock-in system. The second manager says that the employee probably has a reason, and asks. Each manager acts on a belief about people. Douglas McGregor, an American management scholar, gave these beliefs names in his book *The Human Side of Enterprise* (1960).

Theory X is a set of assumptions that people dislike work and must be directed and watched. Under these assumptions, the average person avoids responsibility, wants security above all, and works hard only when pressed by rules, checks and the fear of punishment. A manager who holds these assumptions tends to give detailed orders, supervise closely, and use punishment and reward as the main tools.

The exception to note is that close control is sometimes the right choice. A new employee learning a dangerous task needs clear directions and checking. The weakness appears when a manager applies Theory X to everyone, always. Capable people then feel distrusted, and some live down to the low expectation.

THEORY Y

Theory Y is a set of assumptions that people can enjoy work and seek responsibility. Under these assumptions, effort at work is as natural as rest or play, and people direct themselves toward goals they accept. A manager who holds these assumptions shares goals, delegates tasks, and offers growth, as in the training offered to the cashier.

USING THE THEORIES

Use the two theories with care. They describe what a manager

Critics say that the split is not sharp and that pay motivates some people.

Before moving on: Herzberg separated job features that prevent unhappiness from those that build effort.

Managers act on beliefs about people, and these beliefs have names.

Theory X. A set of assumptions that people dislike work and must be directed and watched.

Theory X assumes that people dislike work and must be directed and watched.

Theory X leads to close control, which can be right when tasks are dangerous or the workers are new.

Theory Y. A set of assumptions that people can enjoy work and seek responsibility.

Theory Y assumes that people can enjoy work and seek responsibility.

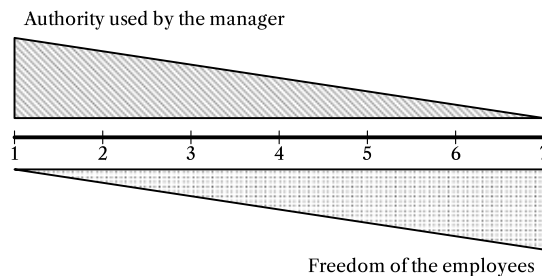
Both theories are assumptions held by managers, not facts.

alone. The trait approach still gives us something: it reminds us that a leader's character matters. Its weakness is that it ignores the situation, and a captain who commands well in a storm may command badly in a boardroom.

Researchers then asked a different question. Instead of asking "What qualities make a leader?", they asked "What should a leader do in this situation?" The situational approach holds that no single way of leading is best, and that the right way depends on the leader, the followers and the situation. A new employee who needs guidance, for example, may need a leader who gives clear directions. A skilled employee of many years may work best with a leader who gives freedom. The situational approach says that a good leader reads the situation and changes behaviour to fit it.

In 1958 two American writers, Robert Tannenbaum and Warren Schmidt, published an article in the *Harvard Business Review* called "How to Choose a Leadership Pattern". They described a continuum, which is a line along which a quantity changes in small steps. At one end, the manager uses a great deal of authority and the employees have little freedom. At the other end, the manager uses little authority and the employees have a great deal of freedom. Between the ends they described seven patterns. In the first, the manager decides and announces the decision. In the second, the manager decides and then tries to persuade the group. In the third, the manager presents a decision and invites questions. In the fourth, the manager presents a tentative decision that the group may change. In the fifth, the manager presents the problem, asks for suggestions and then decides. In the sixth, the manager sets limits and asks the group to decide. In the seventh, the group acts within limits set by the manager.

Figure 17.1 shows the continuum. To read it, begin at point 1 on the left and move along the line to point 7 on the right. The upper wedge shows the authority that the manager uses; it is widest at point 1 and narrows to the right. The lower wedge shows the freedom of the employees; it is narrowest at point 1 and widens to the right. A point on the line is one pattern of behaviour. Points 1, 4 and 7 carry short labels, and the paragraph above names the others.



1 Manager decides and announces
4 Manager proposes, group may change it
7 Group acts within limits set by manager

The situational approach says that effective leadership depends on the leader, the followers and the situation together.

Recall: authority and power, Chapter 14, 'Authority, Delegation and Coordination', section 14.1 'Authority and Power'.

In 1958 Tannenbaum and Schmidt described seven patterns of leader behaviour along one line.

To read the figure, follow the line from left to right: the manager's authority falls and the employees' freedom rises.

16.T3 Meghna and Sons offered a bonus of one month's pay. Suppose instead that employees in a new setting, a hospital pharmacy, already earn comfortable salaries. Which kind of incentive might work better, and why?

WHERE PEOPLE GO WRONG

Some students believe that more pay always raises motivation. Pay is powerful when income is the need that pushes. When it is not, extra pay may only remove a complaint.

Some students believe that a person must finish one level completely before the next appears. Needs overlap, and the order differs between people. The hierarchy is a guide to questions.

Some students believe that Theory X is always bad. It describes assumptions, and close direction is suitable where the task is dangerous or the person is new. The error lies in applying one set of assumptions to everyone.

Some students believe that a bonus and motivation are the same thing. The bonus is outside the person and is offered. Motivation is inside the person. The bonus works only if it meets a need.

IN HISTORY

In 1943 Abraham Maslow, then at Brooklyn College in New York, published *A Theory of Human Motivation* in the journal *Psychological Review*. The paper appeared in volume 50, issue 4, on pages 370 to 396. It arranged human needs in groups that rank one above another, and it said that when the strongest need is met, the next higher need comes forward. Maslow wrote in words, and the stair picture used in this chapter is a later way of showing his idea.

Managers later borrowed the idea, and Douglas McGregor used it in 1960 to explain his two sets of assumptions. Research has not confirmed the exact order of the five levels, and Maslow's paper itself says the order is not fixed for every person. Yet the central question, what does this person want now, remains one of the most useful questions in the book.

REVIEW QUESTIONS

16.1 Define motivation and need in your own words, and give one example of each from a shop.

16.2 Explain the difference between motivation and an incentive. Why can an incentive fail?

16.3 List Maslow's five levels from lowest to highest, and say why the order matters to a manager.

16.4 Place each incentive on Maslow's levels: a monthly salary, safety shoes for the stock clerk, a team outing, the title of senior sales assistant, a chance to design the window display.

16.5 Describe the five parts of the needs-goal model, using an

A common error is to think that money always motivates.

Another error is to treat the five levels as a rigid ladder.

A third error is to treat Theory X as wrong and Theory Y as right in every case.

A last error is to confuse motivation with an incentive.

Maslow's paper of 1943 gave managers a way to ask what people want.

The idea spread from psychology into management, where it is still debated.

This is the idea we call **leadership**. Leadership means the process of influencing people to work willingly toward group goals. Take the friend who planned the trip: she influenced the others, they worked willingly, and the goal belonged to the whole group. Each part of the definition matters. *Influencing* means that the leader changes what other people choose to do. *Willingly* means that people act from their own motivation, and not from fear alone. *Group goals* means that the aim is shared, and is not only the leader's private wish.

Leadership is a process, and a process has parts. The first part is the leader, the person who influences. The second is the followers, the people who are influenced; without followers there is no leadership, because influence needs someone to receive it. The third part is the situation: the task, the time available, and the dangers and chances around the group. The friend who plans a weekend trip faces a calm situation, so her gentle style works. The captain in a storm faces a dangerous one, and he needs a different way of influencing. Leadership changes when any of the three parts changes.

Two exceptions help to avoid mistakes. First, leadership does not depend on a title. An experienced sales assistant whom colleagues ask for advice may lead more than the store manager does. Second, the definition says nothing about whether the goals are good. A leader may influence people toward a harmful aim, and the process is still leadership. Leadership is therefore a description of what happens, and it is not a compliment. When we want to ask whether leadership is good, we look at its goals and its methods.

Now we separate two words that students often confuse. A manager is a person who holds a position in an organisation and carries the authority that comes with it. A leader is a person who influences others, whether or not that person holds a position. Some managers lead well. Some managers do not lead at all, and their employees obey the rules and no more. Some leaders are not managers: the friend on the trip, or the respected assistant at the counter. The best managers do both, because a manager who can also lead gets more from the same people. The table below sets the two ideas side by side. Read each row from left to right: the first column names a feature, and the other two columns say how it appears for a manager and for a leader.

A MANAGER AND A LEADER COMPARED

FEATURE	MANAGER	LEADER
Source of influence	Position and authority	Personal influence, with or without a position
Why people follow	The position requires it	They choose to follow
Main concern	Using resources well to	Moving people toward group

Recall: motivation, Chapter 16, 'Motivation', section 16.1 'What Motivation Is'.

Leadership is the process of influencing people to work willingly toward group goals.

Lead. Word in use. In daily life, "lead" can mean to be in front, as in a race, and "lead" is also a metal. In management, to lead is to influence people so that they follow willingly.

Leadership. The process of influencing people to work willingly toward group goals.

Leadership involves three things together: the leader, the followers and the situation.

Leadership has limits: it does not depend on a title, and it is not always used for good ends.

A manager holds a position, and a leader holds influence; one person can be both, or only one.

WORKED CASE: CLASSIFYING THE CONTROLS AT MEGHNA AND SONS

The worked case applies the three types to the control record.

Each action is placed by asking when it acts.

Rates give a fair comparison between the standard and the result.

The conclusion shows a balanced system with one gap.

Situation. Case file, illustrative. The control record of Meghna and Sons shows: a stock loss standard of 1% (Tk 35,000 of Tk 3,500,000); an actual 2025 loss of Tk 56,000, which is 1.6%; security tags at Tk 30,000 fitted in 2026; and a weekly count of 200 titles. *Question.* Place each action as feedforward, concurrent or feedback control, and say what is still missing.

Reasoning. The standard of 1%, set in advance, guides decisions before the year begins, and in that sense it belongs with feedforward control. The year-end figure of Tk 56,000 is a result found after the work, so it is feedback. The tags act at the moment an item leaves the shop, so they are concurrent. The weekly count of 200 titles comes after each week's trading, so it is feedback, but it is frequent and so gives early signals.

Notice that the record states both taka amounts and rates. The rate (1.6% against 1%) compares loss with the value of stock, so the result can be judged fairly in a year when stock is larger or smaller. The taka amount (Tk 21,000 above the standard) shows how much money is at stake.

Conclusion. The record has all three moments, but its feedforward element is thin: it has a standard and no preventive step, such as training staff in stock handling before they work. A shop that adds one would stop some losses before they begin. *Check.* The deviation was Tk 56,000 - Tk 35,000 = Tk 21,000, and each classification follows from when the action acts.

* * *

SUMMARY

The summary gathers the main ideas.

Controlling is checking results against plans and taking action to correct any difference. The control process has four steps: set a standard, measure performance, compare it with the standard, and correct. A standard is a level of performance set in advance as the measure of what is acceptable. Feedforward control acts before work, concurrent control acts during work, and feedback control acts after work. Effective control is timely, accurate, understood and economical, and it needs checks on those who check.

THINK IT THROUGH

18.T1 Meghna and Sons lost Tk 56,000 against a standard of Tk 35,000. Find the deviation. If the tags remove Tk 15,000 of the yearly loss, how many years do they take to repay their Tk 30,000 cost?

18.T2 A shop owner asks every employee to write down each action in a daily log, and checks the log every evening. Describe how this control could go too far, and one change that keeps the benefit.

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and meetings. For example, a weekly Sunday meeting lets a shop share news before the week begins. (*Chapter XIV*)

Corporate social responsibility. An organisation's duty to act for the good of society as well as for profit. It applies to every organisation, large or small. A bookshop that pays its employees fairly and recycles its waste paper is acting on this duty. (*Chapter VI*)

Creativity. The ability to produce new and useful ideas. Both words matter: an idea that is new but useless is a curiosity, and an idea that is useful but not new is a copy. A quiet corner where customers try a book before buying is an example. (*Chapter XV*)

Decision criterion. A standard used to judge which alternative is best. A criterion is a test that every option must pass or be scored against. For example, a student sets two criteria for a journey: arrive by 8:55 and spend no more than Tk 50. (*Chapter X*)

Decision making. Choosing one course of action from several, in order to solve a problem or use an opportunity. It begins when a manager sees a gap to close or a chance to use. For example, a shop owner picks one site for a second shop from two possible sites. (*Chapter X*)

Decision support system. A computer-based tool that helps managers study data and compare alternatives. It stores data, analyses it and answers what-if questions. It supports the manager and does not decide. For example, sales software that lists the top 20 titles each week helps a shop plan its stock. (*Chapter XI*)

Decision tree. A diagram of choices and chance events, drawn from left to right as branches. Squares mark choices, circles mark chance events, and the branch ends show results in taka. We calculate from right to left, finding the expected value at each circle, and then choose the highest value at the square. (*Chapter XI*)

Delegation. Giving an employee the right to make decisions and take action for a stated task. The task has limits, and the manager still checks the result. For example, an owner lets a store manager approve returns up to Tk 5,000. (*Chapter XIV*)

Departmentalization. The grouping of jobs into departments according to a common basis, such as function, product, customer or territory. A department is a group of jobs under one manager. A shop may group by function into sales, stock and accounts. (*Chapter XIII*)

Division of labour. The dividing of a large job into smaller tasks done by different people. Each worker becomes skilled in a narrow task, but the divided tasks must be joined again, which creates the need for coordination. (*Chapter XIII*)

Effectiveness. How far an organisation reaches its goals, whatever the cost. It asks whether the goal was met and ignores the input used. Example: a cook who serves all 200 guests is effective, even if food was wasted. (*Chapter III*)

Efficiency. How little input an organisation uses to reach a given result. For the same result, less input means higher efficiency. Example: relabelled shelves cut the daily search for stock from 3 hours to 1 hour. (*Chapter III*)

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- World Commission on Environment and Development, *Our Common Future*, 1987

member who breaks it may be laughed at or left out. Norms can help or harm the organisation. (*Chapter XV*)

Human factors. The traits, feelings and relationships of people that affect how they work. A trait is a lasting feature, such as patience. A feeling is a state of mind, such as worry. A relationship is a link between people, such as friendship. They act on the person, the group and the organisation. (*Chapter XV*)

Human relations school. A school of thought holding that workers' feelings and social bonds strongly affect output. It grew from the Hawthorne studies. Example: an employee who feels respected by the group may work harder, even when pay is the same. (*Chapter IV*)

Human skill. The ability to work with, understand and motivate other people. It includes listening and explaining. A manager needs it at every level, because results come through people. A supervisor who notices a tired cashier and explains a new rule well shows human skill. (*Chapter II*)

Incentive. A reward offered to encourage a person to act in a wanted way. It comes from outside the person and works only when it meets a need the person feels. For example, a bonus of one month's pay for meeting a sales target. (*Chapter XVI*)

Informal organization. The network of friendships and social groups that forms among employees on its own. No manager designs it and no document records it. A lunch group that shares book tips is an example. It can help or harm the organisation. (*Chapter XII*)

Innovation. Turning a new idea into a new product, service or method that is put to use. Creativity is the seed and innovation is the crop. A suggestion on paper is creativity; a reading corner built and used by customers is innovation. (*Chapter XV*)

Internal environment. The conditions inside an organisation that its managers can change directly, such as its people, its structure and its culture. When a shop hires a new employee or changes who works the counter, it is changing its internal environment. (*Chapter V*)

Key result area. A part of the organisation's work where results are vital to success. A bookshop might name sales, stock, customer service, returns and profit as its key result areas, and set objectives in each. (*Chapter IX*)

Leadership. The process of influencing people to work willingly toward group goals. It involves a leader, followers and a situation, and it does not depend on a title. Example: a friend who persuades five others to share the costs of a trip. (*Chapter XVII*)

Leadership style. The usual way a leader makes decisions and treats followers. It is a habit and not a rule. Example: a leader who nearly always decides alone and then informs the group has a different style from one who nearly always asks first. (*Chapter XVII*)

Line authority. The right of a manager to direct the work of employees in the chain of command. Orders travel down the line of positions from the top. For example, a store manager directs the

18.3 A standard is a level of performance set in advance as the measure of what is acceptable. Examples: sell at least 120 books a day (quantity); serve each customer within 5 minutes (time); keep the stock loss to Tk 35,000 a year (cost).

18.4 Standard = $0.015 \times 200,000 = \text{Tk } 3,000$. Deviation = $4,200 - 3,000 = \text{Tk } 1,200$ above the standard.

18.5 Training before the first day: feedforward. Watching at the till: concurrent. Reviewing last month's sales: feedback. An alarm on a door: concurrent.

18.6 Feedforward: prevents problems, but depends on forecasts. Concurrent: corrects at once, but costs constant attention. Feedback: easy to apply since results are known, but too late to undo the past.

18.7 Any three of: timely, accurate, understood, economical, focused on key points. Barriers, any two of: resistance, results hard to measure, cost, too much control.

18.8 Planning needs controlling to learn whether the plan works. Controlling needs objectives to supply the standards against which results are compared.

accounts is organizing. It follows planning in the management process. (*Chapter XII*)

Performance appraisal. A review of how well an employee has done against agreed standards. In management by objectives, the standard is the objective agreed at the start. Example: an employee who agreed 14 training hours and received 10 is told the gap. (*Chapter IX*)

Plan. A written or spoken statement of what will be done, by whom and by when, to reach an objective. A plan names the action, the person and the time. Example: "Place the book order by 12 December; the manager will do it." (*Chapter VII*)

Planning premise. An assumption about future conditions on which a plan is built. The manager writes it down so that everyone shares it. Example: a shop assumes that publishers raise prices by 5%, and builds its order budget on that assumption. (*Chapter VII*)

Policy. A general guide for decisions, setting limits within which choices are made. It guides but leaves room for judgement. Example: "We exchange books within 7 days", where the employee decides about a damaged book on the last day. (*Chapter VIII*)

Power. The ability to influence other people's behaviour, whether or not one holds a position. It comes from the person, through knowledge, rewards, punishment or liking, and leaves with the person. For example, a trusted colleague persuades others to follow a plan. (*Chapter XIV*)

Principle of management. A broad guide to action drawn from experience, to be adapted and not applied like a law. It may suit one organisation and fit another badly. Example: dividing work among specialists suits a factory more than a shop of seven people. (*Chapter III*)

Procedure. A fixed series of steps to be followed for a repeated task. It says how to do the work, in a fixed order. Example: ask for the receipt, check the date, check the book, give the new book, record the exchange. (*Chapter VIII*)

Productivity. The amount of output an organisation obtains for each unit of input it uses. It equals output divided by input. Example: 41,000 copies sold by 7 employees is 5,857 copies per employee. (*Chapter III*)

Profession. An occupation that requires long training, a body of knowledge and a code of conduct enforced by an association. Example: medicine, where doctors normally register with a medical council. Managing a small shop does not require a licence. (*Chapter III*)

Referent power. Power that comes from other people's liking and admiration for a leader. It belongs to the relationship, not the position. Example: a sales assistant whom colleagues respect and wish to resemble can influence them without any reward. (*Chapter XVII*)

Resource. Anything an organisation uses to reach its goals, such as people, money, materials and information. A bookshop's resources include its books, cash, shelves, staff and list of school customers.

group example: colleagues at lunch. A team example: staff preparing a school order.

15.3 Stopping ideas: ridicule, saying we tried that before, or silence in reply. Encouraging ideas: listening, giving time and freedom, making failure safe, or providing a path from idea to action.

15.4 Creativity is producing a new and useful idea; innovation is putting the idea into use. The 14 suggestions show creativity, and the reading corner, built in April 2026 at Tk 45,000, shows an idea put into use.

CHAPTER XVI · MOTIVATION

16.p1 Answers vary. A good answer names a goal, such as a skill, a result or the respect of others, that kept the person working without orders.

16.p2 Esteem. The job is steady and safe, so levels 1 and 2 are met, but the clerk feels unnoticed and wants respect for the work.

16.t1 Possible answers: the first assistant may feel a need for esteem and belonging (known by regular customers), while the second may feel mainly a need for income and safety, which the pay already meets. The same pay meets one need and leaves the other untouched.

16.t2 Theory X: it assumes employees must be watched. A result could be distrust, or careless work that matches the low expectation. A Theory Y change: share the till targets with staff and give them responsibility for checking their own sales.

16.t3 Their lower needs are already met, so extra pay may remove no complaint and add little push. Incentives that touch esteem and growth, such as training, a senior title or responsibility for a project, may work better.

16.1 Motivation is the inner push that makes a person choose to act and keep acting toward a goal, such as an assistant who tidies shelves each day to be trusted. A need is a lack a person feels and wants to fill, such as the wish for more income.

16.2 Motivation is inside the person; an incentive is a reward offered from outside. An incentive fails when it does not meet a need the person feels, or when the goal seems out of reach.

16.3 Physiological, safety, social, esteem, self-actualisation. The order matters because the need pushing now is the one an incentive must meet.

16.4 Salary: physiological (and safety). Safety shoes: safety. Team outing: social. Title of senior sales assistant: esteem. Designing the window display: self-actualisation (and esteem).

16.5 Need (wish for income), drive (tension), action (working carefully toward the target), goal (the bonus), need met and tension falls. The loop may restart with a new need.

16.6 Hygiene factors are the conditions around the job (pay, rules, supervision, security); they prevent unhappiness. Motivators are in the work itself (achievement, recognition, responsibility, advancement); they build effort.

and punishment. For example, a clock-in system for every employee. (*Chapter XVI*)

Theory Y. A set of assumptions, named by Douglas McGregor in 1960, that people can enjoy work and seek responsibility. A manager holding it tends to share goals and delegate. For example, offering training for a supervisor post. (*Chapter XVI*)

Time and motion study. Timing each movement of a task to remove wasted effort and set a fair standard. The tool of scientific management. Example: timing a carton at 14 minutes, changing the method, and timing it again at 9 minutes. (*Chapter IV*)

Top manager. A manager at the highest level, who sets the direction of the whole organisation. The top manager decides what the organisation will do and what it will not do. In a bank this is the managing director, and in a small family shop it is the owner-manager. (*Chapter II*)

Transformational leadership. Leading by inspiring followers to look beyond self-interest and to change themselves and the organisation. It differs from a leader who trades pay or praise for work done. Example: an owner who explains why good service matters to the neighbourhood. (*Chapter XVII*)

Triple bottom line. A way of judging an organisation by its profit, its effect on people and its effect on the natural environment. John Elkington proposed it in 1994. It adds two more "last lines" to the single last line of the accounts, which shows profit. (*Chapter VI*)

Uncertainty. A condition in which neither the results nor their chances can be estimated, often because no records or comparisons exist. A manager must rely on judgement. For example, a stall owner opening in a town with no records cannot say what sales to expect. (*Chapter X*)

12.4 People join to meet needs such as company, help, information and belonging. A manager can learn which groups exist, listen to them, share honest information and use their strengths, while watching for harm.

CHAPTER XIII · STRUCTURE: SPAN, DEPARTMENTS AND FORCES

13.p1 The tasks are similar (both routine, in one room) and the employees are near the manager. Either of these, or the small number of other duties, suggests a wider span.

13.p2 Territory, because each branch is a place with its own manager.

13.t1 Too wide: one owner cannot follow 12 people and answer all their questions. Acceptable: all 12 do the same counter work in one place, so the tasks are similar and near, and the rules may be clear.

13.t2 Taller, because a new level has been added. The owner's span falls from 12 to 2 and each manager has 6. A benefit is closer supervision; a cost is the added wages and slower messages through the extra level.

13.t3 Territory. Grouping by product would give each product its own department, and each department would work in all three cities, so the managers would be responsible for a product everywhere and not for a place.

13.t4 One reasonable answer: a span of about 3 to 5 for the senior doctor, since the clinic is small and all work in one place (size and nearness), and a grouping by function (doctors, nurses, reception) because the clinic is small and its work is routine. Other reasoned answers are acceptable.

13.1 His span is 4, because only the 4 report to him directly. The people below him number $4 + (4 \times 3) = 16$.

13.2 The accounts work takes his time, so he has less time to lead. A smaller span lets him give each employee enough attention.

13.3 Function (sales, stock, accounts); product (textbooks, story-books, stationery); customer (walk-in, schools, online); territory (a branch in each city).

13.4 A tall structure has many levels and small spans; a flat structure has few levels and wide spans. A cost of tall: slow messages and high cost of many managers. A cost of flat: managers may be overloaded.

CHAPTER XIV · AUTHORITY, DELEGATION AND COORDINATION

14.p1 The new manager holds the authority, because it belongs to the post. The ten-year employee holds power, because the other employees follow that person's influence. Power can outweigh authority when employees do not yet accept the new manager.

14.p2 Authority is missing. Rafiq has the duty but not the right to act, so every return waits for Nurul and the task cannot be done as delegated. The three ideas must be in balance.

14.t1 Tk 5,400 is above Rafiq's limit of Tk 5,000, so Nurul must approve it. Rafiq's authority covers returns up to Tk 5,000 only.

parts connect becomes the main ability. A first-line manager sees the work directly.

2.3 People (the sales assistants), money (the day's takings), materials (stock and shelves), information (sales records or the list of school customers).

2.4 A functional manager is in charge of one kind of work, such as the accounts department. A general manager is in charge of several kinds of work together, such as the head of a whole branch.

CHAPTER III · THE FUNCTIONS OF MANAGEMENT AND HOW MANAGEMENT IS JUDGED

3.p1 Week 1: $600 \div 3 = 200$ books per employee. Week 2: $800 \div 5 = 160$. Productivity fell, although the total rose.

3.p2 It is effective, because every planned parcel was delivered. It is not efficient, because it used twice the budget.

3.t1 The firm reached its goal of 300 parcels, so it was effective. The vans used twice the distance needed, so input was wasted and it was not efficient. Better planning of routes would help most.

3.t2 $45,100 \div 8 = 5,637.5$, so about 5,638 copies per employee. This is lower than the 5,857 of 2025, because input grew while output stayed at the forecast level.

3.1 Planning, organising, leading, controlling. Examples: set the sales target; assign jobs to the 7 employees; encourage the sales assistants; compare actual sales with the target.

3.2 Before: $1,800 \div 4 = 450$ loaves per baker. After: $2,100 \div 4 = 525$. Change: $75 \div 450 = 0.1667$, a rise of about 16.7%.

3.3 Productivity is a ratio. If input grows faster than output, the ratio falls. Example: 600 books with 3 employees is 200 each; 800 books with 5 employees is 160 each.

3.4 Effectiveness is how far goals are reached, whatever the cost. Efficiency is how little input a result needs. A firm that does a wrong job very cheaply is efficient but not effective.

3.5 Long training, a body of knowledge, and a code of conduct enforced by an association. For a small shop, no licence or governing body is required, so the enforced code and association are missing.

3.6 A law of physics holds every time. A principle is drawn from experience and must be adapted, because it may suit one organisation and fit another badly.

CHAPTER IV · THE EVOLUTION OF MANAGEMENT THOUGHT

4.p1 Unity of command is broken, because the employee has two managers giving different orders. The shop should give each employee one manager, and let the two managers agree on the work before they speak to the employee.

4.t1 The first principle: replace rule-of-thumb methods with a science. The fixed shelf-position list replaced habit with one tested method, and the timing showed that it saved 5 minutes a carton.

4.t2 The saving per carton would be $14 - 11 = 3$ minutes. For 10 car-

10.p2 The price is certain, because the list is fixed for a year and the result is known. The number of customers is uncertainty, because nobody can say the result or estimate its chance. If past records of similar books existed, it would be risk.

10.t1 Supplier A's price is certainty. Supplier B's delivery times are uncertainty, since no records exist to estimate a chance. Step 1 and step 2 come first, but the manager learns what to find out at step 4, evaluating, where the missing delivery facts stop the test.

10.t2 The criteria stay the same, but the figures in them change: the expected sales fall, so criterion A (sales per taka spent) and criterion B (sales less rent) fall for Mirpur. Rent and fitting costs do not change. The owner should re-evaluate the options with the corrected figure and compare again.

10.1 Decision making is choosing one course of action from several, in order to solve a problem or use an opportunity. Example: a shop owner chooses one of two sites for a second shop.

10.2 A problem is a gap between what is and what should be, and an opportunity is a chance to do better than before. An opportunity is easier to miss, because nobody complains about a chance that was lost.

10.3 The steps are: name the problem or opportunity; set the criteria; develop the alternatives; evaluate them; choose; implement and follow up. Criteria come first so that they cannot be bent to favour the option already liked.

10.4 A decision criterion is a standard used to judge which alternative is best. Two examples are the price per copy and the days a supplier takes to deliver.

10.5 (a) Certainty: the lease fixes the amount. (b) Risk: past records allow the chance to be estimated at 4 in 10. (c) Uncertainty: no records exist, so neither the results nor their chances can be estimated.

10.6 It is risk, because the chance can be estimated from records. The estimated chance of a late order is 3 out of 12, which is 1/4.

10.7 She must go back to step 2 and rank the criteria, because no alternative now passes both tests. She must decide whether cost or arrival time counts more.

CHAPTER XI · DECISION TOOLS AND GROUP DECISIONS

11.p1 $0.8 \times 500 = 400$ and $0.2 \times (-1,000) = -200$, so the expected value is $400 - 200 = \text{Tk } 200$.

11.t1 Worth it: an important, unclear decision that affects many people, or one that needs the support of employees to be carried out. Not worth it: an urgent or simple choice, or one where a single person holds both the knowledge and the authority.

11.t2 $0.4 \times 900,000 = 360,000$ and $0.6 \times (-300,000) = -180,000$, so the expected value is Tk 180,000. This is still above Tk 0, so opening still beats waiting on expected value.

11.t3 $900,000p - 300,000(1 - p) = 0$ gives $1,200,000p = 300,000$, so $p = 0.25$. Check: $0.25 \times 900,000 = 225,000$ and $0.75 \times 300,000 =$

5.4 Scanning is regularly collecting and studying information about outside change. Adapting is changing the organisation to fit an outside force. Influencing is acting on the outside force itself. The price-match list is adapting, because the shop changed its own prices.

CHAPTER VI · SOCIAL RESPONSIBILITY, ETHICS AND SUSTAINABILITY

6.p1 (a) is an ethical dilemma, because both options cost someone: dismissing one employee harms that person, and cutting both pays harms both. (b) is only a temptation, because the right answer is known and only the will is tested.

6.t1 The people affected are the employees, the owner, the customer who ordered, and the families of the employees. Step 2 says power creates a duty of equal size, and the owner has power over the employees' safety. Step 4 says the cost of danger must be counted, not left with the employees. The owner should refuse the unsafe night shift and ask the customer for more time.

6.t2 Profit: Tk 750,000 in 2025. People: 7 employees, which is a first fact only. Natural environment: 380 kg of unsold paper sent to recycling in 2025. Two missing facts are, for example, measures of pay and of safety for the employees, and a measure of energy or waste use.

6.1 Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. Example: a bookshop refuses to sell stolen books and sends waste paper to recycling.

6.2 For: society gives organisations permission to operate, so harm to society returns as lost trust. Against: the money belongs to the owners, who did not choose to spend it on society. Owners would more likely make the argument against; a manager or an employee would more likely make the argument for.

6.3 The steps are: 1 social power; 2 responsibility matches power; 3 open dealing with society; 4 social costs counted and priced; 5 help with social problems. The iron law warns that those who do not use power in a way society considers responsible will tend to lose it.

6.4 In a dilemma every option has a cost to someone, such as dismissing one of two employees or cutting both pays. In a temptation one option is plainly wrong and only the will is tested, such as selling stolen books for profit.

6.5 (a) Both: the law requires it and it is right. (b) Neither: it is forbidden and wrong. (c) Ethical but not required by law: it is a kind act that no law demands.

6.6 The rival puts employees and customers at risk, and society bears the cost if harm occurs, so Davis step 4 is ignored. The rival also risks its good name and a stricter law. A manager at our shop should keep our safety standard, not copy the rival, and be ready to explain why to customers.

CHAPTER VII · PLANNING: MEANING, NATURE AND STEPS